



Advanced Graph Theory (MMTHDGT325)

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ADVANCED GRAPH THEORY

Course No: MMTHDGT325

Examination:

(a). Assessment

(b). Theory

Time Duration: 2.5 hrs

Total Credits: 04

Total Marks: 100

Max. Marks: 28

Max. Marks: 72

Min. Pass Marks: 40

Objectives: To expose the student to the various concepts of graph theory to model many types of relations and processes in physical, biological, social and information systems.

UNIT -I

Colorings

Vertex coloring, chromatic number $\chi(G)$, bounds for $\chi(G)$, Brook's theorem, edge coloring, Vizing's theorem, map coloring, six color theorem, five color theorem, every graph is four colourable iff every cubic bridgeless plane map is 4-colorable, every planar graph is 4-colorable iff $\chi'(G) = 3$, Heawood map-coloring theorem, uniquely colorable graphs

UNIT -II

Matchings

Matchings and 1-factors, Berge's theorem, Hall's theorem, 1-factor theorem of Tutte, antifactor sets, f-factor theorem, f-factor theorem implies 1-factor theorem, Erdos-Gallai theorem follows from f-factor theorem, degree factors, k -factor theorem, factorization of K_n .

UNIT -III

Edge graphs and eccentricity sequences

Edge graphs, Whitney's theorem on edge graphs, Beineke's theorem, edge graphs of trees, edge graphs and traversability, total graphs, eccentricity sequences and sets, Lesniak's theorem for trees, construction of trees, neighbourhoods, Lesniak theorem graphs.

UNIT -IV

Groups in graphs and graph spectra

Automorphism groups of graphs, graphs with a given group, Frucht's theorem, Cayley digraph, spectrum of a graph, spectrum of some graphs-regular graph, complement of a graph, edge graph, complete graph, complete bipartite, cycle and path, Laplacian spectrum, energy of a graph, Laplacian energy.

Recommended Books:

1. R. Balakrishnan, K. Ranganathan, A Text Book of Graph Theory, Springer-Verlag, New York
2. B. Bollobas, Extremal Graph Theory, Academic Press.
3. F. Harary, Graph Theory, Addison-Wesley.
4. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science, Prentice Hall
5. S. Pirzada, An Introduction to Graph Theory, Universities Press, Orient Blackswan, 2012
6. W. T. Tutte, Graph Theory, Cambridge University Press.
7. D. B. West, Introduction to Graph Theory, Prentice Hall

Chapter 1

Coloring of Graphs

INTRODUCTION

Graph coloring originated with the celebrated **Four Color Problem**. In 1852, Francis Guthrie observed that four colors appeared sufficient to color the regions of a map so that adjacent regions received different colors. The problem was later popularized by Augustus De Morgan and Arthur Cayley and became one of the central questions in graph theory.

Alfred Kempe proposed an influential proof in 1879. Although the proof was later shown to be incorrect, several of its ideas survived. In 1890, Percy Heawood used related techniques to prove the **Five Color Theorem** and to extend map-coloring ideas to surfaces of higher genus. Later developments shifted attention from planar maps to vertex and edge colorings of general graphs. Brooks' theorem (1941) related the chromatic number $\chi(G)$ to the maximum degree $\Delta(G)$, while Vizing's theorem (1964) showed that the edge chromatic number of a simple graph is either $\Delta(G)$ or $\Delta(G) + 1$.

The Four Color Theorem was finally proved by Appel and Haken in 1976 using computer-assisted methods. The ideas developed around the problem—proper colorings, critical graphs, degree bounds, and structural characterizations—remain fundamental in modern graph theory.

1.1 Vertex Coloring

1.1.1 Basic Definitions

Definition 1.1.1 (Vertex Coloring). Let $G = (V, E)$ be an undirected graph. A **proper vertex coloring** with k colors is a function $c : V \rightarrow \{1, 2, \dots, k\}$ such that $uv \in E$ implies $c(u) \neq c(v)$. A proper coloring using at most k colors is called a **k -coloring**.

The color labels themselves are irrelevant; positive integers are used only for convenience.

Definition 1.1.2 (Chromatic Number). The **chromatic number** of G , denoted by $\chi(G)$, is the minimum number of colors required for a proper coloring; equivalently,

$$\chi(G) = \min\{k : G \text{ admits a proper } k\text{-coloring}\}.$$

Elementary Examples

- If G has no edges and at least one vertex, then $\chi(G) = 1$.
- If G contains at least one edge, then $\chi(G) \geq 2$.
- For the complete graph K_n , $\chi(K_n) = n$.
- For the cycle C_n ($n \geq 3$),

$$\chi(C_n) = \begin{cases} 2, & n \text{ even,} \\ 3, & n \text{ odd.} \end{cases}$$

Proof for cycles. Label the vertices $v_1, \dots, v_n$ consecutively. If n is even, alternating two colors around the cycle gives a proper coloring, and since C_n contains an edge, $\chi(C_n) = 2$. If n is odd, alternating two colors forces v_n and v_1 to receive the same color although they are adjacent, so two colors are impossible; three colors suffice. Hence $\chi(C_n) = 3$. ■

1.1.2 Trees and Bipartite Graphs

Theorem 1.1.3. *Every tree with at least two vertices is 2-chromatic.*

Proof. Let T be a tree with at least two vertices. Since T contains an edge, $\chi(T) \geq 2$. Choose a root $r \in V(T)$ and color each vertex u according to the parity of its distance from r : use color 1 if $d(r, u)$ is even and color 2 if it is odd.

If $xy \in E(T)$, then the unique paths from r to x and r to y differ in length by exactly one. Thus $d(r, x)$ and $d(r, y)$ have opposite parity, so adjacent vertices receive different colors. Hence T is 2-colorable and therefore $\chi(T) = 2$. ■

Theorem 1.1.4. *A graph G is 2-colorable if and only if it contains no odd cycle.*

Proof. If G is 2-colorable, the colors must alternate around every cycle; therefore every cycle has even length.

Conversely, suppose G has no odd cycle. It suffices to consider each connected component separately. Choose a root r in a connected component and define

$$V_1 = \{u : d(r, u) \text{ is even}\}, \quad V_2 = \{u : d(r, u) \text{ is odd}\}.$$

No two vertices in the same set can be adjacent, for otherwise the corresponding shortest paths from r , together with that edge, would create an odd cycle. Hence every edge joins V_1 to V_2 , so coloring V_1 and V_2 with two different colors gives a proper 2-coloring. ■

Theorem 1.1.5. *A nonempty graph G is 2-chromatic if and only if G is bipartite.*

Proof. If G is bipartite with bipartition $V(G) = V_1 \cup V_2$, color V_1 with color 1 and V_2 with color 2. Since G is nonempty, it contains an edge, so $\chi(G) = 2$.

Conversely, if G has a proper 2-coloring, let V_1 and V_2 be the sets of vertices receiving the two colors. No two vertices within the same set are adjacent, so every edge joins V_1 to V_2 . Hence G is bipartite. ■

Remark 1.1.6. Thus, for a nonempty graph,

$$G \text{ is bipartite} \iff G \text{ is 2-colorable} \iff G \text{ has no odd cycle.}$$

Theorem 1.1.7 (A General Upper Bound). *For every graph G , $\chi(G) \leq \Delta(G) + 1$.*

Proof. We induct on $n = |V(G)|$. The case $n = 1$ is immediate. Assume the result holds for graphs with $n - 1$ vertices, and choose $v \in V(G)$. By induction,

$$\chi(G - v) \leq \Delta(G - v) + 1 \leq \Delta(G) + 1.$$

Thus $G - v$ can be colored with at most $\Delta(G) + 1$ colors. Since v has at most $\Delta(G)$ neighbors, at most $\Delta(G)$ colors are forbidden at v , leaving at least one available color. Hence the coloring extends to G , proving $\chi(G) \leq \Delta(G) + 1$. ■

1.1.3 Color-Critical Graphs

Definition 1.1.8 (Color-Critical Graph). A graph G is k -critical if $\chi(G) = k$ and $\chi(G - v) = k - 1$ for every $v \in V(G)$.

Theorem 1.1.9. *If G is k -critical, then:*

- (a) G is connected;
- (b) $\delta(G) \geq k - 1$;
- (c) there do not exist proper subgraphs G_1, G_2 such that $G = G_1 \cup G_2$ and $G_1 \cap G_2$ is complete;
- (d) if $k > 1$, then $G - v$ is connected for every $v \in V(G)$.

Proof. a. Assume G is not connected. Since $\chi(G) = k$, there is a component G_1 of G such that $\chi(G_1) = k$. If v is any vertex of G which is not in G_1 , then G_1 is a component of the subgraph $G - v$. Therefore, $\chi(G - v) = \chi(G_1) = k$. This contradicts the fact that G is k -critical.

b. Assume to the contrary that v be a vertex of G so that $d(v) < k - 1$. Since G is k -critical, the subgraph $G - v$ has a $(k - 1)$ -colouring. As v has at most $k - 2$ neighbours, these neighbours use at most $k - 2$ colours in this $(k - 1)$ -colouring of $G - v$. Now, colour v with the unused colour and this gives a $(k - 1)$ -colouring of G . This contradicts the given assumption that $\chi(G) = k$. Hence every vertex v has degree at least $k - 1$.

c. Let $G = G_1 \cup G_2$, where G_1 and G_2 are subgraphs with $G_1 \cap G_2 = K_r$. Since G is k -critical, therefore G_1 and G_2 both have chromatic number at most $k - 1$. Consider a $(k - 1)$ -colouring of G_1 and a $(k - 1)$ -colouring of G_2 . As $G_1 \cap G_2$ is complete, in the overlap, every vertex in $G_1 \cap G_2$ has a different colour (in each of the $(k - 1)$ -colourings). This implies that colours in the $(k - 1)$ -colouring of G_2 can be rearranged such that it assigns the same colour to each vertex in $G_1 \cap G_2$, as is given by the colouring of G_1 . Combining the two colourings then produces a $(k - 1)$ -colouring of all of G . This is impossible, since $\chi(G) = k$. Thus no subgraphs of the type G_1 and G_2 exist.

d. Assume $G - v$ is disconnected, for some vertex v of G . Then $G - v$ has a subgraph H_1 and H_2 with $H_1 \cup H_2 = G - v$ and $H_1 \cap H_2 = \emptyset$. Let G_1 and G_2 be the subgraphs of G , where G_1 is induced by H_1 and v , while G_2 is induced by H_2 and v . Then $G = G_1 \cup G_2$ and $G_1 \cap G_2 = K_1$ (with K_1 as a single vertex). This contradicts (c) and thus $G - v$ is connected. ■

Corollary 1.1.10. *Every k -chromatic graph has at least k vertices of degree at least $k - 1$.*

Proof. Starting from a k -chromatic graph G , repeatedly delete vertices whenever the chromatic number remains k . This process ends with a k -critical subgraph H . By Theorem 1.1.9, $\delta(H) \geq k - 1$, and since every k -chromatic graph has at least k vertices, H contains at least k vertices of degree at least $k - 1$. Their degrees in G are at least as large. ■

Theorem 1.1.11. *Let*

$$k = \max\{\delta(G') : G' \text{ is a nonempty subgraph of } G\}.$$

Then $\chi(G) \leq k + 1$.

Proof. Let $\chi(G) = r$ and let H be an r -critical subgraph of G . By Theorem 1.1.9, $\delta(H) \geq r - 1$. Since H is a subgraph of G , the definition of k gives $\delta(H) \leq k$. Hence $r - 1 \leq k$, so $\chi(G) = r \leq k + 1$. ■

Theorem 1.1.12. *Let G have degree sequence $d_1 \geq d_2 \geq \dots \geq d_n$. Then*

$$\chi(G) \leq \max_{1 \leq i \leq n} \min\{i, d_i + 1\}.$$

Proof. Let $\chi(G) = k$. By Corollary 1.1.10, G has at least k vertices of degree at least $k - 1$, so $d_k \geq k - 1$. Hence $\min\{k, d_k + 1\} = k$, and therefore

$$\max_{1 \leq i \leq n} \min\{i, d_i + 1\} \geq k = \chi(G).$$

■

Algorithm 1.1.13 (The Greedy Coloring Algorithm). The **greedy coloring algorithm** is a simple method for graph vertex coloring that processes vertices in a specific order and assigns each vertex the smallest possible color not used by its already-colored neighbors.

1. Order the vertices $v_1, v_2, \dots, v_n$
2. For $i = 1$ to n :

Assign v_i the smallest color $c \in \mathbb{N}$ such that no adjacent vertex already has color c .

The algorithm produces a proper coloring but may not use the minimum number of colors (chromatic number $\chi(G)$). The number of colors used depends heavily on the vertex ordering. (See Figures 1.1 and 1.2).

Algorithm 1.1.14. A **Kempe chain** is a maximal connected component of a graph induced by vertices colored with two given colors.

The **Kempe chain argument** is a recoloring technique used in graph coloring where colors are swapped along an alternating path between two colors. By swapping these colors along the entire component, one obtains another proper coloring.

This method was famously used in:

- The proof of the **Five Color Theorem** for planar graphs
- Alfred Kempe's flawed but influential attempt at the **Four Color Theorem**
- Various graph coloring algorithms and proofs

How to use Kempe Chain Argument: Let G be a graph that uses at least two colors for its proper coloring. Choose two colors, say i & j and construct a subgraph H of G induced by the colors i & j . Let K be a connected component in H . Interchange the colors i and j in K . This produces a new coloring of G . In this process, the subgraph K of G is known as Kempe chain and this technique to obtain a new coloring of G is called the Kempe chain argument. (See Figures 1.1.3, 1.1.3).

Theorem 1.1.15 (Brooks' Theorem). *Let G be a connected graph with maximum degree Δ . If G is neither a complete graph nor an odd cycle, then*

$$\chi(G) \leq \Delta.$$

Proof. If $\Delta \leq 2$, then G is a path or a cycle. Paths and even cycles are 2-colorable, while odd cycles are excluded. Hence assume

$$\Delta \geq 3.$$

Case 1: G is not Δ -regular.

Choose a vertex v with $\deg(v) < \Delta$. Order the vertices by decreasing distance from v , so that v is colored last.

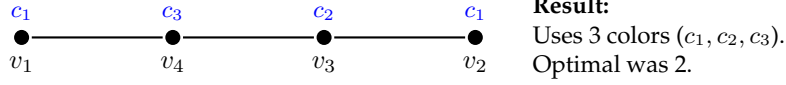
Color greedily using the colors $1, \dots, \Delta$. Every vertex $x \neq v$ has a neighbor closer to v , and this neighbor occurs later in the ordering. Thus, when x is colored, at most $\Delta - 1$ of its neighbors have already been colored. Hence a color is available for x . Finally, since $\deg(v) < \Delta$, a color is also available for v . Therefore

$$\chi(G) \leq \Delta.$$

We may now assume that G is Δ -regular.

1. Greedy Vertex Coloring

Processing Order: $v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4$



2. Kempe Chain (Edge Coloring)

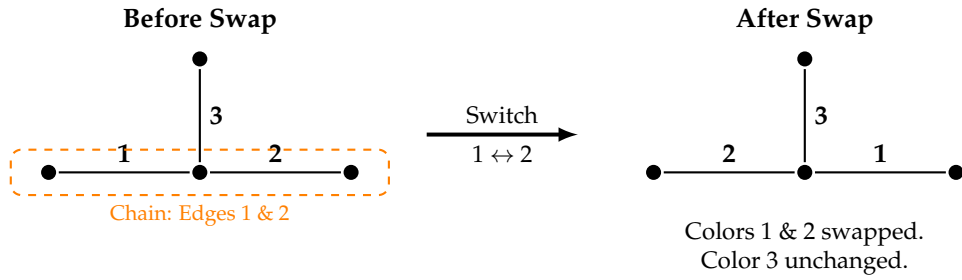


Figure 1.1: Top: Greedy Vertex Coloring. Bottom: Kempe Chain Argument on Edge Coloring.

Case 2: G has a cut-vertex.

Let v be a cut-vertex, and let

$$C_1, \dots, C_k$$

be the components of $G - v$. For each i , consider

$$G_i = G[C_i \cup \{v\}].$$

Since v has neighbors in at least two components,

$$\deg_{G_i}(v) < \Delta.$$

Applying the greedy argument of Case 1 to each G_i , we obtain a Δ -coloring of every G_i . By permuting the colors, we may assume that v receives the same color in every G_i . These colorings then combine to give a proper Δ -coloring of G .

Thus it remains to consider the case in which G is Δ -regular and 2-connected.

We first identify three useful vertices. We claim that there exist vertices u, v, w such that

$$uv, vw \in E(G), \quad uw \notin E(G),$$

and

$$G - \{u, w\}$$

is connected.

If G is 3-connected, choose any induced path $u - v - w$; such a path exists because G is connected and not complete. Since G is 3-connected, deleting u and w leaves a connected graph.

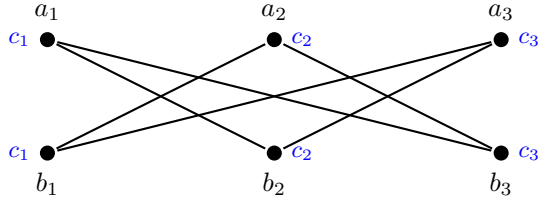
Suppose instead that G is 2-connected but not 3-connected. Then there are vertices v and x such that $G - \{v, x\}$ is disconnected. Hence $G - v$ is connected but has a cut-vertex.

Choose two distinct endblocks B_1 and B_2 of $G - v$, and let b_i be the cut-vertex through which B_i is attached to the rest of $G - v$. Since G itself has no cut-vertex, v must have a neighbor in each set

$$B_i - \{b_i\}.$$

1. Greedy Coloring (Crown Graph)

Order: $a_1, b_1, a_2, b_2, a_3, b_3$



Result:
Used 3 Colors (c_1, c_2, c_3)
(Optimal is 2)

2. Kempe Chain (Edge Coloring)

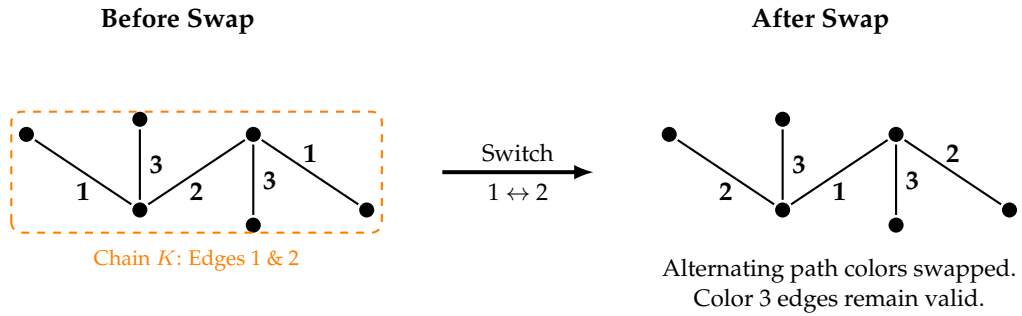


Figure 1.2: Top: Greedy coloring on a Crown Graph (6 vertices) forcing a sub-optimal 3rd color. Bottom: Kempe Chain argument on a larger path, swapping edge colors 1 and 2.

Choose

$$u \in B_1 - \{b_1\}, \quad w \in B_2 - \{b_2\}$$

such that

$$uv, vw \in E(G).$$

Because u and w belong to distinct endblocks, they are not adjacent.

Moreover, u and w are non-cut vertices lying in distinct endblocks. Deleting them does not disconnect $G - v$. Thus

$$G - \{u, v, w\}$$

is connected. Since $\deg(v) = \Delta \geq 3$, the vertex v has a neighbor other than u and w , so adding v back shows that

$$G - \{u, w\}$$

is connected. This proves the claim.

Now give the nonadjacent vertices u and w the same color, say color 1.

Since

$$H = G - \{u, w\}$$

is connected, order the vertices of H by decreasing distance from v , with v last. Color them greedily with the colors $1, \dots, \Delta$.

Every vertex $x \neq v$ has a neighbor closer to v , hence a neighbor that is still uncolored when x is considered. Therefore at most $\Delta - 1$ of its neighbors have already been colored, and a color is available.

Finally, consider v . It has exactly Δ neighbors, but two of them, namely u and w , have the same color. Hence the neighbors of v use at most

$$\Delta - 1$$

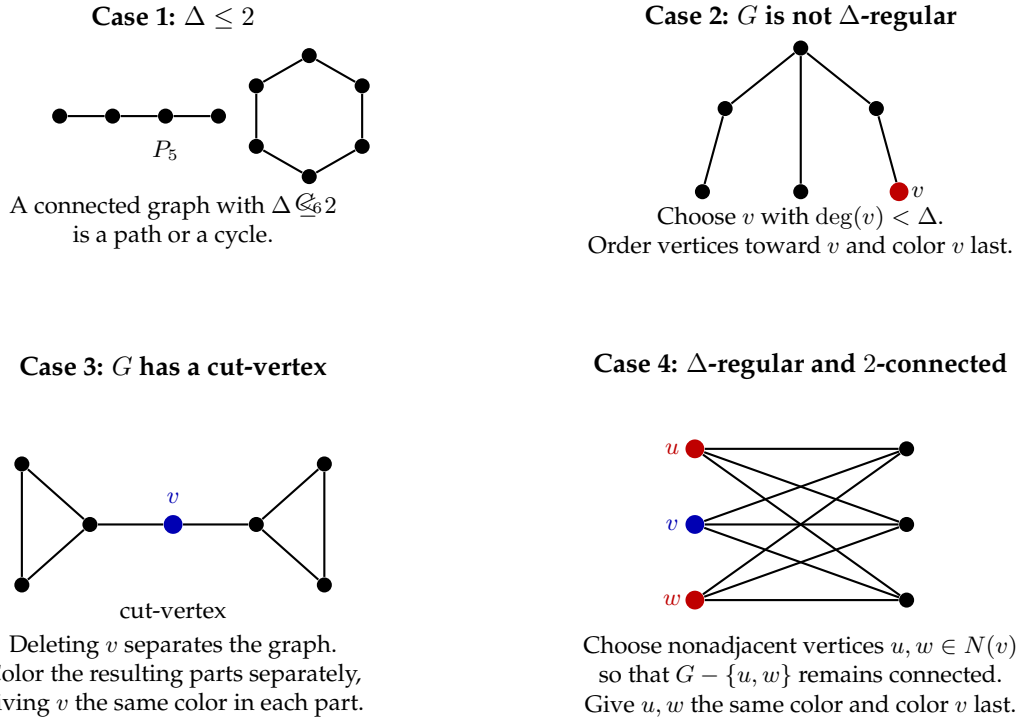


Figure 1.3: Graphical illustration of the main cases in the proof of Brooks' theorem.

distinct colors. Thus one of the Δ colors is available for v .

Consequently, G has a proper Δ -coloring, and therefore

$$\chi(G) \leq \Delta.$$

■

Theorem 1.1.16. For any graph G of order n ,

$$\frac{n}{\chi(G)} \leq \alpha_0(G) \leq n - \chi(G) + 1.$$

Proof. Let $\chi(G) = k$. Then the vertices of G can be partitioned into k colour classes

$$V_1, V_2, \dots, V_k,$$

where each V_i is an independent set.

Let $|V_i| = n_i$. Since $\alpha_0(G)$ is the maximum size of an independent set in G , we have

$$n_i \leq \alpha_0(G) \quad \text{for each } i.$$

Therefore,

$$n = \sum_{i=1}^k n_i \leq \sum_{i=1}^k \alpha_0(G) = k\alpha_0(G).$$

Since $k = \chi(G)$, it follows that

$$\frac{n}{\chi(G)} \leq \alpha_0(G).$$

For the upper bound, let S be a maximum independent set of G , so that

$$|S| = \alpha_0(G).$$

Removing S can reduce the chromatic number by at most one, because all vertices of S can be assigned a single colour. Hence

$$\chi(G - S) \geq \chi(G) - 1.$$

On the other hand, $G - S$ has $n - \alpha_0(G)$ vertices, and therefore

$$\chi(G - S) \leq n - \alpha_0(G).$$

Combining these inequalities gives

$$\chi(G) - 1 \leq \chi(G - S) \leq n - \alpha_0(G).$$

Thus,

$$\chi(G) - 1 \leq n - \alpha_0(G),$$

and hence

$$\boxed{\alpha_0(G) \leq n - \chi(G) + 1}.$$

■

Theorem 1.1.17. For any graph of order n ,

$$2\sqrt{n} \leq \chi + \bar{\chi} \leq n + 1, \text{ and } n \leq \chi\bar{\chi} \leq \left(\frac{n+1}{2}\right)^2, \quad (1.1)$$

where $\chi = \chi(G)$ and $\bar{\chi} = \chi(\bar{G})$.

Proof. Let G be k -chromatic and let $v_1, v_2, \dots, v_k$ be the colour classes of G , where $|V_i| = n_i$. Then $\sum n_i = n$ and $\max n_i \geq n/k$. Since each V_i induces a complete subgraph of $\bar{G}$, $\bar{\chi} \geq \max n_i/n_i$ so that $\bar{\chi} \geq n/n_i$. Thus for each i , $n_i \geq n/\bar{\chi}$. Since the geometric mean of two positive numbers is always less or equal to their arithmetic mean, it follows that $\chi + \bar{\chi} \geq 2\sqrt{n}$.

To prove $\chi + \bar{\chi} \leq n + 1$, we induct on n . Clearly, the equality holds for $n = 1$. We assume that $\chi(G) + \bar{\chi}(G) \leq n$ for all graphs G having fewer than n vertices. Let H and $\bar{H}$ be complementary graphs with n vertices and let v be a vertex of H . Then $G = H - v$ and $\bar{G} = \bar{H} - v$ are complementary graphs with $n - 1$ vertices. Let the degree of v in H be d , so that degree of v in $\bar{H}$ is $n - d - 1$. Clearly, $\chi(H) \leq \chi(G) + 1$ and $\bar{\chi}(H) \leq \bar{\chi}(G) + 1$. If either $\chi(H) < \chi(G) + 1$ or $\bar{\chi}(H) < \bar{\chi}(G) + 1$, then $\chi(H) + \bar{\chi}(H) \leq n + 1$. If $\chi(H) = \chi(G) + 1$ and $\bar{\chi}(H) = \bar{\chi}(G) + 1$, then the removal of v from H , producing G , decreases the chromatic number, so that $d \geq \chi(G)$. Similarly, $n - d - 1 \geq \bar{\chi}(G)$. Thus, $\chi(G) + \bar{\chi}(G) \leq n - 1$. Therefore we always have $\chi(H) + \bar{\chi}(H) \leq n + 1$. Applying now the inequality $4\chi\bar{\chi} \leq (\chi + \bar{\chi})^2$, we get $\chi\bar{\chi} \leq \left(\frac{n+1}{2}\right)^2$. ■

2nd Proof. Let G be a graph of order n . Let $\chi = \chi(G)$ and $\bar{\chi} = \chi(\bar{G})$.

(i) $n \leq \chi\bar{\chi}$: Consider a χ -coloring of G with color classes $V_1, \dots, V_\chi$. Each V_i is an independent set in G . Consider a $\bar{\chi}$ -coloring of $\bar{G}$ with color classes $\bar{V}_1, \dots, \bar{V}_{\bar{\chi}}$. Each $\bar{V}_j$ is an independent set in $\bar{G}$, which means it is a clique in G . Every vertex $v \in V(G)$ belongs to some V_i and some $\bar{V}_j$. If u, v are two distinct vertices such that they are in the same color class V_i in G AND in the same color class $\bar{V}_j$ in $\bar{G}$, then u and v are not adjacent in G (because V_i is an independent set) and u and v are not adjacent in $\bar{G}$ (because $\bar{V}_j$ is an independent set). This is impossible, as any two distinct vertices must be adjacent in either G or $\bar{G}$. Therefore, each vertex must be assigned a unique ordered pair of colors (i, j) where $1 \leq i \leq \chi$ and $1 \leq j \leq \bar{\chi}$. The total number of such unique pairs is $\chi\bar{\chi}$. Since there are n vertices, $n \leq \chi\bar{\chi}$.

(ii) $2\sqrt{n} \leq \chi + \bar{\chi}$: Using the AM-GM inequality: $\frac{\chi + \bar{\chi}}{2} \geq \sqrt{\chi\bar{\chi}}$. From the previous part, $n \leq \chi\bar{\chi}$, so $\sqrt{n} \leq \sqrt{\chi\bar{\chi}}$. Combining these, $\frac{\chi + \bar{\chi}}{2} \geq \sqrt{n}$, which implies $2\sqrt{n} \leq \chi + \bar{\chi}$.

(iii) $\chi + \bar{\chi} \leq n + 1$: This is a standard result, often attributed to Nordhaus and Stewart (1963). The maximum value of $\chi(G) + \chi(\bar{G})$ is $n + 1$, achieved by complete graphs (K_n) and empty graphs $(\bar{K}_n)$. For K_n , $\chi(K_n) = n$ and $\bar{\chi}(K_n) = 1$, so $\chi + \bar{\chi} = n + 1$. For $\bar{K}_n$, $\chi(\bar{K}_n) = 1$ and $\bar{\chi}(\bar{K}_n) = n$, so $\chi + \bar{\chi} = n + 1$.

(iv) $\chi\bar{\chi} \leq \left(\frac{n+1}{2}\right)^2$: From $\chi + \bar{\chi} \leq n + 1$, and using the property that for any real numbers a, b , $4ab \leq (a + b)^2$: $4\chi\bar{\chi} \leq (\chi + \bar{\chi})^2 \leq (n + 1)^2$. Dividing by 4, we get $\chi\bar{\chi} \leq \frac{(n+1)^2}{4} = \left(\frac{n+1}{2}\right)^2$. ■

1.2 Edge Colouring

An edge colouring of a non-empty graph G is a labelling $f : E(G) \rightarrow \{1, 2, \dots\}$ such that adjacent edges are assigned different colours. The labels are called *colours*. A k -edge colouring of G is an edge colouring of G that uses exactly k different colours, and in this case, G is said to be k -edge colourable.

This definition implies that a k -edge colouring of a graph $G(V, E)$ partitions the edge set E into k independent sets $E_1, E_2, \dots, E_k$ such that $E = E_1 \cup E_2 \cup \dots \cup E_k$. These independent sets E_i , for $1 \leq i \leq k$, are called the *colour classes*. The function $f : E(G) \rightarrow \{1, 2, \dots, k\}$ such that $f(e) = i$ for each $e \in E_i$ (where $1 \leq i \leq k$) is called the *colour function*. The minimum number k for which a k -colouring of G exists is called the *edge chromatic number* (or *edge chromatic index*) and is denoted by $\chi'(G)$.

We have the following observations:

1. If H is a subgraph of a graph G , then $\chi'(H) \leq \chi'(G)$.
2. For any graph G , $\chi'(G) \geq \Delta(G)$, where $\Delta(G)$ is the maximum degree of G .

If v is any vertex of G with degree $d(v) = \Delta(G)$, then the $\Delta(G)$ edges incident with v must all have different colours in any edge colouring of G .

3. For a cycle graph C_n of order n :

$$\chi'(C_n) = \begin{cases} 2, & \text{if } n \text{ is even,} \\ 3, & \text{if } n \text{ is odd.} \end{cases}$$

Definition: Let i be a colour used in an edge colouring of a graph G . If there is an edge coloured i incident to a vertex v of G , we say i is *present* at v . If there is no edge coloured i incident to v , we say i is *absent* from v .

Theorem 1.2.1 (König, 1916). For a non-empty bipartite graph G , $\chi'(G) = \Delta(G)$.

Proof. The proof is by induction on the number of edges of G . If G has only one edge, the result is trivial since $\chi'(G) = 1$ and $\Delta(G) = 1$.

Assume that G has more than one edge, and that the theorem holds for all non-empty bipartite graphs having fewer edges than G . Since we know $\Delta(G) \leq \chi'(G)$ (Observation 2), it suffices to prove that G has a $\Delta(G)$ -edge colouring. Let $\Delta(G) = k$.

Consider an arbitrary edge $e = uv$ in G . The graph $G - e$ is bipartite and has fewer edges than G . By the induction hypothesis, $G - e$ has a $\Delta(G - e)$ -edge colouring. Since $\Delta(G - e) \leq \Delta(G) = k$, $G - e$ has a k -edge colouring. We will show how to extend this k -edge colouring of $G - e$ to a k -edge colouring of G by assigning a colour to e .

Since $d_G(u) \leq k$ and $d_G(v) \leq k$, in the k -edge colouring of $G - e$, there is at least one colour absent at vertex u and at least one colour absent at vertex v .

1. **Case 1: There is a colour c that is absent at both u and v .** In this case, we can assign colour c to edge $e = uv$. This results in a k -edge colouring of G .
2. **Case 2: There is no colour absent at both u and v .** This implies that for every colour c , if c is absent at u , it must be present at v , and vice versa. Let i be a colour absent at v (and thus present at u) and j be a colour absent at u (and thus present at v).

Consider the subgraph $H(i, j)$ formed by all edges in $G - e$ coloured either i or j . Let K be the connected component of $H(i, j)$ containing u . We claim that v does not belong to K .

Proof of claim: Assume, for contradiction, that v belongs to K . Then there exists a path P in K between u and v . Since u and v are adjacent in G (via edge e), they belong to different partitions in the bipartite graph G . Therefore, any path between u and v in G must have odd length. The path P is composed of alternately coloured edges i and j . Since colour i is present at u , the first edge of P incident to u must be coloured i . As P has an odd number of edges and alternates colours, its last edge incident to v must also be coloured i . This implies that colour i is present at v , which contradicts our initial assumption that i is absent from v . Thus, v does not belong to the Kempe chain K .

Now, we apply the Kempe chain argument to the component K . We interchange the colours i and j on all edges in K .

- At vertex u : Since $u \in K$ and the colour i was present at u , after the recolouring, colour j becomes present at u and colour i becomes absent at u .
- At vertex v : Since $v \notin K$, the colours of edges incident to v are unaffected by this interchange. Therefore, i remains absent from v .

After this recolouring, colour i is now absent at both u and v . We are now back to Case 1, where we can assign colour i to edge $e = uv$. This yields a k -edge colouring of G .

Therefore, in all cases, G has a $\Delta(G)$ -edge colouring. ■

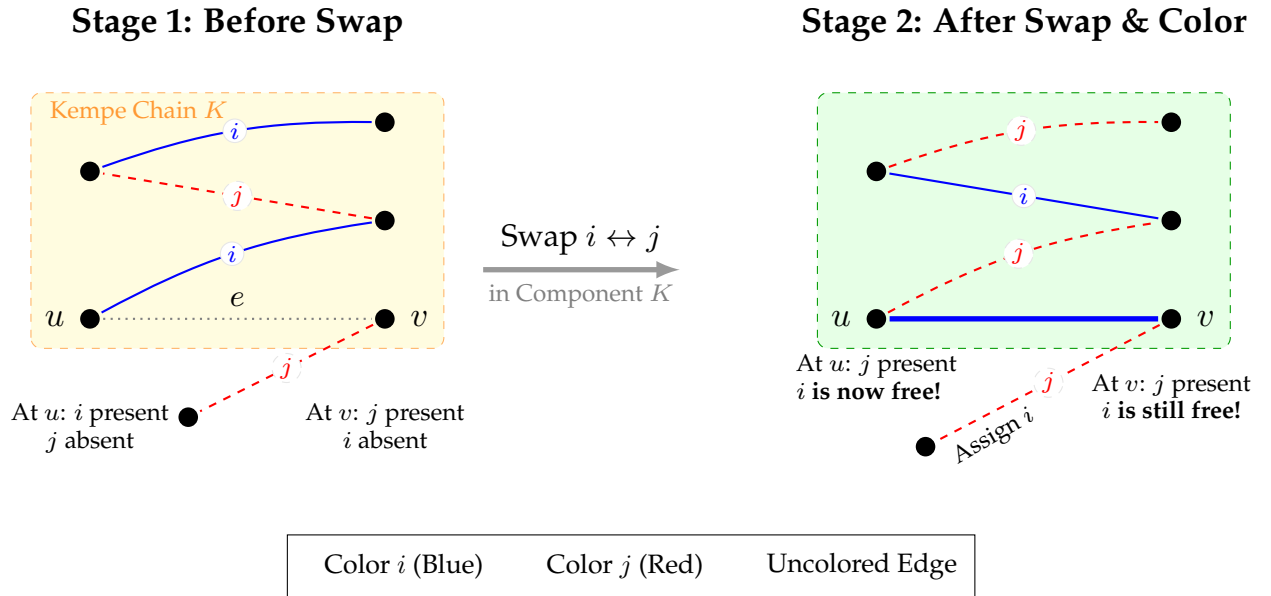


Figure 1.4: Visualization of the Kempe Chain swap. By drawing the highlight background first and using standard arrow syntax, this code is robust and compatible with standard LaTeX installations.

Theorem 1.2.2. If $G = K_n$ is a complete graph with n vertices, $n \geq 2$, then

$$\chi'(G) = \begin{cases} n-1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

Proof. (See Figure 1.5).

Since every vertex of K_n has degree $n-1$, we have $\Delta(K_n) = n-1$. Hence, $\chi'(K_n) \geq n-1$.

Case 1: n is odd. Suppose that K_n can be edge-coloured with $n-1$ colours. Each colour class is a matching, and since n is odd, a matching contains at most $(n-1)/2$ edges. Therefore, $n-1$ colours can colour at most

$$(n-1) \frac{n-1}{2} = \frac{(n-1)^2}{2}$$

edges.

However,

$$|E(K_n)| = \frac{n(n-1)}{2} > \frac{(n-1)^2}{2}.$$

This is a contradiction. Hence $\chi'(K_n) \geq n$.

To show that n colours are sufficient, label the vertices by $0, 1, \dots, n-1$, with arithmetic modulo n . Colour the edge joining i and j with the colour

$$i + j \pmod{n}.$$

If two edges ij and ik incident with the same vertex i received the same colour, then $i + j \equiv i + k \pmod{n}$, and hence $j = k$. Thus adjacent edges receive different colours. Therefore K_n has a proper n -edge colouring.

Hence

$$\chi'(K_n) = n$$

when n is odd.

Case 2: n is even. We already know that $\chi'(K_n) \geq n-1$. It remains to show that $n-1$ colours are sufficient.

Place $n-1$ vertices at the vertices of a regular $(n-1)$ -gon and place the remaining vertex at the centre. Join the central vertex to one outer vertex, and pair the remaining $n-2$ outer vertices symmetrically. These edges form a perfect matching, so they may all be assigned the same colour.

Now rotate this arrangement successively through the $n-1$ positions of the outer polygon. Each rotation gives a new perfect matching, and the resulting $n-1$ perfect matchings together contain every edge of K_n exactly once. Assigning a different colour to each matching gives a proper edge colouring of K_n with $n-1$ colours.

Therefore

$$\chi'(K_n) = n-1$$

when n is even.

Thus,

$$\chi'(K_n) = \begin{cases} n-1, & \text{if } n \text{ is even,} \\ n, & \text{if } n \text{ is odd.} \end{cases}$$

■

Theorem 1.2.3 (Vizing). For any graph G ,

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1.$$

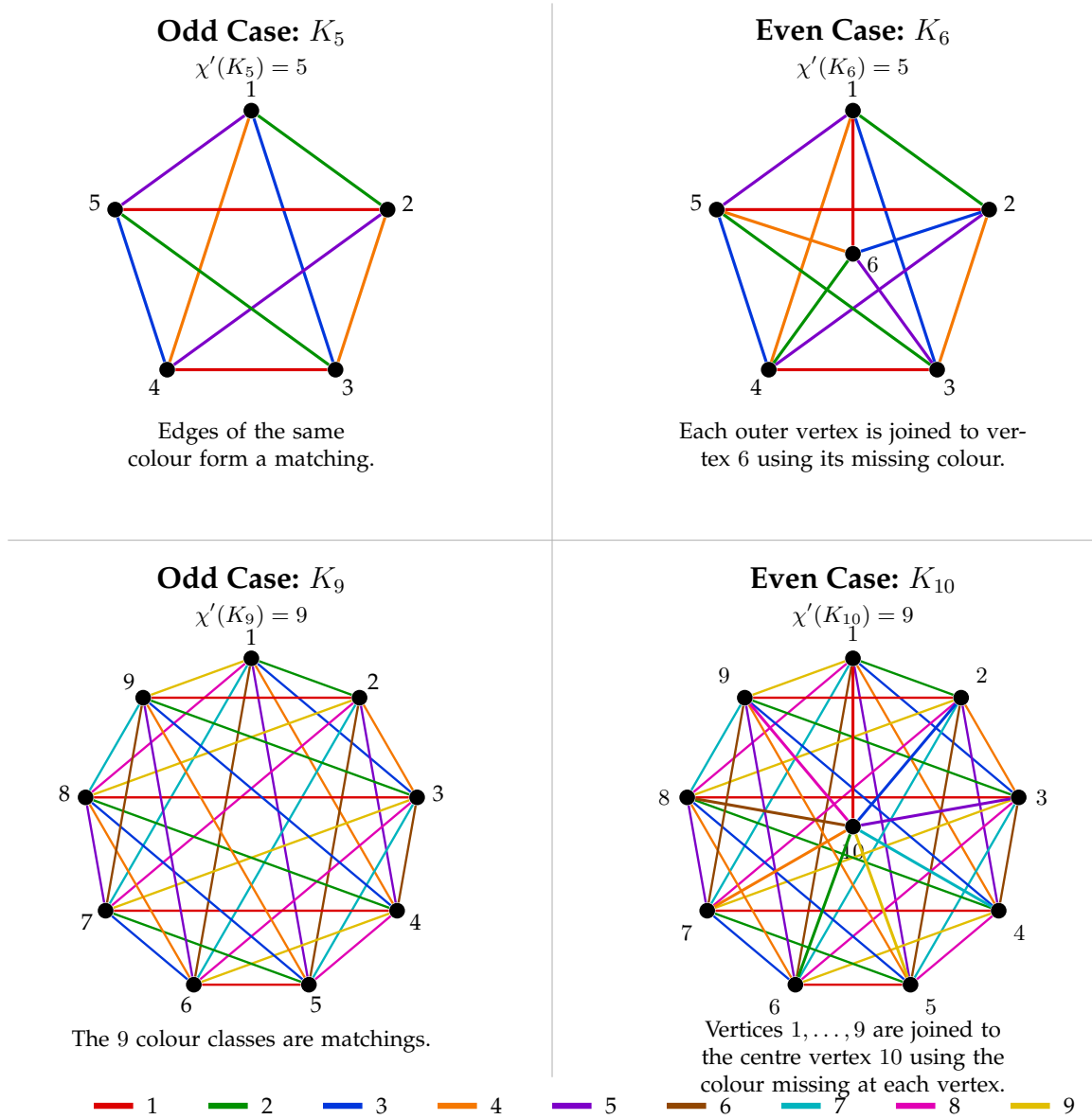
Proof. Let G be a simple graph. We always have $\Delta(G) \leq \chi'(G)$. It remains to prove that $\chi'(G) \leq \Delta(G) + 1$.

We use induction on the number of edges of G . Let $\Delta(G) = k$. If G has only one edge, then $k = 1 = \chi'(G)$. Therefore, assume that G has more than one edge and that the result is true for all graphs having fewer edges than G .

Let $e = v_1v_2$ be an edge of G . By the induction hypothesis, the subgraph $G - e$ has a $(k+1)$ -edge colouring. Let the colours used be $1, 2, \dots, k+1$.

Since $d(v_1) \leq k$ and $d(v_2) \leq k$, out of these $k+1$ colours, at least one colour is absent from v_1 and at least one colour is absent from v_2 . If there is a common colour absent from both v_1 and v_2 , then we use this colour to colour e and obtain a $(k+1)$ -edge colouring of G . Therefore, in this case, $\chi'(G) \leq k+1$.

We now assume that there is a colour, say 1, absent from v_1 but present at v_2 , and a colour, say 2, absent from v_2 but present at v_1 . We start from v_1 and v_2 and construct a sequence of distinct vertices $v_1, v_2, \dots, v_j$, where each v_i , for $i \geq 2$, is adjacent to v_1 .

Figure 1.5: Proper edge colourings of K_5 , K_6 , K_9 , and K_{10} .

Let v_1v_3 be coloured 2. This vertex v_3 exists because colour 2 is present at v_1 . We observe that not all the $k+1$ colours are present at v_3 , and assume that colour 3 is absent from v_3 . If colour 3 is present at v_1 , choose the vertex v_4 so that v_1v_4 is coloured 3. Continuing in this way, we choose a new colour i absent from v_i but present at v_1 , so that v_1v_{i+1} is the edge coloured i .

In this way, we obtain a sequence of vertices $v_1, v_2, \dots, v_{j-1}, v_j$ such that

- (a) v_i is adjacent to v_1 for each $i > 1$;
- (b) colour i is absent from v_i for $i = 1, 2, \dots, j-1$;
- (c) the edge v_1v_{i+1} is coloured i for $i = 1, 2, \dots, j-1$.

Since $d(v_1) \leq k$, condition (a) implies that such a sequence has at most $k+1$ terms; that is, $j \leq k+1$. Assume that $v_1, v_2, \dots, v_j$ is a longest such sequence; that is, it is not possible to find a new colour j , absent from v_j , together with a new neighbour v_{j+1} of v_1 such that v_1v_{j+1} is coloured j .

We first assume that, for some colour j absent from v_j , there is no edge of colour j incident with v_1 . We colour the edge $e = v_1v_2$ by 2, and then recolour the edges v_1v_i by i , for $i = 3, \dots, j-1$. Since colour i is absent from v_i for each $i = 2, \dots, j-1$, this gives a $(k+1)$ -edge colouring of the subgraph $G - v_1v_j$. Since colour j is absent from both v_1 and v_j , we colour v_1v_j by j . This gives a $(k+1)$ -edge colouring of G .

Now assume that whenever colour j is absent from v_j , it is present at v_1 . If v_1v_{j+1} is an edge joining v_1 to a new neighbour v_{j+1} and is coloured j , then we can extend our sequence to $v_1, v_2, \dots, v_j, v_{j+1}$, contradicting the assumption that $v_1, v_2, \dots, v_j$ is a longest sequence. Thus one of the edges $v_1v_3, \dots, v_1v_{j-1}$ must be coloured j ; say v_1v_ℓ , where $3 \leq \ell \leq j-1$.

Now colour $e = v_1v_2$ by 2, and for $i = 3, \dots, \ell-1$, recolour each edge v_1v_i by i , while leaving the colours of the edges v_1v_i for $i = \ell+1, \dots, j$, unchanged. Removing colour j from v_1v_ℓ , we obtain a $(k+1)$ -edge colouring of the edge-deleted subgraph $G - v_1v_\ell$.

Let $H(1, j)$ represent the subgraph of G induced by the edges coloured 1 or j in this partial colouring of G . Since the degree of every vertex in $H(1, j)$ is at most 2, each component of $H(1, j)$ is either a path or a cycle. Since colour 1 is absent from v_1 and colour j is absent from both v_j and v_ℓ , it follows that these three vertices do not all belong to the same connected component of $H(1, j)$. Therefore, if K and L represent the corresponding Kempe chains containing v_j and v_ℓ , respectively, then either $v_1 \notin K$ or $v_1 \notin L$.

Suppose that $v_1 \notin L$. Interchanging colours 1 and j on L , the Kempe chain argument gives a $(k+1)$ -edge colouring of $G - v_1v_\ell$ in which colour 1 is absent from both v_1 and v_ℓ . Colouring v_1v_ℓ by 1 gives a $(k+1)$ -edge colouring of G .

Now suppose that $v_1 \notin K$. Colour the edge v_1v_ℓ by ℓ , recolour the edges v_1v_i by i , for $i = \ell, \dots, j-1$, and remove colour $j-1$ from v_1v_j . Then, from the definition of the sequence $v_1, v_2, \dots, v_j$, we obtain a $(k+1)$ -edge colouring of $G - v_1v_j$ without affecting the two-coloured subgraph $H(1, j)$. Using the Kempe chain argument to interchange the colours on K , we obtain a $(k+1)$ -edge colouring of $G - v_1v_j$ in which colour 1 is absent from both v_1 and v_j . Therefore, colouring v_1v_j by 1 gives a $(k+1)$ -edge colouring of G .

Hence $\chi'(G) \leq k+1 = \Delta(G) + 1$. Together with $\Delta(G) \leq \chi'(G)$, we obtain

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1.$$

■

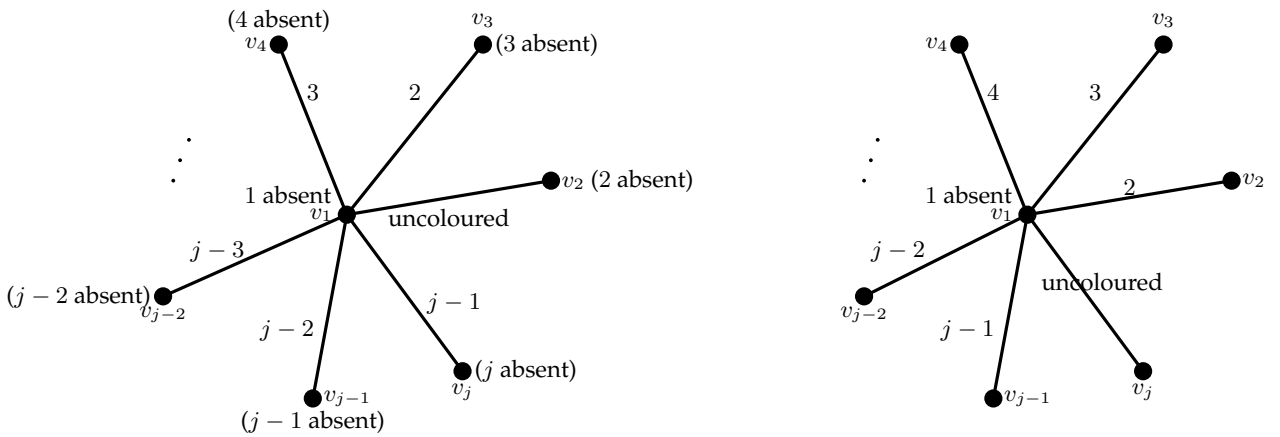


Figure 1.6: Illustration of Vizing's theorem.

Second Proof. The lower bound $\Delta(G) \leq \chi'(G)$ is straightforward. At any vertex v with degree $\Delta(G)$, there are $\Delta(G)$ edges incident with it. Since all these edges must receive distinct colours in a proper edge colouring, at least $\Delta(G)$ colours are necessary. Thus $\chi'(G) \geq \Delta(G)$.

The more challenging part is to prove the upper bound $\chi'(G) \leq \Delta(G) + 1$. This part of the proof typically involves a constructive algorithm or an inductive argument. Here, we provide a high-level constructive proof sketch.

Let $\Delta = \Delta(G)$. We want to show that G can be $(\Delta + 1)$ -edge coloured. We proceed by induction on the number m of edges of G . The base case is trivial for $m = 0$ or $m = 1$.

Assume that any graph with fewer than m edges can be $(\Delta + 1)$ -edge coloured. Consider a graph G with m edges. Let $e = uv$ be an uncoloured edge of G . We try to colour e with a colour $c \in \{1, \dots, \Delta + 1\}$. If we can colour all the edges one by one, we obtain a $(\Delta + 1)$ -edge colouring.

Consider a partial edge colouring c' of G , where some edges may be uncoloured. For any vertex $x \in V(G)$, let $C(x)$ be the set of colours used on edges incident with x , and let

$$M(x) = \{1, \dots, \Delta + 1\} \setminus C(x)$$

be the set of colours missing at x . Since the degree of every vertex is at most Δ and we have $\Delta + 1$ available colours, every vertex x has at least one missing colour; that is, $M(x) \neq \emptyset$.

Now let $e_0 = uv$ be an uncoloured edge. We want to assign a colour to e_0 . If there is a colour $c \in M(u) \cap M(v)$, we can assign colour c to e_0 . If no such colour exists, then $M(u)$ and $M(v)$ are disjoint. Since $|M(u)| \geq 1$ and $|M(v)| \geq 1$, we have $|C(u)| \leq \Delta$ and $|C(v)| \leq \Delta$, while $|M(u)| \geq (\Delta + 1) - \Delta = 1$ and $|M(v)| \geq (\Delta + 1) - \Delta = 1$.

If $M(u) \cap M(v) = \emptyset$, we try to rearrange the colours to make a colour available for uv . Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there is an edge vw coloured a . Since $b \notin M(u)$, there is an edge ux coloured b . We consider an (a, b) -alternating path or cycle starting from v . Such a path has edges whose colours alternate between a and b .

The proof involves a technique called colour shifting or fan colouring. Let $a \in M(u)$ and $b \in M(v)$. Since $a \notin M(v)$, there exists a vertex w such that the edge vw is coloured a . Consider the path P starting at w whose edges alternate between colours a and b . If this path does not reach u , we can interchange colours a and b along P . This frees colour a at v and allows us to colour uv with a .

The essential point is that interchanging the two colours along an alternating path does not create a conflict. The argument considers the possible (a, b) -alternating paths and cycles arising from the vertices involved. Since $\Delta + 1$ colours are available while the maximum degree is only Δ , there is sufficient flexibility to rearrange the colours and make a colour available for the uncoloured edge.

By systematically applying this procedure, colouring one uncoloured edge at a time and using colour shifting whenever necessary, we obtain a proper $(\Delta + 1)$ -edge colouring of G . Hence

$$\chi'(G) \leq \Delta(G) + 1.$$

Together with $\Delta(G) \leq \chi'(G)$, this proves the result. ■

Theorem 1.2.4. *Let G be a graph of order n and let $\overline{G}$ be the complement of G . Then*

(a) *if n is even, then*

$$n - 1 \leq \chi'(G) + \chi'(\overline{G}) \leq 2(n - 1),$$

and

$$0 \leq \chi'(G)\chi'(\overline{G}) \leq (n - 1)^2;$$

(b) *if n is odd, then*

$$n \leq \chi'(G) + \chi'(\overline{G}) \leq 2n - 3,$$

and

$$0 \leq \chi'(G)\chi'(\overline{G}) \leq (n - 1)(n - 2).$$

Further, the bounds are best possible for every positive integer $n \neq 2$.

Proof. Let G be a graph of order n and let $\overline{G}$ be its complement. Clearly,

$$\Delta(G) \geq n - 1 - \Delta(\overline{G}),$$

and hence

$$\Delta(G) + \Delta(\overline{G}) \geq n - 1.$$

Since $\chi'(G) \geq \Delta(G)$ and $\chi'(\overline{G}) \geq \Delta(\overline{G})$, it follows that

$$\chi'(G) + \chi'(\overline{G}) \geq n - 1.$$

Part 1: Even n . Let n be even. Then

$$\chi'(G) + \chi'(\overline{G}) \geq n - 1$$

and

$$\chi'(G)\chi'(\overline{G}) \geq 0.$$

The lower bounds are attained when $G = K_n$ or $G = \overline{K_n}$.

For the upper bounds, both G and $\overline{G}$ are subgraphs of K_n . Since $\chi'(K_n) = n - 1$ for even n , we have

$$\chi'(G) \leq n - 1 \quad \text{and} \quad \chi'(\overline{G}) \leq n - 1.$$

Therefore,

$$\chi'(G) + \chi'(\overline{G}) \leq 2(n - 1)$$

and

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)^2.$$

These upper bounds are attained for suitable graphs, including the complete bipartite graph $K_{1,n-1}$ when $n \neq 2$.

Part 2: Odd n . Let n be odd. We claim that

$$\chi'(G) + \chi'(\overline{G}) \geq n.$$

For example, if $G = K_n$, then $\chi'(K_n) = n$, while $\overline{G}$ is the empty graph and has edge chromatic number 0. Thus

$$\chi'(G) + \chi'(\overline{G}) = n.$$

Similarly, the same value is obtained when G is the empty graph.

Clearly,

$$\chi'(G)\chi'(\overline{G}) \geq 0,$$

and this lower bound is also attained for complete or empty graphs.

We now prove the upper bounds

$$\chi'(G) + \chi'(\overline{G}) \leq 2n - 3$$

and

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)(n - 2).$$

For $n = 1$ and $n = 3$, the inequalities are immediate. Thus assume that $n \geq 5$. If

$$\Delta(G) + \Delta(\overline{G}) \leq 2n - 5,$$

then by Vizing's theorem,

$$\chi'(G) + \chi'(\overline{G}) \leq \Delta(G) + \Delta(\overline{G}) + 2 \leq 2n - 3.$$

Otherwise, we consider the following cases.

- (i) Suppose $\Delta(G) = n - 1$ and $\Delta(\overline{G}) = n - 2$. Then G has a pendant vertex v . Also, $\Delta(G - v) \geq n - 2$ and hence $\chi'(G - v) \geq n - 2$. Since $\overline{G - v}$ is a subgraph of K_{n-1} ,

$$\chi'(\overline{G - v}) \leq \chi'(K_{n-1}) = n - 2.$$

As $\overline{G}$ is the disjoint union of an isolated vertex and a subgraph of K_{n-1} , we obtain the required bound.

- (ii) Suppose $\Delta(G) = n - 2$ and $\Delta(\overline{G}) = n - 2$. Again, G has a pendant vertex v , and as before,

$$\chi'(G - v) \leq n - 2 \quad \text{and} \quad \chi'(\overline{G}) \leq n - 2.$$

Similarly, $\chi'(\overline{G}) = n - 2$. Thus

$$\chi'(G) + \chi'(\overline{G}) = 2n - 4 < 2n - 3.$$

(iii) Suppose $\Delta(G) = n - 1$ and $\Delta(\overline{G}) = n - 3$. In this case, G has a vertex v of degree 2, and so $\chi'(G - v) = n - 2$.

Let vu and vw be the edges incident with v in G , where $d(u) = n - 1$. In an $(n - 2)$ -edge colouring of $G - v$, change the colour i of an edge uu' to a new colour $n - 1$. Now colour vu by i and vw by $n - 1$. This gives an $(n - 1)$ -edge colouring of G . By Vizing's theorem,

$$\chi'(\overline{G}) \leq n - 2.$$

Hence

$$\chi'(G) + \chi'(\overline{G}) \leq 2n - 3.$$

Since

$$\chi'(G) + \chi'(\overline{G}) \leq 2n - 3,$$

the possible largest product under the stated bounds is

$$\chi'(G)\chi'(\overline{G}) \leq (n - 1)(n - 2).$$

The upper bounds are attained, for example, by suitable complete bipartite graphs such as $K_{1,n-1}$.

Thus all the stated inequalities follow. ■

A more complete argument for the upper bound when n is odd involves considering the possible maximum degrees of G and $\overline{G}$. By Vizing's theorem,

$$\chi'(G) \leq \Delta(G) + 1 \quad \text{and} \quad \chi'(\overline{G}) \leq \Delta(\overline{G}) + 1.$$

Hence

$$\chi'(G) + \chi'(\overline{G}) \leq \Delta(G) + \Delta(\overline{G}) + 2.$$

For every graph G of order n ,

$$\Delta(G) + \Delta(\overline{G}) \geq n - 1.$$

Also,

$$\Delta(G) + \Delta(\overline{G}) \leq 2n - 2.$$

The sharper bound $2n - 3$ for odd n requires a more detailed examination of the possible degree configurations in G and $\overline{G}$.

Similarly, the maximum value

$$\chi'(G)\chi'(\overline{G}) = (n - 1)(n - 2)$$

is obtained when the two edge chromatic numbers take the values $n - 1$ and $n - 2$, respectively. The determination of such sharp bounds belongs to the general study of graph invariants of complementary graphs.

1.3 Region Colouring (Map Colouring)

A region colouring of a planar graph is a labelling of its regions $f : R(G) \rightarrow \{1, 2, \dots\}$; the labels called *colours*, such that no two adjacent regions get the same colour. A k -region colouring of a planar graph G consists of k different colours and G is then called k -region colourable. From the definition, it follows that the k -region colouring of a planar graph G partitions the region set R into k independent sets $R_1, R_2, \dots, R_k$, so that $R = R_1 \cup R_2 \cup \dots \cup R_k$. The independent sets are called the *colour classes*, and the function $f : R(G) \rightarrow \{1, 2, \dots, k\}$ such that $f(r) = i$, for each $r \in R_i, 1 \leq i \leq k$, is called the *colour function*. The minimum number k for which there is a k -region colouring of the planar graph G is called the *region-chromatic number* of G , and is denoted by $\chi''(G)$. The colouring of regions is also called *map colouring*, because of the fact that in an atlas different countries are coloured such that countries with common boundaries are shown in different colours.

Theorem 1.3.1. 1. A planar graph G is k -region colourable if and only if its dual G^* is k -vertex colourable.

2. If G is a plane connected graph without loops, then G has a k -vertex colouring if and only if its dual G^* has a k -region colouring.

Proof. a. Let the regions and edges of G be respectively denoted by $r_1, \dots, r_t$ and $e_1, \dots, e_m$. Let the vertices of G^* be $r_1^*, \dots, r_t^*$ and edges be $e_1^*, \dots, e_m^*$. Then the vertices and edges of G^* are in one-to-one correspondence with the regions and edges of G , and two vertices r^* and s^* in G^* are joined by an edge e^* if and only if the corresponding regions r and s in G have the corresponding edge e as a common edge on their boundary.

Let G be k -region colourable. We colour the vertices in G^* such that each vertex r^* in G^* gets the same colour as assigned to the region r in G . Since the vertices r^* and s^* are only adjacent in G^* if the corresponding regions r and s are adjacent in G , G^* is k -vertex colourable.

Conversely, let G^* be k -vertex colourable. Now colour the regions of G such that the region r in G gets the same colour as the vertex r^* in G^* . This gives a k -region colouring of G , since the regions r and s are adjacent in G only if the corresponding vertices r^* and s^* are adjacent in G^* .

b. Since G has no loops its dual G^* has no bridges, and therefore G^* is planar. Thus by (a), G^* is k -region colourable if and only if the double dual G^{**} is k -vertex colourable. Since G is connected, G is isomorphic to G^{**} and hence the result follows. ■

Theorem 1.3.2. A planar graph is bicolourable if and only if it is an Euler graph.

Proof. Suppose G is bicolourable. Around each vertex, the colours of the incident faces alternate. Hence every vertex has even degree, so G is an Euler graph.

Conversely, suppose G is an Euler graph. Then every vertex of G has even degree, so every face of the dual graph G^* has even length. Hence G^* is bipartite. Therefore its vertices can be coloured with two colours, which gives a two-colouring of the faces of G .

Thus G is bicolourable if and only if it is an Euler graph. ■

Second Proof. Suppose G is bicolourable, that is, its dual G^* is 2-vertex-colourable. So G^* is bipartite, so every face of G^* has even length. Thus, every vertex of the dual of G^* has even degree. Since the bidual G^{**} is isomorphic to G , every vertex of G has even degree. Hence G is an Euler graph.

Conversely, suppose G is an Euler graph. Since $G^{**} \cong G$, every vertex of G^{**} has even degree. Thus every face of G^* has even length, and hence G^* is bipartite. Therefore the faces of G can be coloured with two colours, so G is bicolourable. ■

Theorem 1.3.3. Every planar graph is 6-colourable.

Proof. We use induction on the number of vertices. Every planar graph has a vertex v of degree at most 5. By induction, $G - v$ is 6-colourable. Since v has at most 5 neighbours, at most 5 colours appear on them, so one of the 6 colours is available for v . Hence G is 6-colourable. ■

Theorem 1.3.4 (Heawood). Every planar graph is 5-colourable.

Proof. (See Figure 1.7). Let G be a planar graph with n vertices. We use induction on n , the order of G . The result is obvious for $n \leq 5$. So, let $n \geq 6$. Assume the result is true for all planar graphs with fewer than n vertices.

Let G' be the graph obtained from G by deleting the vertex v and removing all the edges incident with v . The graph G' with order $n - 1$ is clearly planar and by induction hypothesis is 5-colourable. Let the colours used to colour G' be c_1, c_2, c_3, c_4, c_5 . We know for a planar graph with $n \geq 6$ vertices,

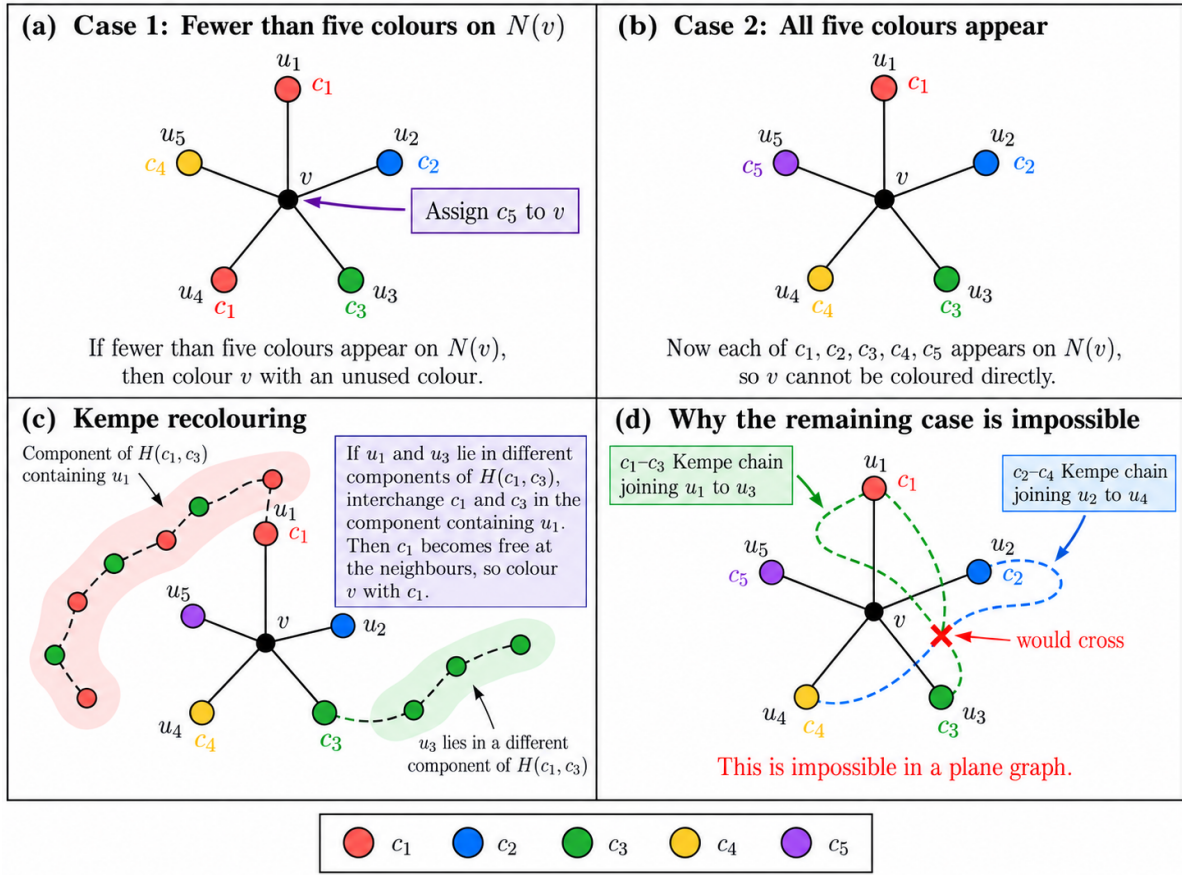


Figure 1.7: Heawood 5-color Theorem

there exists a vertex, say v , such that $d(v) \leq 5$. Thus v has at most five neighbours in G and all of these neighbours are the already coloured vertices in G' . If in G' less than five colours are used to colour these neighbours, then the 5-colouring of G is obtained by using the colouring for G' on all vertices, and by colouring v with the colour not used to colour the neighbours of v .

Let all the five colours be used in G' , to colour the neighbours of v . This implies that there are exactly five neighbours of v , say u_1, u_2, u_3, u_4 and u_5 . Assume without loss of generality that u_i is coloured with c_i , for each i , $1 \leq i \leq 5$.

Consider all the vertices of G' that are coloured with colour c_1 and c_3 . If u_1 and u_3 are in different components of the Kempe subgraph $H(c_1, c_3)$ induced by those vertices coloured c_1 and c_3 , then using the Kempe chain argument to interchange the colours c_1 and c_3 in the component containing u_1 leaves c_1 unused on the set $\{u_1, u_2, u_3, u_4, u_5\}$. We colour v by c_1 to get a 5-colouring of G . Finally, if u_1 and u_3 are in the same component of $H(c_1, c_3)$, and u_2 and u_4 are in the same component of $H(c_2, c_4)$, then any component of $H(c_2, c_4)$ can not contain both u_2 and u_4 since otherwise the $c_1 - c_3$ Kempe chain from u_1 to u_3 crosses the $c_2 - c_4$ Kempe chain from u_2 to u_4 . This is impossible as the graph is a plane graph. Thus, we always have a pair of colors, say c_i and c_j , such that the induced graph $H(c_i, c_j)$ has a component K which does not contain both u_i and u_j . Interchanging the colors on any such component releases a color for the last uncolored vertex. ■

Theorem 1.3.5. *Every planar graph is four colourable if and only if every cubic bridgeless planar graph is 4-colourable.*

Proof. We observe that every planar graph is four colourable if and only if every bridgeless planar graph (without cut edges) is four colourable. This is because if G has a bridge e and G' is obtained from G by contracting e , then $\chi''(G') = \chi''(G)$ (as the elementary contraction of identifying the end vertices of a bridge affects neither the number of regions in the planar graph nor the adjacency

of any of the regions). We now prove that every bridgeless planar graph is 4-colourable if and only if every cubic bridgeless planar graph is 4-colourable. If every bridgeless planar graph is 4-colourable, then evidently every cubic bridgeless planar graph is 4-colourable. Conversely, let all cubic bridgeless planar graphs be 4-colourable. Let G be a bridgeless planar graph. We obtain G' from G by performing the following operations. In case G has a vertex v of degree two, let xv and yv be the edges incident with v . Subdivide xv at u and yv at w . Take two new vertices a and b and add the edges au, aw, bu, bw and ab . Now remove v (and uv and uw). In doing so, we have replaced v by a $K_4 - e$ and we see that each new vertex has degree three.

If G has a vertex v so that $d(v) \geq 4$, let $vx_1, vx_2, \dots, vx_d$ be the edges incident with v . Subdivide each vx_i producing a new vertex v_i , for $1 \leq i \leq d$. We then remove v and add the new edges $v_1v_2, v_2v_3, \dots, v_{d-1}v_d, v_dv_1$. Here we have replaced v by C_d and again the degree of each new vertex is three.

In both cases the graph G' formed is a bridgeless cubic graph. If these $K_4 - e$ and C_d introduced are contracted, we get the original graph G . Hence G is 4-colourable, since we have assumed that G' is 4-colourable. ■

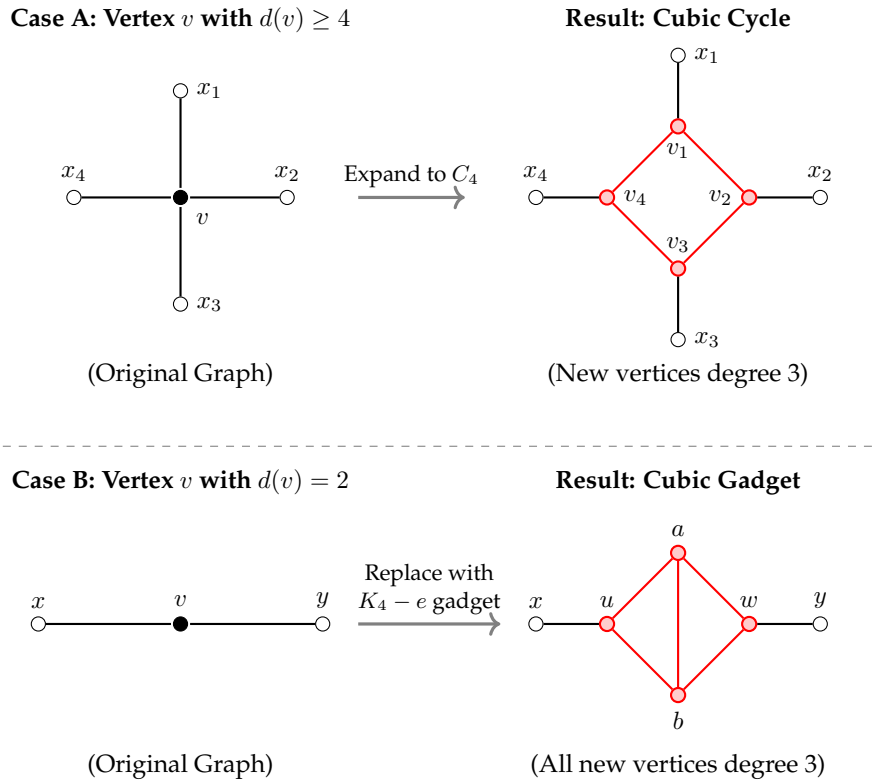


Figure 1.8: Visualisation of the operations described in Theorem 1.3.5. Top: Expanding a vertex of degree 4 into a cycle. Bottom: Replacing a vertex of degree 2 with a specialized gadget.

Theorem 1.3.6 (Tait's Theorem). *A cubic bridgeless, planar graph G is 4 region colorable if and only if, G is 3-edge colorable.*

Proof. Suppose G is 4-region colorable. Let the 4-region coloring of G be given, where the colors are the four elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$. We now define an edge coloring of G . We define the color of an edge as the sum of colors of regions adjacent to it, noting that each edge lies on the boundary of at most two regions (because the given graph is cubic and bridgeless). Thus, we are done.

Conversely: [Refer to Figure above]. Suppose that G is 3-edge colorable. Let a 3-edge coloring of G be given using the colors $c_1 = (0, 1), c_2 = (1, 0), c_3 = (1, 1)$. This produces a partition of $E(G)$ into three perfect matchings E_1, E_2, E_3 where E_i is the set of edges colored c_i . Let G_1 be a spanning subgraph of G with edge set $E(G_1) = E_1 \cup E_2$ and G_2 be the spanning subgraph of G with edge set

$E(G_2) = E_2 \cup E_3$. Then, both G_1 and G_2 are 2-regular spanning subgraphs of G so each of G_1 and G_2 is the disjoint union of even cycles. For $i = 1, 2$ each region of G_i is the union of regions of G . For each cycle C in the graph G_i , every region of G either lies interior or exterior to C . Furthermore, each edge of G belongs to a cycle C' of G_i and is on the boundary of two distinct regions of G_i , one of which lies interior to C' and the other exterior to C' .

We now define a 4-region coloring of G . We assign to region R of G the color (a_1, a_2) with $a_i \in \{0, 1\}$ for $i = 1, 2$, where $a_i = 0$ if R lies interior to even number of cycles in G_i and $a_i = 1$ if R lies interior to odd number of cycles in G_i .

Next, we shall show that this 4-region coloring of G is proper. Let R_1 and R_2 be two adjacent regions with common edge e . Then clearly either e is in G_1 or in G_2 or in both. Thus, e lies on a cycle in G_1 or a cycle in G_2 or both. Let C be a cycle in G_1 containing the edge e . Exactly, one of R_1 and R_2 lies interior to C , while for every other cycle C' , either both lie interior to C' or both exterior to C' . Hence, the first coordinate of the colours of R_1 and R_2 are different. Therefore, the colors of every two adjacent regions of G differ in the first coordinate or the second coordinate or both. Hence, the 4-region coloring of G is proper. ■

Theorem 1.3.7 (Tait's Theorem). *A cubic bridgeless planar graph G is 4-region colorable if and only if G is 3-edge colorable.*

Proof. We prove the result in both directions.

Suppose first that G is 4-region colorable.

Let the four region colors be identified with the four elements of $\mathbb{Z}_2 \times \mathbb{Z}_2$, namely

$$(0, 0), \quad (0, 1), \quad (1, 0), \quad (1, 1).$$

We define a color for each edge of G as follows. If an edge e separates two regions having colors a and b , assign to e the color

$$a + b,$$

where the addition is taken in $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Since the region coloring is proper, the two regions adjacent to an edge have different colors. Therefore, $a \neq b$, and hence

$$a + b \neq (0, 0).$$

Thus every edge receives one of the three nonzero elements

$$(0, 1), \quad (1, 0), \quad (1, 1).$$

We now show that the three edges incident with any vertex receive three different colors.

Let v be a vertex of G . Since G is cubic, exactly three edges are incident with v . Let the three regions occurring around v have colors a, b, c . Because the region coloring is proper, these three colors are pairwise distinct.

The three incident edges receive the colors

$$a + b, \quad b + c, \quad c + a.$$

These three colors are nonzero and pairwise distinct. For example, if

$$a + b = b + c,$$

then, since every element of $\mathbb{Z}_2 \times \mathbb{Z}_2$ is its own additive inverse, cancellation gives $a = c$, which is impossible. The same argument applies to the other pairs.

Hence the three edges incident with every vertex receive three distinct colors. Therefore, G is 3-edge colorable.

Conversely, suppose that G is 3-edge colorable.

Let the three edge colors be

$$c_1 = (0, 1), \quad c_2 = (1, 0), \quad c_3 = (1, 1).$$

Let

$$E_i = \{e \in E(G) : e \text{ has color } c_i\}, \quad i = 1, 2, 3.$$

Since G is cubic and the edge coloring is proper, exactly one edge of each color is incident with every vertex. Hence each set E_i is a perfect matching.

Now define two spanning subgraphs G_1 and G_2 of G by

$$E(G_1) = E_1 \cup E_2$$

and

$$E(G_2) = E_2 \cup E_3.$$

At each vertex, exactly two edges belong to G_1 , one from E_1 and one from E_2 . Therefore, every vertex has degree 2 in G_1 . Similarly, every vertex has degree 2 in G_2 .

Thus both G_1 and G_2 are 2-regular spanning subgraphs. Consequently, each of them is a disjoint union of cycles.

Moreover, the cycles are even. Indeed, in a cycle of G_1 , the edge colors must alternate between c_1 and c_2 , so the cycle has even length. Similarly, the colors on every cycle of G_2 alternate between c_2 and c_3 , and hence those cycles are also even.

Because G is planar, every cycle divides the plane into an interior region and an exterior region.

We now use the cycles of G_1 and G_2 to construct a 4-region coloring of G .

Let R be any region of G . Assign to R the ordered pair

$$(a_1, a_2),$$

where $a_1, a_2 \in \{0, 1\}$ are defined as follows:

$$a_1 = \begin{cases} 0, & \text{if } R \text{ lies inside an even number of cycles of } G_1, \\ 1, & \text{if } R \text{ lies inside an odd number of cycles of } G_1, \end{cases}$$

and

$$a_2 = \begin{cases} 0, & \text{if } R \text{ lies inside an even number of cycles of } G_2, \\ 1, & \text{if } R \text{ lies inside an odd number of cycles of } G_2. \end{cases}$$

Thus every region receives one of the four possible colors

$$(0, 0), \quad (0, 1), \quad (1, 0), \quad (1, 1).$$

It remains to prove that this region coloring is proper.

Let R_1 and R_2 be two adjacent regions of G , and let e be their common boundary edge.

Every edge of G belongs to at least one of the subgraphs G_1 and G_2 . More precisely,

$$e \in E_1 \Rightarrow e \in G_1,$$

$$e \in E_2 \Rightarrow e \in G_1 \cap G_2,$$

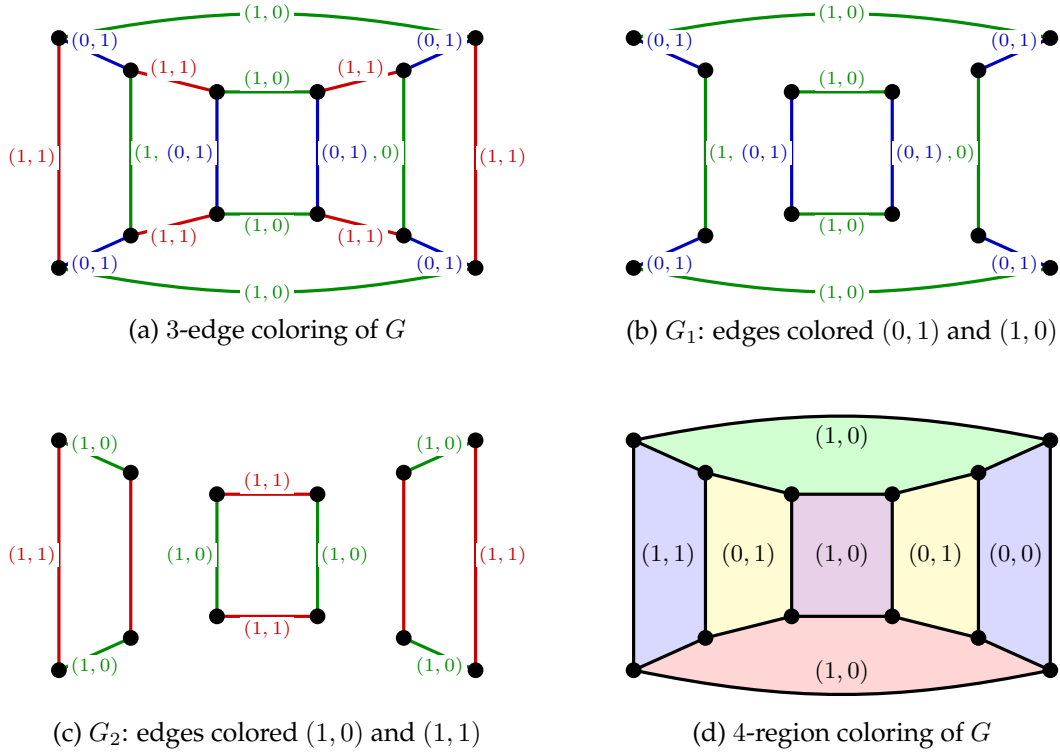
and

$$e \in E_3 \Rightarrow e \in G_2.$$

Suppose first that $e \in G_1$. Since G_1 is a disjoint union of cycles, the edge e lies on exactly one cycle C of G_1 .

The two regions R_1 and R_2 lie on opposite sides of the edge e . Hence one of them lies inside the cycle C and the other lies outside C .

For every other cycle C' of G_1 , the regions R_1 and R_2 lie on the same side of C' . Therefore, when we pass from R_1 to R_2 across the edge e , the number of cycles of G_1 containing the region changes by exactly one.

Figure 1.9: Tait's Theorem: $\chi'(G) = 3 \implies G$ is 4-region colorable.

Hence the parity changes. Therefore, the first coordinates of the colors of R_1 and R_2 are different:

$$a_1(R_1) \neq a_1(R_2).$$

Similarly, if $e \in G_2$, then e lies on a cycle of G_2 , and the same argument shows that

$$a_2(R_1) \neq a_2(R_2).$$

Thus we have three cases.

If $e \in E_1$, then $e \in G_1$, so the first coordinates of the colors of R_1 and R_2 are different.

If $e \in E_3$, then $e \in G_2$, so the second coordinates are different.

If $e \in E_2$, then e belongs to both G_1 and G_2 , so both coordinates are different.

Therefore, in every case, two adjacent regions receive different colors. Hence the constructed 4-region coloring is proper.

Thus G is 4-region colorable.

Therefore, a cubic bridgeless planar graph G is 4-region colorable if and only if it is 3-edge colorable. ■

Figure 1.10: Tait's Theorem

1.4 Heawood Map-Colouring Theorem

Let S_p denote the orientable surface of genus p . Topologically, S_p may be regarded as a sphere with p handles attached. The chromatic number of S_p , denoted by $\chi(S_p)$, is defined to be the maximum chromatic number among all graphs that can be embedded on S_p . In particular, S_0 is the sphere, and the Four Colour Theorem asserts that $\chi(S_0) = 4$.

Definition 1.4.1 (Genus of a Graph). The *genus* of a graph G , denoted by $p(G)$, is the minimum non-negative integer p such that G can be embedded on the orientable surface S_p .

Thus $p(G)$ is the minimum number of handles that must be added to a sphere in order to obtain a surface on which G can be embedded. In particular, $p(G) = 0$ if and only if G is planar.

If a graph is embedded on S_p with n vertices, m edges, and f faces, then Euler's formula takes the form

$$n - m + f = 2 - 2p.$$

Theorem 1.4.2 (Genus of the Complete Graph). *For every integer $n \geq 3$, the orientable genus of the complete graph K_n is*

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Proof. Let K_n be embedded on an orientable surface S_p of genus p . Let m and f denote the numbers of edges and faces, respectively. Since K_n has n vertices, we have $m = n(n-1)/2$.

Every face boundary contains at least three edges, and each edge is incident with two faces. Hence $3f \leq 2m$, so that $f \leq 2m/3$.

By Euler's formula, $n - m + f = 2 - 2p$. Therefore

$$2 - 2p = n - m + f \leq n - m + \frac{2m}{3} = n - \frac{m}{3}.$$

Using $m = n(n-1)/2$, we get

$$2 - 2p \leq n - \frac{n(n-1)}{6}.$$

On simplification, this gives $12p \geq (n-3)(n-4)$. Hence $p \geq (n-3)(n-4)/12$. Since p is an integer,

$$p(K_n) \geq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

For the reverse inequality, Ringel and Youngs proved that K_n can be embedded on an orientable surface of genus $\lceil (n-3)(n-4)/12 \rceil$. Therefore

$$p(K_n) \leq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Combining the two inequalities, we obtain

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

■

Detailed proof of the Ringel-Youngs Theorem is as follows.

Theorem 1.4.3 (Ringel–Youngs). *For every integer $n \geq 3$, the orientable genus of the complete graph K_n is*

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Proof. Let K_n be embedded on an orientable surface S_p of genus p . Since K_n has n vertices and $m = n(n-1)/2$ edges, let f denote the number of faces of the embedding.

Every face boundary contains at least three edges, and every edge is incident with two faces. Hence $3f \leq 2m$. By Euler's formula, $n - m + f = 2 - 2p$, and therefore

$$2 - 2p \leq n - m + \frac{2m}{3} = n - \frac{m}{3}.$$

Substituting $m = n(n-1)/2$ and simplifying, we obtain $12p \geq (n-3)(n-4)$. Since p is an integer,

$$p(K_n) \geq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

It remains to prove that this lower bound is attainable. This is the main part of the Ringel–Youngs theorem.

For an embedding of a graph, define its *face excess* by

$$\varepsilon = \sum_F (|F| - 3),$$

where $|F|$ denotes the length of the boundary of the face F . Since the sum of the lengths of all face boundaries is $2m$, we have $3f + \varepsilon = 2m$. Consequently, $f = (2m - \varepsilon)/3$. Substitution in Euler's formula gives

$$p = \frac{(n-3)(n-4) + 2\varepsilon}{12}.$$

Thus, to obtain an embedding of minimum genus, it is sufficient to construct an embedding of K_n for which

$$\varepsilon = \frac{1}{2} \left(12 \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil - (n-3)(n-4) \right).$$

The value of ε depends only on the residue class of n modulo 12. The required values are

$n \pmod{12}$	0	1	2	3	4	5	6	7	8	9	10	11
ε	0	3	5	0	0	5	3	0	2	3	3	2.

Ringel and Youngs constructed embeddings having exactly these face excesses by means of *current graphs*. We describe the construction.

A current graph is an oriented graph whose directed edges are labelled by elements of a cyclic group $\mathbb{Z}_q$. If an edge receives the current a in one direction, it receives the current $-a$ in the opposite direction. At each ordinary vertex the currents are chosen to satisfy Kirchhoff's law: the sum of the currents entering the vertex is 0 in $\mathbb{Z}_q$.

The boundary of a face of the current graph is read successively, and the sequence of currents so obtained is called its *log*. The current graphs used by Ringel and Youngs are chosen so that every nonzero element of the current group occurs in the log in the required way. From such a log a rotation system is obtained as follows.

If

$$(a_1, a_2, \dots, a_r)$$

is the log at the vertex 0, then the rotation at a vertex $i \in \mathbb{Z}_q$ is obtained by adding i to every numerical entry:

$$i : (i + a_1, i + a_2, \dots, i + a_r),$$

where the additions are taken modulo q . Thus the single current graph generates simultaneously the rotations at all the numbered vertices.

Because every nonzero group element occurs in the log, every difference $j - i$, with $i \neq j$, occurs. Hence the derived graph contains every edge joining two numbered vertices. In other words, the numbered vertices induce a complete graph.

At an ordinary vertex of degree three, Kirchhoff's law implies that the corresponding faces in the derived embedding are triangles. Thus, when there are no exceptional vertices, the derived rotation system gives a triangular embedding of a complete graph.

This immediately settles the residue classes $n \equiv 0, 3, 4, 7 \pmod{12}$. In these cases $(n-3)(n-4)$ is divisible by 12, so that $\varepsilon = 0$, and Ringel–Youngs current graphs give triangular embeddings of K_n . For such an embedding, $3f = 2m$, and hence

$$p = \frac{(n-3)(n-4)}{12}.$$

For the remaining residue classes a triangular embedding of K_n is arithmetically impossible, because $(n-3)(n-4)$ is not divisible by 12. Ringel and Youngs therefore introduced certain exceptional vertices, called *vortices*, into the current graph. A vortex does not satisfy the ordinary Kirchhoff condition and produces one or more large faces in the derived embedding. New vertices are inserted into these faces and joined to the vertices occurring on their boundaries.

The resulting derived graph is either K_n itself or a graph of the form $K_n - L$, where L is a small graph consisting of the few edges between the newly introduced vertices that have not yet been obtained. The missing edges of L are then inserted by local modifications of the embedding. These modifications consist of diagonal changes in quadrilateral faces and, when necessary, the addition of handles. Each added handle is used to insert the required missing adjacencies without creating crossings.

The constructions are arranged so that the resulting embeddings have precisely the following face excesses:

$n \pmod{12}$	ε
0, 3, 4, 7	0
8, 11	2
1, 6, 9, 10	3
2, 5	5.

Consequently, in every residue class the constructed embedding satisfies

$$p = \frac{(n-3)(n-4) + 2\varepsilon}{12} = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Thus, for every $n \geq 3$, Ringel and Youngs obtained an embedding of K_n on an orientable surface of genus $\lceil (n-3)(n-4)/12 \rceil$. Therefore

$$p(K_n) \leq \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Combining this with the previously established lower bound gives

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

■

The following inequality is due to Heawood [113].

Theorem 1.4.4. *The chromatic number of the orientable surface of positive genus p has the upper bound*

$$\chi(S_p) \leq \frac{7 + \sqrt{1 + 48p}}{2}, \quad p > 0.$$

Proof. Let G be an (n, m) graph embedded on S_p . Since any graph can be augmented to a triangulation of the same genus by adding edges without reducing χ , we assume G to be a triangulation. Let d' be the average degree of the vertices of G . Then n, m and f (the number of regions) are related by the equations

$$d'n = 2m = 3f.$$

Solving for m and f in terms of n and using Euler's equation $n - m + f = 2 - 2p$, we get

$$d' = \frac{12(p-1)}{n} + 6. \tag{1.2}$$

As $d' \leq n-1$, this gives $n-1 \geq \frac{12(p-1)}{n} + 6$. Solving for n and taking the positive square root, we obtain $n \geq \frac{7 + \sqrt{1 + 48p}}{2}$.

Let $H(p) = \frac{7 + \sqrt{1 + 48p}}{2}$. Then we show that $H(p)$ colours are sufficient to colour the vertices of G . If $n = H(p)$, obviously we have sufficient colours. In case $n > H(p)$, we have $\frac{1}{n} < \frac{1}{H(p)}$ then equation (1.2) gives obtain

$$d' < \frac{12(p-1)}{H(p)} + 6 \quad (1.3)$$

Now,

$$\begin{aligned} 2H(p) &= 7 + \sqrt{1 + 48p} \\ \text{or } (2H(p) - 7)^2 &= 1 + 48p \\ \text{or } 4H^2(p) - 28H(p) + 49 &= 1 + 48p \\ \text{or } H(p) [H(p) - 7] &= 12(p-1) \\ \text{or } H(p) - 7 &= \frac{12(p-1)}{H(p)} \\ \text{i.e. } H(p) - 1 &= \frac{12(p-1)}{H(p)} + 6 \end{aligned}$$

Hence, equation 1.3 gives, $d' < H(p) - 1$. Therefore, when $n > H(p)$, there is a vertex v of degree at most $H(p) - 2$. Identify v and any adjacent vertex by an elementary contraction to obtain a new graph G' . If $n' = n - 1 = H(p)$, then G' can be coloured in $H(p)$ colours. If $n' > H(p)$, repeat the argument and evidently we get an $H(p)$ colourable graph. It is then easy to see that the colouring of this graph induces a colouring of the preceding one in $H(p)$ colours, and so forth, so that G itself is $H(p)$ -colourable. Hence the result follows. ■

Theorem 1.4.5 (Heawood Map-Colouring Theorem). *For every positive integer p , the chromatic number of the orientable surface S_p of genus p is*

$$\chi(S_p) = \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor.$$

Proof. By the upper-bound theorem, every graph G embeddable on S_p satisfies $\chi(G) \leq \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor$. Hence $\chi(S_p) \leq \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor$.

It remains to establish the reverse inequality. Recall that the orientable genus of the complete graph K_n is

$$p(K_n) = \left\lceil \frac{(n-3)(n-4)}{12} \right\rceil.$$

Let n be the largest integer such that K_n is embeddable on S_p . Then $p(K_n) \leq p$, whereas $p(K_{n+1}) > p$. Since p is an integer, the genus formula gives

$$\frac{(n-3)(n-4)}{12} \leq p < \frac{(n-2)(n-3)}{12}.$$

Thus $n^2 - 7n + 12 \leq 12p < n^2 - 5n + 6$. The first inequality yields $n \leq \frac{7 + \sqrt{1 + 48p}}{2}$, while the second yields $n > \frac{5 + \sqrt{1 + 48p}}{2}$. Therefore

$$\frac{5 + \sqrt{1 + 48p}}{2} < n \leq \frac{7 + \sqrt{1 + 48p}}{2}.$$

Since the two bounds differ by 1 and n is an integer,

$$n = \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor.$$

Now K_n is embeddable on S_p and $\chi(K_n) = n$. Hence $\chi(S_p) \geq n = \left\lfloor \frac{7+\sqrt{1+48p}}{2} \right\rfloor$.

Combining the two inequalities, we obtain

$$\chi(S_p) = \left\lfloor \frac{7 + \sqrt{1 + 48p}}{2} \right\rfloor.$$

■

1.5 Uniquely Colorable Graphs

Definition 1.5.1 (Uniquely Colorable Graph). A graph $G = (V, E)$ is said to be *uniquely k -colorable* if there is a unique partition of V into k independent sets. Equivalently, every proper k -coloring of G determines the same partition of V into color classes, apart from a permutation of the colors.

Theorem 1.5.2. *If G is uniquely k -colorable, then every vertex of G has degree at least $k - 1$.*

Proof. Let c be the unique k -coloring of G , and suppose that some vertex v has degree less than $k - 1$. Then at most $k - 2$ colors occur among the neighbours of v . Hence, among the k available colors, there is a color different from $c(v)$ which is not used on any neighbour of v . Recoloring v with this color gives another proper k -coloring of G , and the resulting partition into color classes is different from the original one. This contradicts the uniqueness of the k -coloring. Therefore, $d(v) \geq k - 1$ for every $v \in V(G)$. ■

Theorem 1.5.3. *In a uniquely k -colorable graph G , the subgraph induced by the union of any two color classes is connected.*

Proof. Let C_i and C_j be two distinct color classes in the unique k -coloring of G , and let $H = G[C_i \cup C_j]$. Suppose that H is disconnected, and let H_1 be one of its components. Interchange the colors i and j on all vertices of H_1 , leaving the colors of all other vertices unchanged. Since H_1 is a component of H , no edge of H joins a vertex of H_1 to a vertex of another component of H . Hence the interchange preserves proper coloring and produces another proper k -coloring of G inducing a different partition of the vertices into color classes. This contradicts the assumption that G is uniquely k -colorable. Therefore, $G[C_i \cup C_j]$ is connected. ■

Theorem 1.5.4. *A uniquely k -colorable graph is $(k - 1)$ -connected.*

Proof. Let G be uniquely k -colorable. If G is complete, the result is immediate. Suppose that G is not complete. Assume, to the contrary, that G is not $(k - 1)$ -connected. Then there exists a vertex cut X with $|X| \leq k - 2$ such that $G - X$ is disconnected. Since at most $k - 2$ vertices occur in X , at least two colors, say c_1 and c_2 , are not used on any vertex of X . Let C_1 and C_2 denote the corresponding color classes. By the preceding theorem, the subgraph $G[C_1 \cup C_2]$ is connected. As no vertex of X has color c_1 or c_2 , the entire subgraph $G[C_1 \cup C_2]$ lies in $G - X$ and therefore in a single component, say G_1 . Let G_2 be another component of $G - X$, and choose $v \in V(G_2)$. Its color is different from c_1 and c_2 . Consider the subgraph induced by the color of v together with c_1 , and interchange these two colors on the component containing v . This produces another proper k -coloring of G with a different color-class partition, contradicting uniqueness. Hence G is $(k - 1)$ -connected. ■

Theorem 1.5.5. *Every planar graph is four-colorable if and only if every Hamiltonian planar graph is four-colorable.*

Proof. If every planar graph is four-colorable, then every Hamiltonian planar graph is four-colorable, since every Hamiltonian planar graph is planar. Conversely, suppose that every Hamiltonian planar graph is four-colorable. Using the standard fact that every planar graph can be embedded as a subgraph of a Hamiltonian planar graph, let G be an arbitrary planar graph and let G' be a Hamiltonian planar graph containing G as a subgraph. By hypothesis, G' is four-colorable. Restricting such a coloring to G gives a proper four-coloring of G . Hence every planar graph is four-colorable. ■

Chapter 2

Matchings and Factors

Historical Note and Motivation

The theory of matchings began with **Julius Petersen's** 1891 study of regular graphs and evolved through **Philip Hall's** 1935 matching criteria for bipartite graphs. The field was revolutionized by **W.T. Tutte** in 1947, who characterized 1-factors in general graphs and later extended this to f -factors. These structural foundations were complemented by **Claude Berge's** algorithmic approach using augmenting paths and the **Erdős-Gallai** characterization of degree sequences. The study culminates in the factorization of K_n , a problem rooted in 19th-century tournament designs.

This chapter explores the fundamental tension between **local constraints** (vertex degrees) and **global structure** (factors). By mastering these topics, we learn how to determine when local demands can be universally satisfied, a concept essential for solving real-world assignment and scheduling problems. Theoretically, this section highlights the power of reduction, demonstrating how complex f -factor problems can be solved using the simpler mechanics of 1-factors.

2.1 Matchings

Definition 2.1.1 (Matching, Maximal, and Maximum). A **matching** in a graph is a set of independent edges. That is, a subset M of the edge set E of a graph $G(V, E)$ is a matching if no two edges of M have a common vertex.

- A matching M is said to be **maximal** if there is no matching M' strictly containing M . That is, M is maximal if it cannot be enlarged (you cannot add another edge without breaking the matching property).
- A matching M is said to be **maximum** if it has the largest possible cardinality. That is, M is maximum if there is no matching M' such that $|M'| > |M|$.

Definition 2.1.2 (Saturated and Unsaturated Vertices). Let M be a matching in a graph G .

- A vertex v in G is said to be **M -saturated** (or saturated by M) if there is an edge $e \in M$ incident with v .
- A vertex which is not incident with any edge of M is said to be **M -unsaturated**.

In other words, given a matching M in a graph G , the vertices belonging to the edges of M are M -saturated and the vertices not belonging to the edges of M are M -unsaturated.

Consider the graph shown in Figure 2.1 below. Clearly, $M = \{v_1v_2, v_3v_7, v_4v_5\}$ is a matching. The vertices $v_1, v_2, v_3, v_4, v_5, v_7$ are M -saturated, but v_6 is M -unsaturated.

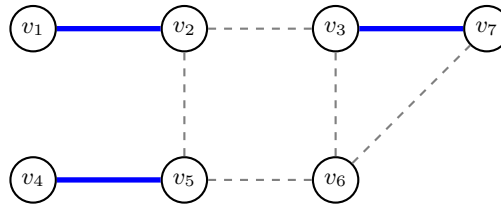


Figure 2.1: Visual representation of the matching M . The blue edges form the matching.

Definition 2.1.3 (Perfect Matching). A matching M in a graph G is said to be a **perfect matching** if M saturates every vertex of G .

Definition 2.1.4 (Complete Matching in Bipartite Graphs). If G is a bipartite graph with bipartition $V = V_1 \cup V_2$ and $|V_1| = n_1$, $|V_2| = n_2$, then a matching M of G saturating all the vertices in V_1 is called a **complete matching** (or V_1 matched into V_2).

Definition 2.1.5 (Alternating Path). Let H be a subgraph of a graph G . An H -**alternating path** (cycle) in G is a path (cycle) whose edges are alternately in $E(G) \setminus E(H)$ and $E(H)$.

Specifically, if M is a matching in a graph G , then an M -**alternating path** (cycle) in G is a path (cycle) whose edges are alternately in $E(G) \setminus M$ and M . That is, in an M -alternating path, the edges alternate between M -edges and non- M -edges.

Definition 2.1.6 (Augmenting Path). An M -alternating path whose end vertices are M -unsaturated is said to be an M -**augmenting path**.

Example 2.1.7. Examples to explain definitions given above.

- **Matching:** A set of edges without common vertices.
- **Maximal Matching:** A matching to which no more edges can be added.
- **Maximum Matching:** A matching with the largest possible number of edges.



Figure 2.2: **Maximal Matching.** Size = 1. We cannot add (A, B) or (C, D) because B and C are already busy.

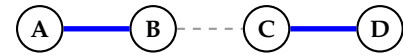


Figure 2.3: **Maximum Matching.** Size = 2. This is the largest possible matching for this graph.

Saturated vs. Unsaturated Vertices

A vertex is **Saturated** if it touches a matched edge (blue). Otherwise, it is **Unsaturated**.

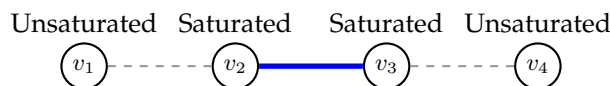


Figure 2.4: Vertices v_2, v_3 are saturated by the blue edge. v_1, v_4 are unsaturated.

Perfect Matching

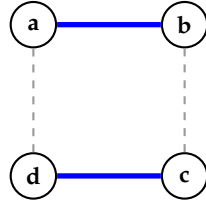
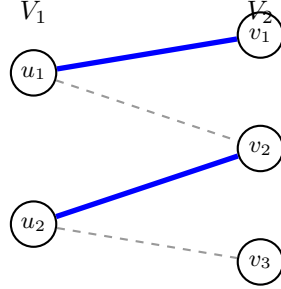
A matching that saturates **every** vertex in the graph.

Complete Matching (Bipartite)

In a bipartite graph $G = (V_1 \cup V_2, E)$, a matching is **complete from V_1 to V_2** if every vertex in V_1 is saturated.

Alternating and Augmenting Paths

- **Alternating Path:** Edges alternate between Non-Matching and Matching.
- **Augmenting Path:** An alternating path that starts and ends at unsaturated vertices.

Figure 2.5: A Perfect Matching on C_4 . All 4 vertices are covered.Figure 2.6: Complete Matching of V_1 into V_2 . All vertices in V_1 are blue-connected. Vertex v_3 is left over, which is fine.Figure 2.7: **Alternating Path** (1 – 2 – 3). Starts Unsat(1) → Sat(3).Figure 2.8: **Augmenting Path** (1 – 2 – 3 – 4). Starts Unsat(1) → Ends Unsat(4).

Theorem 2.1.8 (Berge). *A matching M of a graph G is maximum if and only if G contains no M -augmenting path.*

Proof. Suppose first that M is a maximum matching and that $P = v_1v_2 \cdots v_k$ is an M -augmenting path. Then the edges of P alternate between edges outside M and edges in M , with the first and last edges outside M . Thus $v_2v_3, v_4v_5, \dots, v_{k-2}v_{k-1} \in M$, while $v_1v_2, v_3v_4, \dots, v_{k-1}v_k \notin M$.

Replacing the edges of M on P by the remaining edges of P produces a matching

$$M_1 = [M - \{v_2v_3, \dots, v_{k-2}v_{k-1}\}] \cup \{v_1v_2, \dots, v_{k-1}v_k\},$$

with $|M_1| = |M| + 1$, contradicting the maximality of M . Hence no M -augmenting path exists.

Conversely, suppose that G contains no M -augmenting path. If M were not maximum, there would exist a matching M' with $|M'| > |M|$. Let H be the spanning subgraph of G whose edge set is $M \triangle M'$.

Since each vertex is incident with at most one edge of M and at most one edge of M' , every vertex of H has degree at most 2. Hence each component of H is an isolated vertex, a path, or a cycle, and its edges alternate between M and M' .

Every cycle contains the same number of edges from M and M' . Since $|M'| > |M|$, some path component must contain one more edge from M' than from M . Such a path begins and ends with edges of M' , and therefore its end vertices are unsaturated by M . Hence it is an M -augmenting path, a contradiction.

Therefore, M is maximum. ■

Theorem 2.1.9 (Hall's Marriage Theorem). *A bipartite graph $G = (X \cup Y, E)$ has a matching that covers all vertices in X if and only if for every $S \subseteq X$, $|N(S)| \geq |S|$.*

Proof. Assume a matching M covers all vertices in X . For any $S \subseteq X$, each vertex in S is matched by M to a distinct vertex in Y . Let $M(S)$ be this set of matched vertices in Y . Then $|M(S)| = |S|$. Since every vertex in S is connected to its matched partner, $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |M(S)| = |S|$.

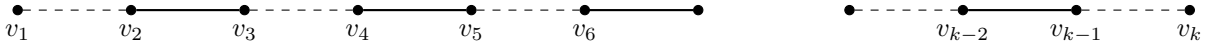


Figure 2.9:

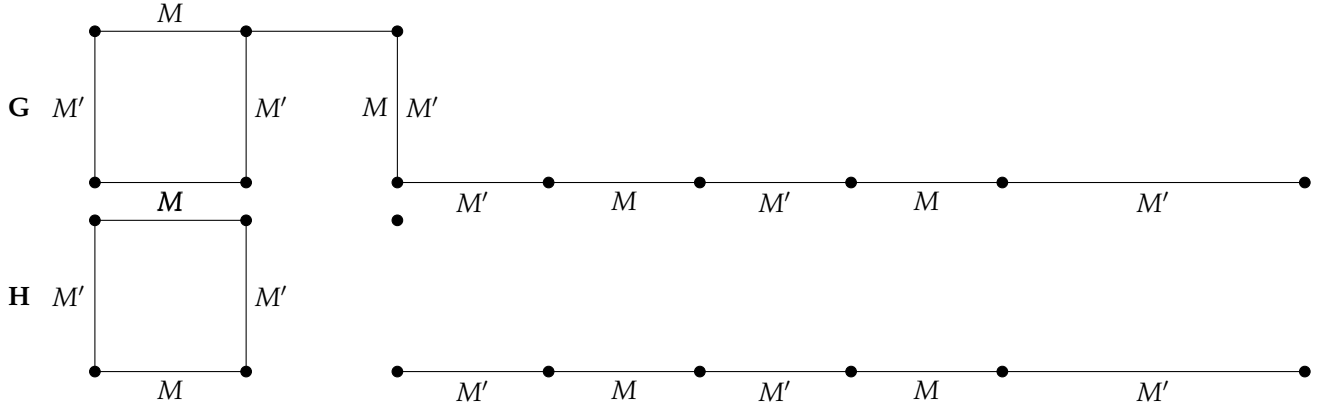


Figure 2.10:

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction that a matching covering X exists. Suppose no such matching exists. Let M be a maximum matching in G . Since M does not cover X , there must be at least one unmatched vertex $u \in X$. By Berge's Lemma, since M is a maximum matching, there are no M -augmenting paths in G . Let A be the set of all vertices reachable from u by an M -alternating path. Let $X' = A \cap X$ and $Y' = A \cap Y$. Since u is unmatched and starts an M -alternating path, $u \in X'$. For any $y \in Y'$, it must be connected to some $x \in X'$ by an edge not in M . M must match this x to some $y' \in Y'$, otherwise we could extend the path to x to form an M -augmenting path from u to x , a contradiction (since x would be unmatched). Specifically, any vertex $x \in X'$ (other than u) must be matched by M to some $y \in Y'$. If x were unmatched, then an M -alternating path from u to x would be an M -augmenting path, a contradiction. Similarly, any vertex $y \in Y'$ must be matched by M to some $x \in X'$. If y were matched to $x \notin X'$, then combining the path from u to y (ending with an M -edge) with the edge (y, x) (which is in M) would form an M -alternating path from u to x ending with an M -edge. This would imply $x \in X'$, a contradiction. Therefore, all edges of M that are incident to A must connect a vertex in $X' \setminus \{u\}$ to a vertex in Y' . This implies that M provides a perfect matching between $X' \setminus \{u\}$ and Y' . Thus, $|X'| - 1 = |Y'|$.

Now consider the set $S = X'$. By construction of A , all neighbors of X' must be in Y' . Thus $N(X') \subseteq Y'$. From our previous deduction, $|Y'| = |X'| - 1$. Therefore, $|N(X')| \leq |Y'| = |X'| - 1$. This means $|N(X')| < |X'|$, which contradicts the given condition that $|N(S)| \geq |S|$ for all $S \subseteq X$. Our initial assumption that no matching covering X exists must be false. Hence, such a matching exists. ■

2nd Proof using Augmenting Paths. Assume a matching M covers all vertices in X . For any $S \subseteq X$, each $x \in S$ is matched to a distinct $y_x \in Y$. These y_x form the set $M(S)$. Clearly, $|M(S)| = |S|$ and $M(S) \subseteq N(S)$. Thus, $|N(S)| \geq |S|$.

Assume $|N(S)| \geq |S|$ for all $S \subseteq X$. We prove by contradiction. Suppose no matching covers X . Let M be a maximum matching. Since it doesn't cover X , there's an unmatched $u \in X$. By the given theorem (Berge's Lemma), since M is maximum, there are no M -augmenting paths.

Let A be the set of vertices reachable from u by M -alternating paths. Let $X_A = A \cap X$ and $Y_A = A \cap Y$. (See Figure 2.11). Since there are no M -augmenting paths, u is the only vertex in A that is M -unsaturated. Furthermore, if $y \in Y_A$ is matched by M , its matched partner x' must be in X_A . (If $x' \notin X_A$, then extending the M -alternating path from u to y with the edge $(y, x') \in M$ would make x' reachable, implying $x' \in X_A$, a contradiction). Thus, M perfectly matches Y_A with $X_A \setminus \{u\}$. Therefore, $|Y_A| = |X_A| - 1$.

Now, let $S = X_A$. We claim $N(S) \subseteq Y_A$. Take any $x \in S$ and its neighbor $y \in N(x)$. If $y \notin Y_A$,

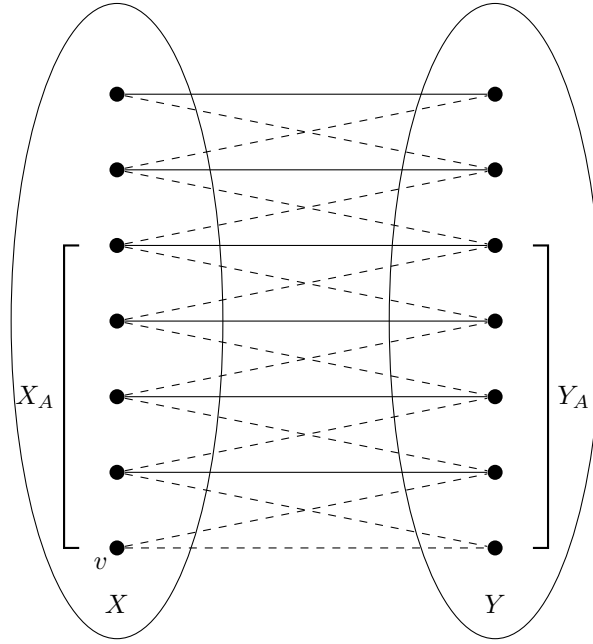


Figure 2.11: Bipartite graph illustrating sets $X_A \subseteq X$ and $Y_A \subseteq Y$ relative to a matching.

then the edge (x, y) cannot be in M (because if it were, then x would be matched to y , but $x \in X_A$ and $y \notin Y_A$ implies x is not matched within A , a contradiction to M matching $X_A \setminus \{u\}$ to Y_A). If $(x, y) \notin M$, then combining the M -alternating path from u to x with (x, y) would make y reachable, implying $y \in Y_A$, a contradiction. Thus, $N(S) \subseteq Y_A$.

Combining these, $|N(S)| \leq |Y_A| = |X_A| - 1 = |S| - 1$. So, $|N(S)| < |S|$ for $S = X_A$. This contradicts our initial assumption. Therefore, a matching covering X must exist. ■

Theorem 2.1.10 (Hall's SDR Theorem). *Let $\mathcal{A} = \{A_i : 1 \leq i \leq r\}$ be a finite family of nonempty finite sets. Then $\mathcal{A}$ has a system of distinct representatives if and only if, for every subset $I \subseteq \{1, 2, \dots, r\}$,*

$$\left| \bigcup_{i \in I} A_i \right| \geq |I|.$$

Equivalently, the union of any k members of $\mathcal{A}$ contains at least k elements, for $1 \leq k \leq r$.

Proof. Construct a bipartite graph $G = (X \cup Y, E)$ by taking $X = \{1, 2, \dots, r\}$ and $Y = \bigcup_{i=1}^r A_i$, and joining $i \in X$ to $y \in Y$ precisely when $y \in A_i$.

A system of distinct representatives for $\mathcal{A}$ is exactly a matching in G that saturates every vertex of X . Indeed, choosing a representative $x_i \in A_i$ corresponds to choosing the edge ix_i , and the representatives are distinct precisely when the chosen edges form a matching.

By Hall's Marriage Theorem, such a matching exists if and only if $|N(S)| \geq |S|$ for every subset $S \subseteq X$. But, by the construction of G ,

$$N(S) = \bigcup_{i \in S} A_i.$$

Hence Hall's condition is equivalent to

$$\left| \bigcup_{i \in S} A_i \right| \geq |S|$$

for every $S \subseteq X$.

Therefore, $\mathcal{A}$ has an SDR if and only if the union of any k members of $\mathcal{A}$ contains at least k elements. ■

Theorem 2.1.11. Let G be a bipartite graph with bipartite sets V_1 and V_2 of orders n_1 and n_2 , respectively, where $n_1 \leq n_2$. Suppose that G satisfies Hall's condition

$$|S| \leq |N(S)|$$

for every subset $S \subseteq V_1$. Then G has at least

$$r(\delta, n_1) = \prod_{i=1}^{\min\{\delta, n_1\}} (\delta - i + 1)$$

complete matchings, where

$$\delta = \min\{d(v) : v \in V_1\}.$$

Proof. We proceed by induction on n_1 .

If $n_1 = 1$, then $V_1 = \{v_1\}$ and $d(v_1) = \delta$. Each edge incident with v_1 gives a complete matching. Hence there are δ complete matchings. Since $r(\delta, 1) = \delta$, the result holds.

Assume that the result is true for bipartite graphs whose first partite set has $n_1 - 1$ vertices. Let $v_1 \in V_1$. Since $d(v_1) \geq \delta$, there are at least δ choices for a vertex $u_1 \in N(v_1)$.

For each such choice of u_1 , consider the graph $G' = G - \{v_1, u_1\}$. Its first partite set is $V_1' = V_1 - \{v_1\}$ and has order $n_1 - 1$. On deleting u_1 , a vertex of V_1' can lose at most one incident edge. Therefore the minimum degree of the vertices of V_1' in G' is at least $\delta - 1$.

By the induction hypothesis, G' has at least $r(\delta - 1, n_1 - 1)$ complete matchings. Each such matching, together with the edge $v_1 u_1$, gives a complete matching of G . Since there are at least δ choices for u_1 , the number of complete matchings of G is at least

$$\delta r(\delta - 1, n_1 - 1).$$

If $\delta \geq n_1$, then $\delta - 1 \geq n_1 - 1$, and hence

$$\delta r(\delta - 1, n_1 - 1) = \delta \prod_{i=1}^{n_1-1} (\delta - i) = \delta(\delta - 1) \cdots (\delta - n_1 + 1) = r(\delta, n_1).$$

If $\delta < n_1$, then $\delta - 1 < n_1 - 1$, and therefore

$$\begin{aligned} \delta \cdot \prod_{i=1}^{\min(\delta-1, n_1-1)} (\delta - i) &= \delta \cdot \prod_{i=1}^{\delta-1} (\delta - i) \\ &= \delta((\delta - 1)(\delta - 2) \cdots 1) \\ &= \delta(\delta - 1)! \\ &= \delta! \\ &= \prod_{i=1}^{\delta} (\delta - i + 1). \end{aligned}$$

Thus, in this case also, the number of complete matchings is at least $r(\delta, n_1)$.

Hence the result follows by induction. ■

Theorem 2.1.12. Every k -regular bipartite graph with $k > 0$ has a perfect matching.

Proof. Let $G = (X \cup Y, E)$ be a k -regular bipartite graph, where X and Y are the two partite sets.

Since every vertex has degree k , counting the edges from the side X gives

$$|E| = \sum_{x \in X} d(x) = k|X|.$$

Similarly, counting the same edges from the side Y gives

$$|E| = \sum_{y \in Y} d(y) = k|Y|.$$

Hence

$$k|X| = k|Y|.$$

Since $k > 0$, we obtain

$$|X| = |Y|.$$

We now verify Hall's condition. Let $S \subseteq X$, and let $N(S)$ denote the set of all vertices of Y adjacent to at least one vertex of S .

Every vertex of S has degree k , and all of its incident edges join a vertex of S to a vertex of $N(S)$. Therefore, the number of edges between S and $N(S)$ is

$$|E(S, N(S))| = k|S|.$$

On the other hand, each vertex of $N(S)$ has total degree k . Therefore, at most $k|N(S)|$ edges can join S to $N(S)$. Thus

$$|E(S, N(S))| \leq k|N(S)|.$$

Combining the two relations, we get

$$k|S| \leq k|N(S)|.$$

Since $k > 0$,

$$|S| \leq |N(S)|.$$

Thus Hall's condition holds for every subset $S \subseteq X$. By Hall's Marriage Theorem, G has a matching that saturates every vertex of X .

Finally, since $|X| = |Y|$, a matching that saturates all vertices of X also saturates all vertices of Y . Hence the matching is a perfect matching. ■

2.2 Factors of Graphs

Definition 2.2.1 (Factor). Let G be a graph. A **factor** F of G is a spanning subgraph of G . That is, $V(F) = V(G)$ and $E(F) \subseteq E(G)$.

Definition 2.2.2 ($\mathcal{F}$ -Factor and H -Factor). Let $\mathcal{F} = \{H_1, H_2, \dots, H_k\}$ be a collection of graphs. A factor F is called an **$\mathcal{F}$ -factor** if every connected component of F is isomorphic to some element of $\mathcal{F}$.

In the specific case where $\mathcal{F} = \{H\}$ (a singleton set), the factor F is called an **H -factor**. In this case, every component of the spanning subgraph is isomorphic to H .

Degree Constraints: f -Factors and k -Factors

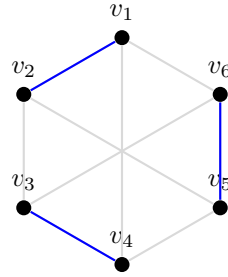
Let G_1 and G_2 be graphs. Recall the *Cartesian product* of G_1 and G_2 , denoted by $G_1 \square G_2$, is the graph with vertex set

$$V(G_1 \square G_2) = V(G_1) \times V(G_2).$$

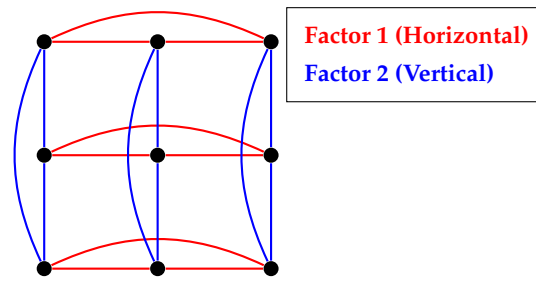
Two vertices (u_1, u_2) and (v_1, v_2) are adjacent in $G_1 \square G_2$ if and only if either $u_1 = v_1$ and $u_2 v_2 \in E(G_2)$, or $u_2 = v_2$ and $u_1 v_1 \in E(G_1)$.

Example 2.2.3. Consider the graph $G = K_3 \square K_3$ (also known as the Rook's graph on a 3×3 grid). This graph has 9 vertices and is 4-regular. It admits a K_3 -factorization consisting of two factors, F_1 and F_2 .

- **Factor 1 (Red):** Three disjoint triangles formed horizontally.
- **Factor 2 (Blue):** Three disjoint triangles formed vertically.

 **K_2 -Factor**

The blue edges form a spanning subgraph where every component is isomorphic to K_2 .

Figure 2.12: Visualisation of an H -factor where $H = K_2$.**Graph $G = F_1 \cup F_2$** Figure 2.13: Visualisation of G decomposed into two K_3 -factors.

Since $E(G) = E(F_1) \cup E(F_2)$ and $E(F_1) \cap E(F_2) = \emptyset$, this constitutes a factorization.

Definition 2.2.4 (Vertex Function). A **vertex function** of a graph $G(V, E)$ is a mapping $f : V \rightarrow \mathbb{N}_0$, where $\mathbb{N}_0$ denotes the set of non-negative integers.

Definition 2.2.5 (f -Factor). A spanning subgraph F of a graph G is called an **f -factor** if the degree of every vertex v in F corresponds to the value assigned by the vertex function. That is:

$$\deg_F(v) = f(v) \quad \forall v \in V.$$

Definition 2.2.6 (k -Factor). A **k -factor** is a special case of an f -factor where $f(v) = k$ for all $v \in V$. Thus, a k -factor is a k -regular spanning subgraph.

- A **1-factor** is a perfect matching (vertices have degree 1).
- A **2-factor** is a disjoint union of cycles covering all vertices.

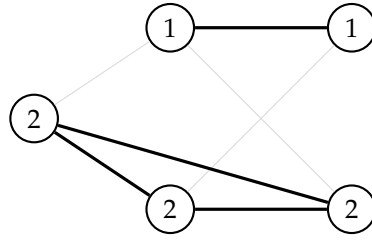
Factorisation

Definition 2.2.7 (Factorisation / Decomposition). Let G be a graph and let $\{F_1, F_2, \dots, F_q\}$ be a set of factors of G . These factors constitute a **factorisation** (or decomposition) of G if their edge sets form a partition of $E(G)$. We write:

$$G = F_1 \oplus F_2 \oplus \dots \oplus F_q$$

This means every edge of G belongs to exactly one factor F_i .

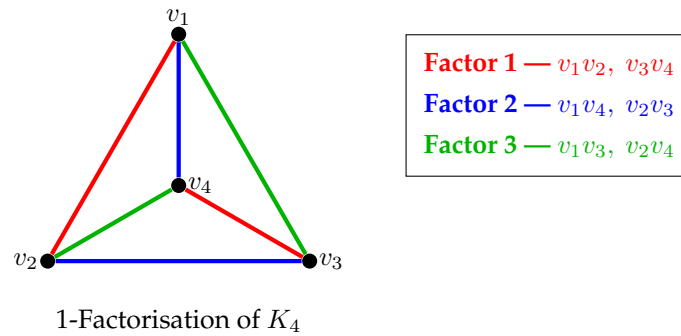
Definition 2.2.8 (Isomorphic Factorisation). If each factor F_i in a factorisation is isomorphic to the same graph F , the factorisation is called an **isomorphic factorisation** or an **F -decomposition** of G .

Example of an f -factor

Numbers inside nodes represent the target degree $f(v)$. The thick edges form the subgraph satisfying these constraints.

Figure 2.14: An f -factor where some vertices require degree 1 and others degree 2.

Definition 2.2.9 (k -Factorisation). A factorisation is called a **k -factorisation** if every factor F_i is a k -factor. A graph admitting such a decomposition is said to be **k -factorable**.

Figure 2.15: A 1-factorisation of the complete graph K_4 . The edges are partitioned into three distinct perfect matchings (Red, Blue, Green).

Definition 2.2.10. A component of a graph is odd or even according to whether it has an odd or even number of vertices. The number of odd components of a graph G is denoted by $k_0(G)$.

Tutte's One-Factor Theorem: A graph $G = (V, E)$ has a perfect matching if and only if for every subset $S \subseteq V$, the number of odd components in $G - S$ is less than or equal to $|S|$. That is, $k_0(G - S) \leq |S|$ for all $S \subseteq V$.

Proof. A Video lecture available at

<https://youtu.be/EW4oA9HZzgg?si=8QGKe7Sfpdem9Rg5>

or visit YouTube via <https://ajaz.vercel.app/>

■

Theorem 2.2.11 (Tutte). A graph G has a 1-factor if and only if

$$k_0(G - S) \leq |S|, \text{ for all } S \subseteq V. \quad (8.8.1)$$

Proof. Let G be a graph having 1-factor M . Let S be an arbitrary subset of V and let $O_1, O_2, \dots, O_k$ be the odd components of $G - S$. For each i , the vertices in O_i can be adjacent only to other vertices in O_i and to vertices in S . Since G has a 1-factor (perfect matching), at least one vertex from each O_i is to be matched with a different vertex in S . Thus the number of vertices in S is at least k , the number of odd components. Hence $|S| \geq k$, so that $k_0(G - S) \leq |S|$ (Fig. T1).

Conversely, assume that the condition (8.8.1) holds. Let G have no 1-factor. Join pairs of non-adjacent vertices of G and continue this process till we get a maximal supergraph G^* of G having no 1-factor. Since joining two odd components by an edge results in an even component, therefore

$$k_0(G^* - S) \leq k_0(G - S). \quad (8.8.2)$$

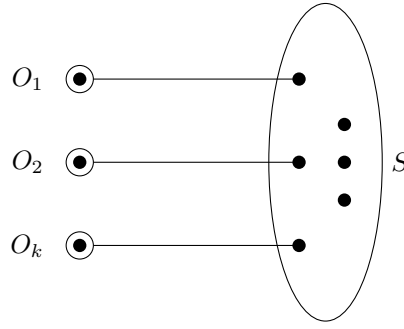


Figure T1

Thus for G^* also, we have $k_0(G^* - S) \leq |S|$.

First, we note that G has even order. This is because the condition (8.8.1) must hold for $S = \phi$, which gives $k_0(G) = 0$. Therefore, in this case G has no odd component, so G itself is an even-ordered graph. Also, for every pair of non-adjacent vertices u and v of G^* , $G^* + uv$ has a 1-factor, and any such 1-factor definitely contains the edge uv .

Now, let K be the set of those vertices of G^* which are having degree $n - 1$. Clearly, $K \neq V$, since otherwise $G^* = K_n$ has a perfect matching.

Claim that each component of $G^* - K$ is complete. Assume, on the contrary, that in $G^* - K$, there exists a component G_1 which is not complete. Then in G_1 , there are vertices x , y and z such that $xy, yz \in E(G^*)$ and $xz \notin E(G^*)$. Also, $y \in V(G_1)$ so $y \notin K$ and so $d(y|G^*) < n - 1$ and thus there exists a vertex w of G^* with $yw \notin E(G^*)$. Evidently, $w \notin K$ (Fig. T2).

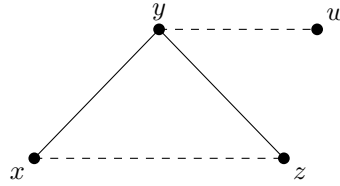
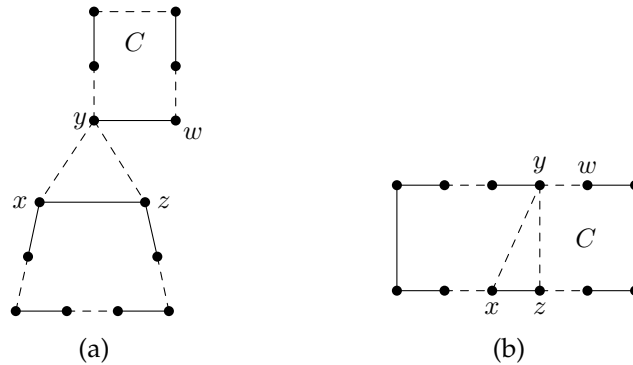


Figure T2

Now, by the choice of G^* , both $G^* + xz$ and $G^* + yw$ have 1-factors, say M_1 and M_2 respectively. M_1 must contain xz and M_2 must contain yw . Let H be the subgraph of $G^* + \{xz, yw\}$ induced by the edges that are in M_1 and M_2 but not in both (that is, in the symmetric difference of M_1 and M_2). Since M_1 and M_2 are 1-factors, each vertex of G^* is saturated by both M_1 and M_2 , and H is a disjoint union of even cycles in which the edges alternate between M_1 and M_2 . We have two cases to consider.

Case 1 Let xz and yw belong to different components of H , as shown in Figure T3 (a). If yw belongs to the cycle C , then the edge of M_1 in C together with the edges of M_2 not belonging to C form a 1-factor of G^* , contradicting the choice of G^* .



Ordinary lines correspond to the edges of M_1 and bold lines to the edges of M_2 .

Figure T3

Case 2 Let xz and yw belong to the same component C of H . Since each component of H is

a cycle, C is a cycle (Fig. T3 (b)). By symmetry of x and z , it can be assumed that the vertices x, y, w and z appear in that order on C . Then the edges of M_1 belonging to the $yw \dots x$ section of C , together with the edge yz , and the edges of M_2 not in the $yw \dots z$ section of C form a 1-factor of G^* , again contradicting the choice of G^* .

Thus each component of $G^* - K$ is complete.

By conditions (8.8.2), $k_0(G^* - K) \leq |K|$. Therefore one vertex of each of the odd components of $G^* - K$ is matched to a vertex of K . This is because each vertex of K is adjacent to every other vertex of G^* . Also the remaining vertices in each of the odd and even components of $G^* - K$ can be matched among themselves (Fig. T4). The total number of vertices thus matched is even. As $|V(G^*)|$ is even, the remaining vertices of K can be matched among themselves, thus giving a 1-factor of G^* , which is a contradiction.

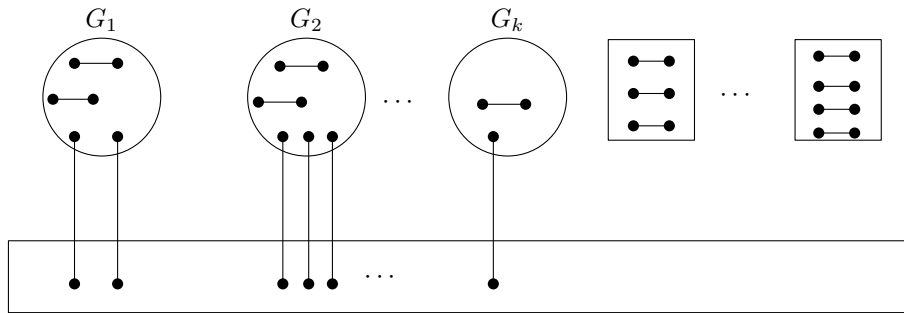


Figure T4

■

Remark 2.2.12. Among the various proofs of Theorem 8.8, one proof due to Anderson [4] uses Hall's theorem.

Tutte's characterisation of graphs admitting 1-factors has been used to obtain several results including those for regular graphs and many of these are reported in the review article by Akiyama and Kano [1]. We present some of these results. The first is due to Peterson [181].

Theorem 2.2.13 (Peterson). *Every cubic graph without cut edges has a 1-factor*

Proof. Let G be a cubic graph with vertex set $V(G)$. Let G have no cut edges and let $S \subseteq V$. Let $O_1, O_2, \dots, O_r$ be the odd components in $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S . Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S . Then we have;

$$\sum_{v \in V(O_i)} \deg(v) = 3n_i = 2m_i + k_i.$$

This gives, $k_i = 3n_i - 2m_i$, $1 \leq i \leq r$. As n_i is odd, therefore, k_i must be odd. Moreover, since there are no cut edges in the graph, we have $k_i \neq 1$, that is, $k_i \geq 3$, that is, there are at least 3 edges from each O_i to S . Hence, we have

$$3r \leq \sum_{s \in S} \deg(s) = 3|S|.$$

Therefore,

$$k_o(G - S) = r \leq |S|$$

for each $S \subseteq V(G)$. The proof then follows by Tutte's theorem.

■

Theorem 2.2.14 (Generalization of Theorem 2.2.13). *Let G be a k -regular graph having even number of vertices and is $(k - 1)$ -edge connected. Then G has a 1-factor.*

Proof. Let G be a k -regular, $(k-1)$ -edge-connected graph with even order and let S be any subset of V . Let $O_1, O_2, \dots, O_r$ be the odd components of $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S . Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k-1)$ -connected, we have

$$k_i \geq k-1, \quad (2.1)$$

Now, G is k -regular and vertices of any O_i are either adjacent within O_i or adjacent to some vertex of S , therefore,

$$\sum_{v \in V(O_i)} d(v) = kn_i = 2m_i + k_i \quad (2.2)$$

and vertices in S can be adjacent within S or adjacent to some vertex of O_i , $1 \leq i \leq r$ or may be adjacent to some other vertex of G , so

$$\sum_{v \in S} d(v) = k|S| \geq \sum_{i=1}^r k_i + 2|E(S)|. \quad (2.3)$$

Now, (2.2) gives $k_i = kn_i - 2m_i$, so that if k and k_i have the same parity, therefore, $k_i \geq k-1$ implies $k_i \geq k$. Summing this over r odd components, we obtain by using (2.3)

$$rk \leq \sum_{i=1}^r k_i \leq k|S| + 2|E(S)| \leq k|S|.$$

Therefore, $k_0(G - S) = r \leq |S|$ and by Tutte's theorem, G has a 1-factor. ■

Definition 2.2.15 (Anti-factors). A subset $S \subseteq V$ of a graph $G(V, E)$ such that $k_0(G - S) > |S|$ is called an antifactor set. An antifactor set S such that no proper subset of S is an antifactor set is called a minimal antifactor set.

Theorem 2.2.16. Let G be a k -regular graph of even order that is also $(k-1)$ -edge connected. If G_1 is obtained from G by deleting any $k-1$ edges, then G_1 has a 1-factor.

Proof. Let G be a k -regular, $(k-1)$ -edge-connected graph with even order. Let $K \subseteq V(G)$ and let $V(G_1) = V(G) - K$ and let S be any subset of V . Let $O_1, O_2, \dots, O_r$ be the odd components of $G - S$. Note that the vertices of O_i are either adjacent within O_i or adjacent to some vertex of S . Let $|O_i| = n_i$, m_i be the number of edges in G having both ends in O_i and let k_i be the number of edges in G having one end in O_i and the other end in S for $1 \leq i \leq r$. Then as G is $(k-1)$ -connected, we have

a_i = number of edges from K to S

b_i = number of edges within O_i

m_i = number of edges from O_i to S

$$a_i + m_i \geq k-1, \text{ then } \sum_{i=1}^r a_i + \sum_{i=1}^r m_i \geq k$$

Moreover, $\sum_{i=1}^r a_i + \frac{1}{2} \sum_{i=1}^r b_i \leq k-1$, which gives, $2 \sum_{i=1}^r a_i + \sum_{i=1}^r b_i \leq 2(k-1)$.

Also, $\sum_{i=1}^r m_i \leq k|S|$. Adding, we get,

$$2 \sum_{i=1}^r a_i + \sum_{i=1}^r b_i + \sum_{i=1}^r m_i \leq 2(k-1) + k|S| = k(|S| + 2) - 2.$$

Finally,

$$\begin{aligned}
 kr &\leq \sum_{i=1}^r a_i + \sum_{i=1}^r m_i \\
 &\leq 2 \sum_{i=1}^r a_i + \sum_{i=1}^r b_i + \sum_{i=1}^r m_i \\
 &\leq k(|S| + 2) - 2 \\
 &< k(|S| + 2).
 \end{aligned}$$

Thus, $r < |S| + 2$, a contradiction. ■

Corollary 2.2.17. *A $(k - 1)$ -regular simple graph on $2k$ vertices has a 1-factor.*

Proof. Assume G is a $(k - 1)$ -regular simple graph with $2k$ vertices having no 1-factor. Then G has an antifactor set S and by Lemma 8.4, $k_0(G - S) \geq |S| + 2$. Therefore,

$$|S| + (|S| + 2) \leq 2k \implies |S| \leq k - 1.$$

Let $|S| = k - r$. Then $r \neq 1$. For if $r = 1$, then $|S| = k - 1$ and therefore $k_0(G - S) = k + 1$. Thus each odd component of $G - S$ is a singleton and therefore each such vertex is adjacent to all the $k - 1$ vertices of S , as G is $(k - 1)$ -regular. This implies that every vertex of S is of degree at least $k + 1$, a contradiction.

Thus, $|S| = k - r$, with $2 \leq r \leq k - 1$.

If G_1 is any component of $G - S$ and $v \in V(G_1)$, then v can be adjacent to at most $|S|$ vertices of S . Since G is $(k - 1)$ -regular, v is adjacent to at least $(k - 1) - (k - r) = r - 1$ vertices of G_1 . So, $|V(G_1)| \geq r$.

Counting the vertices of S and the components, we obtain:

$$(|S| + 2)r + |S| \leq 2k$$

Substituting $|S| = k - r$:

$$(k - r + 2)r + (k - r) \leq 2k$$

Expanding and simplifying:

$$\begin{aligned}
 kr - r^2 + 2r + k - r &\leq 2k \\
 kr - r^2 + r - k &\leq 0
 \end{aligned}$$

Multiplying by -1 and factoring:

$$\begin{aligned}
 r^2 - r(k + 1) + k &\geq 0 \\
 (r - 1)(r - k) &\geq 0
 \end{aligned}$$

Since $2 \leq r \leq k - 1$, the term $(r - 1)$ is positive and $(r - k)$ is negative. Their product is negative, which violates the condition $(r - 1)(r - k) \geq 0$. ■

Definition 2.2.18. A *vertex function* for a general graph $G(V, E)$ is a function $f : V \rightarrow \mathbb{N}_0$ such that $0 \leq f(v) \leq d(v)$, for every $v \in V$.

For any function $f : V \rightarrow \mathbb{N}_0$ and for any $U \subseteq V$, we set

$$f(U) = \sum_{v \in U} f(v).$$

Definition 2.2.19. Three pair-wise disjoint (possibly empty) subsets S, T and U of the vertex set V of a general graph G , such that $V = S \cup T \cup U$, is called a *decomposition* $D = (S, T, U)$ of G . The components, say $C_1, C_2, \dots, C_k$, of the vertex-induced subgraph $\langle U \rangle$ of G are called the components of U . For each C_i , let

$$h_i = f(C_i) + q_G[T, C_i],$$

where f is a given vertex function of G , and $q_G[T, C_i]$ denotes the number of edges of G , whose one end is in T and the other end is in C_i . Then C_i is said to be an odd or even component of U , depending upon h_i being odd or even, and the number of odd components of U is denoted by $k'_0(D, f)$.

The f -deficiency of D for the given G is defined as

$$\delta(D, f) = k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T].$$

Construction 1 (Construction of Graph G' for f -factor Theorem). Let G be a general graph with the vertex function f . This defines for each vertex v_i , its degree $d_i = d(v_i|G)$, $f_i = f(v_i)$ and $k_i = d_i - f_i$. For each vertex v_i of G , take two subsets X_i and Y_i of vertices of G' , where $|X_i| = d_i$ and $|Y_i| = k_i$. Let the vertices of Y_i be labelled y_j^i , $1 \leq j \leq k_i$, arbitrarily. Now, label the vertices of X_i in the following manner.

Label the edges of G arbitrarily as $1, 2, \dots, m$. Since $|X_i| =$ number of link edges l_i incident with v_i plus twice the number of loops ℓ_i^0 incident with v_i , a vertex $x_j^i \in X_i$ is taken corresponding to each link edge j incident with v_i and two vertices $x_h^i, x_{h'}^i$ are taken corresponding to each loop h incident at v_i . As for edges, if link j joins v_i and v_k , then $x_j^i x_j^k$ is made an edge. Corresponding to loop h incident with x_h^i , the edge $x_h^i x_{h'}^i$ is taken. In addition, all the edges of the complete bipartite subgraph with vertex partition $\{X_i, Y_i\}$ are taken for each i . These constitute the graph G' . The induced subgraph $\langle X_i \cup Y_i \rangle$ of G' is called the *star graph* at i and is denoted by $St(i)$. For multi graph G , it is a complete bipartite graph, but for general graphs it will contain edges in $\langle X_i \rangle$ which we call *loop edges*. The other edges of $St(i)$ are called *star edges* and the remaining edges of G' are called as *link edges*.

Illustration: This construction is illustrated in Figure 2.16. In the general graph G , $d(v_1) = 5$, $d(v_2) = 3$ and $d(v_3) = 2$. Let $f = (3, 2, 1)$ be the vertex function, so that $f(v_1) = 3$, $f(v_2) = 2$ and $f(v_3) = 1$. Therefore, $k_1 = 2$, $k_2 = 1$ and $k_3 = 1$. Here $|X_1| = 5$, $|X_2| = 3$ and $|X_3| = 2$, and $|Y_1| = 2$, $|Y_2| = 1$ and $|Y_3| = 1$.

Label the edges of G as $1, 2, 3, 4$ and 5 as shown. Then $Y_1 = \{y_1^1, y_2^1\}$, $Y_2 = \{y_1^2\}$ and $Y_3 = \{y_1^3\}$, and we have $X_1 = \{x_1^1, x_{1'}^1, x_2^1, x_3^1, x_4^1\}$, $X_2 = \{x_2^2, x_3^2, x_5^2\}$ and $X_3 = \{x_4^3, x_5^3\}$. The vertex set of G' is $X_i \cup Y_i$, $1 \leq i \leq 3$. The edges of G' are $x_1^1 x_{1'}^1, x_2^1 x_2^2, x_3^1 x_3^2, x_4^1 x_4^3, x_5^2 x_5^3$ (shown by dotted lines) and the star edges (shown by bold lines). The dotted lines in G form an f -factor with $f = (3, 2, 1)$.

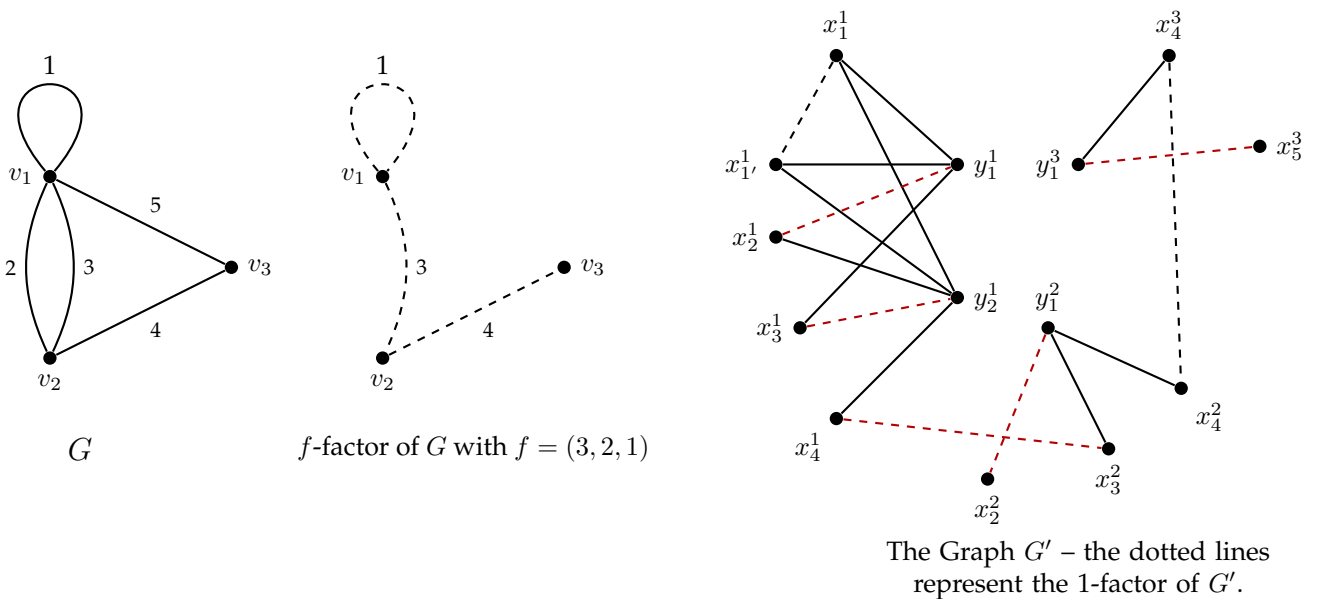


Figure 2.16: Dark dotted edges in G' represent edges induced by f -factor of G for a 1-factor of G' and red dotted edges are additional edges required to produce a 1-factor of G' .

Example 2.2.20 (Construction of G' for a Cycle C_4). Let G be a cycle C_4 with vertices v_1, v_2, v_3, v_4 . Let $f(v_i) = 1$ for all i . Thus, for every vertex v_i :

$$|X_i| = d(v_i) = 2 \quad \text{and} \quad |Y_i| = d(v_i) - f(v_i) = 1.$$

The construction is visualized below (Figure 2.17). The graph G' consists of 4 "corner" gadgets. Finding a 1-factor in G' corresponds to finding a 1-factor in G .

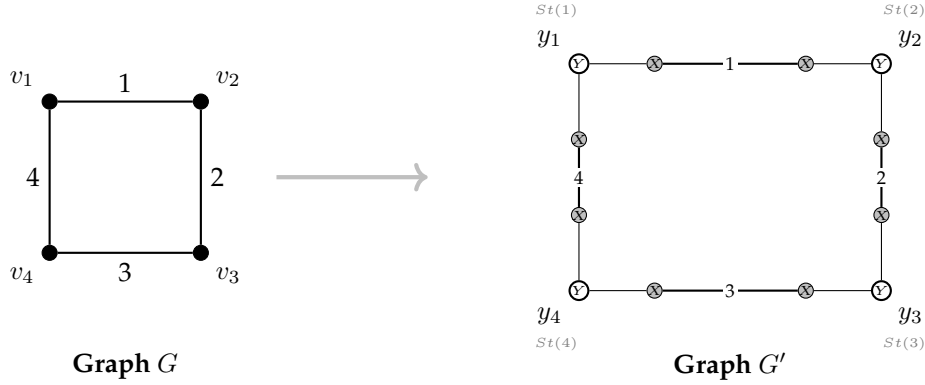


Figure 2.17: Transformation of C_4 into G' with $f(v) = 1$.

Theorem 2.2.21. A general graph G has an f -factor if and only if G' has a 1-factor

Proof. We prove the necessary and sufficient conditions separately.

($\Rightarrow$) **Necessity** Assume G has an f -factor F with link edges L and loop edges L^0 . Let g_i denote the number of link edges and f_i^0 the number of loop edges of F incident at vertex v_i . Consequently, the degree in the factor is $f_i = g_i + 2f_i^0$.

Recall that for the constructed graph G' , the size of the sets X_i and Y_i are defined as:

$$|X_i| = d_i = l_i + 2l_i^0 \quad \text{and} \quad |Y_i| = k_i = d_i - f_i,$$

where l_i is the number of link edges incident with v_i and l_i^0 is the number of loops incident with v_i in the original graph G . We can expand the expression for $|Y_i|$ as follows:

$$\begin{aligned} |Y_i| &= (l_i + 2l_i^0) - (g_i + 2f_i^0) \\ &= (l_i - g_i) + 2(l_i^0 - f_i^0). \end{aligned}$$

Let L' and $(L^0)'$ be the set of link and loop edges in G' corresponding to L and L^0 . The matching consisting of these edges keeps all vertices of Y_i unsaturated. Furthermore, it leaves unsaturated a set of vertices in X_i with cardinality:

$$(l_i - g_i) + 2(l_i^0 - f_i^0) = k_i = |Y_i|.$$

We can now match the vertices of Y_i with these remaining unsaturated vertices of X_i using the star edges $St(i)$. Let S' be this set of edges. Clearly, the union $L' \cup (L^0)' \cup S'$ constitutes a 1-factor in G' .

($\Leftarrow$) **Sufficiency** Conversely, assume G' has a 1-factor H . Since the vertices in Y_i can only be matched to vertices in X_i (by the construction of the star graphs), there are exactly

$$d_i - |Y_i| = d_i - (d_i - f_i) = f_i$$

vertices of X_i that are *not* matched by H to vertices in Y_i . Let us denote this set of vertices as X_i^* . Note that $f_i = g_i + 2f_i^0$.

Clearly, $2f_i^0$ vertices from X_i^* correspond to loop edges of the induced subgraph $\langle X_i \rangle$ and are thus matched in pairs amongst themselves. The remaining g_i vertices in X_i^* are matched by the link edges of H to vertices in X_j belonging to other star graphs $St(j)$ of G' .

If F is the set of edges in G corresponding to these loop edges and link edges selected by H , then the degree of v_i in the subgraph induced by F is:

$$d_{\langle F \rangle}(v_i) = g_i + 2f_i^0 = f_i.$$

Thus, F is an f -factor of G . ■

Theorem 2.2.22 (f -factor theorem). *A general graph G has an f -factor if and only if it has no decomposition D with $\delta(D, f) > 0$.*

OR

If f is a vertex function for a graph G , then either G has an f -factor or an f -barrier but not both.

Proof. Refer to "Introduction to Graph Theory" by S Pirzada. ■

Definition 2.2.23. Suppose $D = (S, T, U)$ is a decomposition of V and $x \in S, y \in T$. The decomposition $D_1 = (S \setminus \{x\}, T, U \cup \{x\})$ is said to be obtained by an x -**transfer from** S , and the decomposition $D_2 = (S, T \setminus \{y\}, U \cup \{y\})$ is said to be obtained by a y -**transfer from** T .

Lemma 2.2.24. *Let $D = (S, T, U)$ be a decomposition of $V(G)$. An x -transfer from S does not reduce the deficiency of D if $f(x) = d(x)$ or $d(x) - 1$, and a y -transfer from T does not reduce the deficiency of D if $f(y) = 0$ or 1 .*

Proof. Let $D' = (S - \{x\}, T, U \cup \{x\})$ be obtained from D by an x -transfer. Then

$$\delta(D', f) - \delta(D, f) \geq f(x) - d(x).$$

Thus the difference is at least 0 when $f(x) = d(x)$, and at least -1 when $f(x) = d(x) - 1$. Since the deficiencies of any two decompositions have the same parity, the latter difference cannot be -1 . Hence

$$\delta(D', f) \geq \delta(D, f).$$

Similarly, if $D'' = (S, T - \{y\}, U \cup \{y\})$ is obtained by a y -transfer, then

$$\delta(D'', f) - \delta(D, f) \geq -f(y).$$

If $f(y) = 0$, this is nonnegative, while if $f(y) = 1$, it is at least -1 . Again parity excludes the value -1 . Therefore

$$\delta(D'', f) \geq \delta(D, f).$$

Hence neither transfer reduces the deficiency under the stated hypotheses. ■

Theorem 2.2.25. *The f -factor theorem implies the 1-factor theorem of Tutte.*

Proof. Take $f(v) = 1$ for every $v \in V(G)$. Then an f -factor is exactly a 1-factor.

Assume that $k_0(G - S) \leq |S|$ for every $S \subseteq V(G)$. Suppose, to the contrary, that G has no 1-factor. By the f -factor theorem, there exists a decomposition $D = (S, T, U)$ with positive deficiency.

Since $f(y) = 1$ for every $y \in T$, Lemma 2.2.24 allows us to transfer every vertex of T into U without decreasing the deficiency. Hence we obtain the decomposition

$$(S, \emptyset, V(G) \setminus S)$$

with positive deficiency.

For this decomposition, since $f(S) = |S|$, we have

$$\delta(D, 1) = k_0(G - S) - |S|.$$

Thus $k_0(G - S) > |S|$, contrary to the assumed condition. Therefore G has a 1-factor. Hence the f -factor theorem implies Tutte's 1-factor theorem. ■

Theorem 2.2.26 (Erdős–Gallai Theorem). Let $d_1 \geq d_2 \geq \dots \geq d_n \geq 0$ be a nonincreasing sequence of integers. Then $(d_1, d_2, \dots, d_n)$ is graphical if and only if $\sum_{i=1}^n d_i$ is even and, for every $k = 1, 2, \dots, n$,

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min\{d_i, k\}.$$

Corollary 2.2.27. *The Erdős–Gallai theorem on degree sequences follows from the f -factor theorem.*

Proof. Let $d_1 \geq d_2 \geq \dots \geq d_n \geq 0$ and suppose that $\sum_{i=1}^n d_i \equiv 0 \pmod{2}$. To obtain this condition, we observe that d is graphic if and only if the complete graph K_n has a d -factor. Let $V(K_n) = \{v_1, \dots, v_n\}$ and define $f(v_i) = d_i$ for $1 \leq i \leq n$. Then $(d_1, \dots, d_n)$ is graphic if and only if K_n has an f -factor.

By the f -factor theorem, K_n has an f -factor if and only if $\delta(D, f) \leq 0$ for every decomposition $D = (S, T, U)$ of $V(K_n)$, where

$$\delta(D, f) = k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T].$$

Let $r = |T|$. Since every vertex of K_n has degree $n - 1$, we have $d(T) = r(n - 1)$. Also, every vertex of S is adjacent to every vertex of T , so $q[S, T] = |T||S| = r|S|$. Hence

$$\begin{aligned} \delta(D, f) &= k'_0(D, f) - f(S) + f(T) - d(T) + q[S, T] \\ &= k'_0(D, f) - f(S) + f(T) - r(n - 1) + r|T| \\ &= k'_0(D, f) + f(T) - f(S) - r(n - 1 - |S|) \end{aligned}$$

Now $K_n[U]$ is complete. Thus, if $U \neq \emptyset$, it has only one component, and consequently $k'_0(D, f) \in \{0, 1\}$. Moreover, $f(V(K_n)) = \sum_{i=1}^n d_i$ is even, and therefore $\delta(D, f)$ is even because $\delta(D, f)$ and $\sum f(V(G))$ always have the same parity. Hence the condition $\delta(D, f) \leq 0$ is equivalent to

$$f(T) \leq f(S) + r(n - 1 - |S|).$$

Since $U = V(K_n) \setminus (S \cup T)$, we have $n - 1 - |S| = (r - 1) + |U|$. Therefore,

$$f(T) \leq r(r - 1) + f(S) + r|U|.$$

For fixed T , the sets S and U partition $V(K_n) \setminus T$. For each $v \notin T$, placing v in S contributes $f(v)$, whereas placing v in U contributes r . Hence the minimum possible value of $f(S) + r|U|$ is $\sum_{v \notin T} \min\{f(v), r\}$. Thus the f -factor condition is equivalent to

$$f(T) \leq r(r - 1) + \sum_{v \notin T} \min\{f(v), r\}$$

for every $T \subseteq V(K_n)$ with $|T| = r$.

Taking $T = \{v_1, \dots, v_r\}$ gives $f(T) = \sum_{i=1}^r d_i$ and $\sum_{v \notin T} \min\{f(v), r\} = \sum_{i=r+1}^n \min\{d_i, r\}$. Hence

$$\sum_{i=1}^r d_i \leq r(r-1) + \sum_{i=r+1}^n \min\{d_i, r\},$$

which is the Erdős–Gallai inequality. ■

Theorem 2.2.28 (*k*-factor Theorem). *Let $d = [d_i]_1^n$ be a graphical sequence. If the sequence $d' = [d_i - k]_1^n$ is also graphical, then there exists a realization of d that contains a k -factor.*

Proof. Let G_1 be a realization of $d = [d_i]_1^n$ and let G_2 be a realization of $d' = [d_i - k]_1^n$, both on the same vertex set $V = \{v_1, \dots, v_n\}$. Thus $d_{G_1}(v_i) = d_i$ and $d_{G_2}(v_i) = d_i - k$ for every i . Choose G_1 and G_2 so that $|E(G_1) \cap E(G_2)|$ is as large as possible. Put $E_1 = E(G_1)$ and $E_2 = E(G_2)$. We claim that $E_2 \subseteq E_1$. Once this is proved, the graph with edge set $E_1 \setminus E_2$ has degree $d_i - (d_i - k) = k$ at every vertex, and hence is a k -factor of G_1 .

We first record a switching observation. Suppose that $v_i v_j \in E_2 \setminus E_1$ and $v_i v_r, v_j v_s \in E_1 \setminus E_2$. Then $v_r v_s \in E_1 \setminus E_2$. Indeed, if $v_r v_s \in E_2$, then in G_2 we may replace the edges $v_i v_j$ and $v_r v_s$ by $v_i v_r$ and $v_j v_s$. This preserves every vertex degree and increases $|E_1 \cap E_2|$, a contradiction. If $v_r v_s \notin E_1$, then in G_1 we may replace $v_i v_r$ and $v_j v_s$ by $v_i v_j$ and $v_r v_s$. Again all degrees are preserved and the number of common edges increases. Hence $v_r v_s \in E_1 \setminus E_2$.

Now suppose, to the contrary, that $E_2 \setminus E_1 \neq \emptyset$. For each vertex u , let $t(u) = d_{E_2 \setminus E_1}(u)$. Since $d_{G_1}(u) - d_{G_2}(u) = k$, we have $d_{E_1 \setminus E_2}(u) = t(u) + k$. Choose a vertex v_1 for which $t(v_1)$ is maximum, and write $t = t(v_1) > 0$. Choose v_j such that $v_1 v_j \in E_2 \setminus E_1$. Since $t(v_j) \geq 1$, we have $d_{E_1 \setminus E_2}(v_j) = t(v_j) + k \geq k + 1$. Hence there exists a vertex w such that $v_j w \in E_1 \setminus E_2$.

Let $A = N_{E_1 \setminus E_2}(v_1)$. Since $d_{E_1 \setminus E_2}(v_1) = t + k$, we have $|A| = t + k$. First suppose that $w \notin A$. For every $a \in A$, the switching observation applied to $v_1 v_j \in E_2 \setminus E_1$ and $v_1 a, v_j w \in E_1 \setminus E_2$ gives $aw \in E_1 \setminus E_2$. Thus w is incident with the $t + k$ edges joining it to the vertices of A , as well as the edge $v_j w$. Therefore $d_{E_1 \setminus E_2}(w) \geq t + k + 1$, and hence $t(w) \geq t + 1$, contradicting the maximality of t .

Now suppose that $w \in A$. For every $a \in A \setminus \{w\}$, the same switching observation gives $aw \in E_1 \setminus E_2$. In addition, both $v_1 w$ and $v_j w$ belong to $E_1 \setminus E_2$. Hence $d_{E_1 \setminus E_2}(w) \geq (t + k - 1) + 2 = t + k + 1$, so again $t(w) \geq t + 1$, contradicting the maximality of t .

Therefore $E_2 \setminus E_1 = \emptyset$, and hence $E_2 \subseteq E_1$. Consequently, the graph $H = (V, E_1 \setminus E_2)$ satisfies $d_H(v_i) = d_i - (d_i - k) = k$ for every i . Hence H is a k -factor of G_1 . ■

2.3 Factorization of Complete Graphs

Definition 2.3.1. A **factorization** of a graph G is a decomposition of the edges of G into edge-disjoint spanning subgraphs, known as factors. If the graph is decomposed into k -factors (where every vertex has degree k), it is called a **k -factorization**.

For the complete graph K_n , the type of factorization possible depends on the parity of n .

Case 1: n is Even (1-Factorization)

Consider a complete graph K_{2n} . It has $2n$ vertices and is $(2n - 1)$ -regular.

- A 1-factor is a perfect matching (regular of degree 1).
- Since the total degree of each vertex is $2n - 1$, K_{2n} can be decomposed into exactly $2n - 1$ perfect matchings.

Case 2: n is Odd (2-Factorization)

Consider a complete graph K_{2n+1} . It has $2n + 1$ vertices and is $2n$ -regular.

- A 2-factor is a spanning subgraph where every vertex has degree 2 (a disjoint union of cycles).
- In the context of K_n , we are often interested in decomposing it into Hamiltonian cycles.
- Since the total degree is $2n$, K_{2n+1} can be decomposed into n Hamiltonian cycles.

Theorem 2.3.2. *For every positive integer n , the complete graph K_{2n} is 1-factorable.*

Proof. Let the vertex set be $V = \{0, 1, \dots, 2n-2, 2n-1\}$. We designate the vertex $2n-1$ as the fixed center and treat the remaining vertices $\{0, 1, \dots, 2n-2\}$ as elements of $\mathbb{Z}_{2n-1}$ (the integers modulo $2n-1$).

We construct $2n-1$ factors, denoted $F_0, F_1, \dots, F_{2n-2}$. For each $i \in \{0, \dots, 2n-2\}$, define the set of edges F_i as:

$$F_i = \{\{2n-1, i\}\} \cup \{\{i-k, i+k\} \pmod{2n-1} \mid k = 1, \dots, n-1\}.$$

1. F_i is a 1-factor: The fixed vertex $2n-1$ is matched to vertex i . The remaining $2n-2$ vertices form the polygon and are paired distinctly. The relation $i-k \equiv i+k \pmod{2n-1}$ implies $2k \equiv 0$, which is impossible for $1 \leq k \leq n-1$ since $2n-1$ is odd. Thus, every vertex has degree exactly 1 in F_i .

2. Decomposition:

- Every edge incident to the center $2n-1$ is of the form $\{2n-1, x\}$ and appears uniquely in factor F_x .
- For any edge $\{u, v\}$ not incident to $2n-1$, the sum of the indices modulo $2n-1$ is invariant for a fixed factor. Specifically, $u + v \equiv (i-k) + (i+k) \equiv 2i \pmod{2n-1}$. Since $2n-1$ is odd, $\gcd(2, 2n-1) = 1$, meaning 2 is invertible. Thus, $i \equiv (u+v) \cdot 2^{-1} \pmod{2n-1}$ is uniquely determined.

Therefore, every edge of K_{2n} belongs to exactly one factor, constituting a 1-factorization. ■

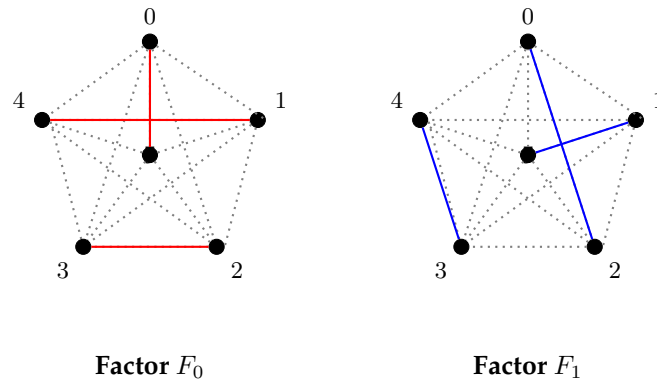


Figure 2.18: 1-Factorization of K_6 ($n = 3$). The solid colored lines represent the specific 1-factor, while the dotted lines represent the remaining edges of the complete graph.

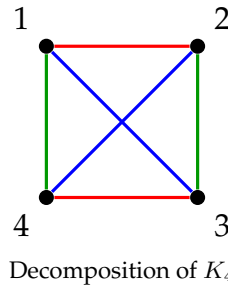
Theorem 2.3.3. *For every positive integer n , the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.*

Example 2.3.4 (1-Factorization of K_4). Let $n = 4$. The graph K_4 has 4 vertices and degree 3. It can be decomposed into $4 - 1 = 3$ perfect matchings (1-factors).

The Factors:

- F_1 (Red): Matches $\{(1, 2), (3, 4)\}$
- F_2 (Blue): Matches $\{(1, 3), (2, 4)\}$

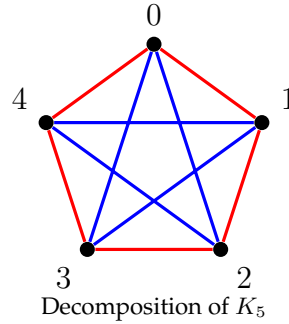
- F_3 (Green): Matches $\{(1, 4), (2, 3)\}$



Example 2.3.5 (2-Factorization of K_5). Let $n = 5$. The graph K_5 has 5 vertices and degree 4. It can be decomposed into $(5 - 1)/2 = 2$ Hamiltonian cycles.

The Factors:

- H_1 (Red): The outer pentagon cycle $\{(0, 1), (1, 2), (2, 3), (3, 4), (4, 0)\}$.
- H_2 (Blue): The inner star cycle $\{(0, 2), (2, 4), (4, 1), (1, 3), (3, 0)\}$.



Theorem 2.3.6. For every positive integer n , the complete graph K_{2n+1} is 2-factorable into Hamiltonian cycles.

Proof. Let the vertex set be $V = \{0, 1, \dots, 2n\}$. We designate the vertex $2n$ as the fixed center and treat the remaining set $\{0, \dots, 2n - 1\}$ as elements of $\mathbb{Z}_{2n}$.

We construct n Hamiltonian cycles $C_0, C_1, \dots, C_{n-1}$ as follows:

1. **The Base Cycle C_0 :** Define the cycle C_0 by the sequence of vertices:

$$C_0 = (2n, 0, 1, 2n - 1, 2, 2n - 2, \dots, n, 2n).$$

The path within $\mathbb{Z}_{2n}$ (excluding the fixed vertex $2n$) zigzags through the vertices. The lengths of the edges in this path correspond to the distances $1, 2, \dots, n$ in the modular arithmetic of $\mathbb{Z}_{2n}$.

2. **The General Cycles:** For each $i \in \{0, \dots, n - 1\}$, let C_i be the cycle obtained by adding $i \pmod{2n}$ to every vertex in $\{0, \dots, 2n - 1\}$, while keeping the vertex $2n$ fixed.

3. **Verification:**

- **Edges incident to $2n$:** In cycle C_i , the vertex $2n$ connects to i and $n + i$. As i ranges from 0 to $n - 1$, these pairs $\{i, n + i\}$ are disjoint and cover all vertices in $\mathbb{Z}_{2n}$. Thus, every edge incident to $2n$ appears exactly once.
- **Edges within $\mathbb{Z}_{2n}$:** The base path in C_0 contains exactly one edge of each "length" class $d \in \{1, \dots, n\}$. By the properties of cyclic groups, rotating a set of edges that contains one representative of each length generates all possible edges within the complete graph induced by $\mathbb{Z}_{2n}$.

Thus, the collection $\{C_0, \dots, C_{n-1}\}$ forms a 2-factorization of K_{2n+1} . ■

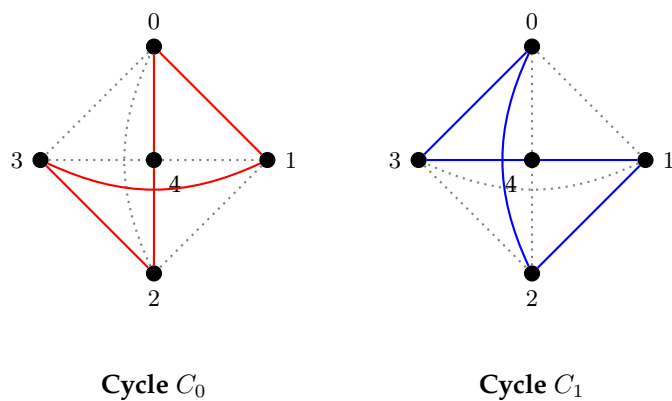


Figure 2.19: Decomposition of K_5 . We curve the diagonal edges (0-2 and 1-3) so they do not overlap with the central vertex 4. All 10 edges of K_5 are present in both figures (either solid or dotted).

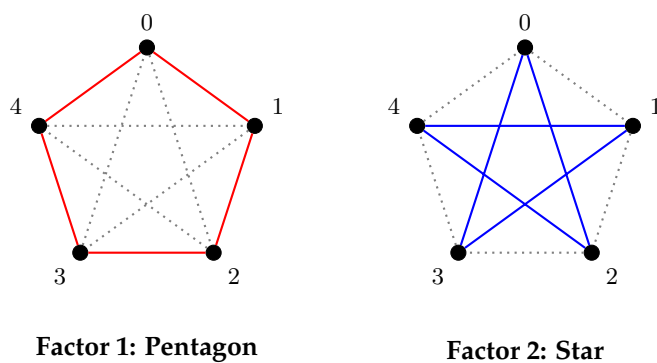


Figure 2.20: Decomposition of K_5 into 2 Hamiltonian Cycles. The dotted lines in each figure represent the edges belonging to the other factor.

Chapter 3

Edge Graphs

Please refer to Chapter 9 of the Book "An Introduction to Graph Theory" by S Pirzada.

Chapter 4

Spectral Theory of Graphs

INTRODUCTION

Spectral graph theory studies a graph through the eigenvalues of matrices associated with it. For a simple graph, the adjacency matrix records which vertices are adjacent, while the Laplacian matrix combines adjacency information with vertex degrees. Their spectra provide algebraic invariants that reveal important structural information about the graph.

The subject has strong connections with linear algebra, combinatorics, and mathematical chemistry. Early spectral ideas in chemistry arose from Hückel's work on molecular orbitals, while the mathematical development of spectral graph theory was advanced by Collatz and Sinogowitz in the 1950s. Graph energy, introduced by Ivan Gutman in 1978, gave another important spectral invariant motivated by chemical applications.

In this chapter we develop the adjacency spectrum and Laplacian spectrum of a graph, compute the spectra of several standard graph families, examine cospectral graphs, regular graphs, complements, and line graphs, and conclude with graph energy and Laplacian energy. The emphasis is on deriving the results directly from the underlying matrices so that the connection between graph structure and spectral information remains clear.

Video Resources

A detailed lecture series for this chapter is available at <https://www.youtube.com/@Prof.DrAijaz/playlists>. Additional study material is available at <https://aijaz.vercel.app/>.

4.1 Spectrum of Graphs

Let $G = (V, E)$ be a simple undirected graph with vertex set $V = \{v_1, \dots, v_n\}$ and m edges. Spectral graph theory studies the relationship between the structure of G and the eigenvalues of matrices associated with it, especially the adjacency matrix and the Laplacian matrix.

Definition 4.1.1 (Adjacency Matrix). The **adjacency matrix** of G is the $n \times n$ matrix $A(G) = [a_{ij}]$ defined by

$$a_{ij} = \begin{cases} 1, & \text{if } v_i \text{ is adjacent to } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

Since G is simple, $a_{ii} = 0$ for every i . Since G is undirected, $a_{ij} = a_{ji}$, and therefore $A(G)$ is symmetric.

Definition 4.1.2 (Spectrum). The **adjacency spectrum** of G , denoted by $\text{Spec}(G)$, is the multiset of eigenvalues of $A(G)$, each repeated according to its algebraic multiplicity. Equivalently, the eigenvalues are the roots of the characteristic polynomial $\phi(G, \lambda) = \det(\lambda I - A(G))$. Because $A(G)$ is real and symmetric, its eigenvalues are real and may be ordered as $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.

Theorem 4.1.3 (Real Eigenvalues). *Every eigenvalue of the adjacency matrix $A(G)$ is real.*

Proof. Since G is undirected, $a_{ij} = a_{ji}$ for all i, j . Hence $A(G)$ is a real symmetric matrix. By the spectral theorem, every eigenvalue of a real symmetric matrix is real. Moreover, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of $A(G)$. ■

Theorem 4.1.4 (Sum of Eigenvalues). *If $\lambda_1, \dots, \lambda_n$ are the adjacency eigenvalues of a simple graph G , then $\sum_{i=1}^n \lambda_i = 0$.*

Proof. For any square matrix, the sum of the eigenvalues, counted with algebraic multiplicity, equals the trace. Therefore $\sum_{i=1}^n \lambda_i = \text{tr}(A) = \sum_{i=1}^n a_{ii}$. Since G is simple, $a_{ii} = 0$ for every i . Hence $\text{tr}(A) = 0$, and the required sum is zero. ■

Lemma 4.1.5. *Let $A = [a_{ij}]$ be the adjacency matrix of a simple graph G . Then:*

- (A) *the sum of all 2×2 principal minors of A is $-|E(G)|$;*
- (B) *the sum of all 3×3 principal minors of A is twice the number of triangles in G .*

Proof. For $i < j$, the principal minor determined by rows and columns i, j is

$$\det \begin{pmatrix} 0 & a_{ij} \\ a_{ji} & 0 \end{pmatrix} = -a_{ij}^2 = -a_{ij},$$

because $a_{ij} \in \{0, 1\}$ and $a_{ij} = a_{ji}$. Thus this minor equals -1 exactly when $v_i v_j \in E(G)$ and equals 0 otherwise. Summing over all unordered pairs $\{i, j\}$ therefore gives $-|E(G)|$.

Now let $i < j < k$. The corresponding 3×3 principal minor is

$$\det \begin{pmatrix} 0 & a_{ij} & a_{ik} \\ a_{ji} & 0 & a_{jk} \\ a_{ki} & a_{kj} & 0 \end{pmatrix} = a_{ij}a_{jk}a_{ki} + a_{ik}a_{kj}a_{ji} = 2a_{ij}a_{jk}a_{ki}.$$

This value is 2 precisely when all three edges $v_i v_j$, $v_j v_k$, and $v_k v_i$ are present, that is, when v_i, v_j, v_k form a triangle. Otherwise the value is 0. Hence the sum of all 3×3 principal minors is twice the number of triangles in G . ■

Theorem 4.1.6 (Closed Walks and Spectral Moments). *Let N_k be the total number of closed walks of length k in G . Then $N_k = \sum_{i=1}^n \lambda_i^k$. In particular, $\sum_{i=1}^n \lambda_i^2 = 2m$.*

Proof. A standard property of the adjacency matrix states that the (i, j) -entry of A^k equals the number of walks of length k from v_i to v_j . In particular, $(A^k)_{ii}$ counts the closed walks of length k that begin and end at v_i . Therefore $N_k = \sum_{i=1}^n (A^k)_{ii} = \text{tr}(A^k)$.

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A , then $\lambda_1^k, \dots, \lambda_n^k$ are the eigenvalues of A^k . Since the trace is the sum of the eigenvalues, $N_k = \text{tr}(A^k) = \sum_{i=1}^n \lambda_i^k$.

For $k = 2$, every closed walk has the form u, v, u , where $uv \in E(G)$. Each edge contributes exactly two such walks, one beginning at each endpoint. Hence $N_2 = 2m$, and therefore $\sum_{i=1}^n \lambda_i^2 = 2m$. ■

Theorem 4.1.7 (Spectrum of Bipartite Graphs). *A graph G is bipartite if and only if its adjacency spectrum is symmetric about 0. Equivalently, λ is an eigenvalue of multiplicity r if and only if $-\lambda$ is an eigenvalue of multiplicity r .*

Proof. Suppose first that G is bipartite with bipartition $V(G) = V_1 \cup V_2$. Order the vertices so that the vertices of V_1 come first. Then

$$A = \begin{pmatrix} 0 & B \\ B^T & 0 \end{pmatrix}.$$

Suppose

$$A \begin{pmatrix} u \\ v \end{pmatrix} = \lambda \begin{pmatrix} u \\ v \end{pmatrix}.$$

This is equivalent to $Bv = \lambda u$, $B^T u = \lambda v$. Now replace v by $-v$. Then

$$A \begin{pmatrix} u \\ -v \end{pmatrix} = \begin{pmatrix} -Bv \\ B^T u \end{pmatrix} = -\lambda \begin{pmatrix} u \\ -v \end{pmatrix}.$$

Thus, whenever λ is an eigenvalue, so is $-\lambda$. The transformation $(u, v)^T \mapsto (u, -v)^T$ is invertible, so it gives a bijection between the eigenspaces for λ and $-\lambda$. Since A is symmetric and therefore diagonalizable, the two eigenvalues have the same algebraic multiplicity.

Conversely, suppose that the spectrum of G is symmetric about 0. Then for every nonnegative integer r , $\text{tr}(A^{2r+1}) = \sum_{i=1}^n \lambda_i^{2r+1} = 0$, because the terms corresponding to λ and $-\lambda$ cancel.

If G contained an odd cycle of length $2r+1$, then going once around that cycle, in either direction and starting at any of its vertices, would produce closed walks of length $2r+1$. Hence $\text{tr}(A^{2r+1}) > 0$, contradicting the preceding equality. Therefore G contains no odd cycle. A graph is bipartite if and only if it contains no odd cycle, so G is bipartite. ■

Theorem 4.1.8 (Spectrum of K_n). For $n \geq 2$, $\text{Spec}(K_n) = \{(n-1)^{(1)}, (-1)^{(n-1)}\}$.

Proof. The adjacency matrix of K_n is $A = J - I$, where J is the all-ones matrix. The vector $\mathbf{1} = (1, \dots, 1)^T$ satisfies $J\mathbf{1} = n\mathbf{1}$, so n is an eigenvalue of J . If $x \perp \mathbf{1}$, then the sum of the entries of x is zero, and hence $Jx = 0$. Thus $\text{Spec}(J) = \{n^{(1)}, 0^{(n-1)}\}$. Since $A = J - I$, subtracting I decreases each eigenvalue by 1. Therefore A has eigenvalue $n-1$ once and eigenvalue -1 with multiplicity $n-1$. ■

Theorem 4.1.9 (Spectrum of $K_{r,s}$). For $r, s \geq 1$, $\text{Spec}(K_{r,s}) = \{(\sqrt{rs})^{(1)}, (-\sqrt{rs})^{(1)}, 0^{(r+s-2)}\}$.

Proof. Order the vertices so that the r vertices in one partite set come first and the s vertices in the other partite set come last. Then

$$A(K_{r,s}) = \begin{pmatrix} 0 & J_{r \times s} \\ J_{s \times r} & 0 \end{pmatrix}.$$

The row space is generated by one row of each block type, so $\text{rank}(A) = 2$. Hence 0 is an eigenvalue with multiplicity $r+s-2$.

Let the two nonzero eigenvalues be α and β . Since $\text{tr}(A) = 0$, we have $\alpha + \beta = 0$, so $\beta = -\alpha$. Also, $K_{r,s}$ has rs edges, and the second spectral moment gives $\alpha^2 + \beta^2 = 2rs$. Substituting $\beta = -\alpha$ gives $2\alpha^2 = 2rs$, so $\alpha^2 = rs$. Thus the two nonzero eigenvalues are $\sqrt{rs}$ and $-\sqrt{rs}$. ■

Definition 4.1.10 (Cyclic Permutation Matrix). For $n \geq 2$, let Q_n be the $n \times n$ cyclic permutation matrix defined by $Q_n e_j = e_{j+1}$ ($1 \leq j < n$), $Q_n e_n = e_1$, where $e_1, \dots, e_n$ are the standard basis vectors of $\mathbb{R}^n$.

Lemma 4.1.11. For $n \geq 2$, the eigenvalues of Q_n are $1, \omega, \omega^2, \dots, \omega^{n-1}$, $\omega = e^{2\pi i/n}$.

Proof. Since Q_n cyclically permutes the standard basis vectors, applying it n times returns every basis vector to its original position. Thus $Q_n^n = I$. If λ is an eigenvalue of Q_n , then $\lambda^n = 1$, so every eigenvalue is an n th root of unity.

Conversely, the characteristic polynomial of Q_n is $\det(\lambda I - Q_n) = \lambda^n - 1$. Its roots are exactly $1, \omega, \omega^2, \dots, \omega^{n-1}$. Hence these are precisely the eigenvalues of Q_n . ■

Theorem 4.1.12 (Spectrum of a Cycle). For $n \geq 3$, the eigenvalues of C_n are $\lambda_k = 2 \cos\left(\frac{2\pi k}{n}\right)$, $k = 0, 1, \dots, n-1$.

Proof. Order the vertices cyclically. With this ordering, $A(C_n) = Q_n + Q_n^T = Q_n + Q_n^{n-1}$. Let x be an eigenvector of Q_n corresponding to the eigenvalue ω^k . Since $Q_n^T = Q_n^{-1}$, we also have $Q_n^T x = \omega^{-k}x$. Therefore $A(C_n)x = (\omega^k + \omega^{-k})x$. Using Euler's identity,

$$\omega^k + \omega^{-k} = e^{2\pi i k/n} + e^{-2\pi i k/n} = 2 \cos\left(\frac{2\pi k}{n}\right).$$

As $k = 0, 1, \dots, n-1$, these give all n eigenvalues of C_n . ■

Alternative proof. Label the vertices of C_n by $0, 1, \dots, n-1$ in cyclic order and put $\omega = e^{2\pi i/n}$. For $k = 0, 1, \dots, n-1$, define $x_k = (1, \omega^k, \omega^{2k}, \dots, \omega^{(n-1)k})^T$. The two neighbors of vertex r are $r-1$ and $r+1$, with indices taken modulo n . Hence $(Ax_k)_r = \omega^{k(r-1)} + \omega^{k(r+1)} = \omega^{kr}(\omega^{-k} + \omega^k)$. Since $(x_k)_r = \omega^{kr}$, $(Ax_k)_r = 2 \cos\left(\frac{2\pi k}{n}\right)(x_k)_r$. Thus x_k is an eigenvector of $A(C_n)$ with eigenvalue $2 \cos(2\pi k/n)$. The vectors $x_0, \dots, x_{n-1}$ are linearly independent, so these account for all eigenvalues. ■

Lemma 4.1.13. Let $X = (x_1, \dots, x_n)^T$ be an eigenvector of $A(P_n)$ corresponding to the eigenvalue λ . Then λ is also an eigenvalue of C_{2n+2} . Two linearly independent eigenvectors of $A(C_{2n+2})$ corresponding to λ are $Y = (x_1, \dots, x_n, 0, -x_n, \dots, -x_1, 0)^T$ and $Z = (0, x_1, \dots, x_n, 0, -x_n, \dots, -x_1)^T$.

Proof. Write the path as $P_n = v_1 v_2 \cdots v_n$. The equation $A(P_n)X = \lambda X$ gives

$$x_2 = \lambda x_1, \quad x_{i-1} + x_{i+1} = \lambda x_i \quad (2 \leq i \leq n-1), \quad x_{n-1} = \lambda x_n.$$

These are exactly the coordinate equations that must hold for an eigenvector on the path.

Now place the entries of $Y = (x_1, \dots, x_n, 0, -x_n, \dots, -x_1, 0)^T$ cyclically around C_{2n+2} . At the first n positions, the required equations are precisely the path equations above. At the next position the entry is 0, and its two neighboring entries are x_n and $-x_n$, whose sum is 0. On the following n positions, the same path equations appear with all signs reversed. At the final zero, the neighboring entries are $-x_1$ and x_1 , whose sum is again 0. Therefore $A(C_{2n+2})Y = \lambda Y$.

The vector Z is obtained from Y by a cyclic shift. Since the adjacency matrix of a cycle is unchanged by such a shift, Z also satisfies $A(C_{2n+2})Z = \lambda Z$.

It remains to verify that Y and Z are linearly independent. We must have $x_1 \neq 0$, because $x_1 = 0$ would imply $x_2 = 0$ from $x_2 = \lambda x_1$, and then the recurrence would force every x_i to be zero, contradicting $X \neq 0$. The first coordinate of Y is therefore nonzero, whereas the first coordinate of Z is zero. Hence Y and Z are linearly independent. ■

Theorem 4.1.14 (Spectrum of a Path). The eigenvalues of P_n are $\lambda_k = 2 \cos\left(\frac{k\pi}{n+1}\right)$, $k = 1, 2, \dots, n$.

Proof. By the preceding lemma, every eigenvalue λ of P_n is an eigenvalue of C_{2n+2} with geometric multiplicity at least 2. Since the adjacency matrix of a cycle is real symmetric, geometric and algebraic multiplicities are equal.

The eigenvalues of C_{2n+2} are $2 \cos\left(\frac{k\pi}{n+1}\right)$, $k = 0, 1, \dots, 2n+1$. For $k = 0$ and $k = n+1$, the corresponding eigenvalues are 2 and -2 , and each occurs only once. Every remaining eigenvalue occurs twice, because $\cos\left(\frac{k\pi}{n+1}\right) = \cos\left(\frac{(2n+2-k)\pi}{n+1}\right)$. Therefore every eigenvalue of P_n must belong to the list $2 \cos\left(\frac{k\pi}{n+1}\right)$, $k = 1, \dots, n$. These n numbers are distinct because $k\pi/(n+1)$ lies strictly between 0 and π , where the cosine function is strictly decreasing. Since $A(P_n)$ is an $n \times n$ real symmetric matrix, it has exactly n real eigenvalues counted with multiplicity. Hence the displayed numbers are precisely the eigenvalues of P_n . ■

Example 4.1.15 (Spectrum of P_4). For P_4 , the path-spectrum formula gives

$$\text{Spec}(P_4) = \left\{ 2 \cos \frac{\pi}{5}, 2 \cos \frac{2\pi}{5}, 2 \cos \frac{3\pi}{5}, 2 \cos \frac{4\pi}{5} \right\}.$$

Using the exact cosine values,

$$\text{Spec}(P_4) = \left\{ \frac{1 + \sqrt{5}}{2}, \frac{\sqrt{5} - 1}{2}, -\frac{\sqrt{5} - 1}{2}, -\frac{1 + \sqrt{5}}{2} \right\}.$$

The eigenvalues occur in opposite pairs, which agrees with the fact that every path is bipartite.

Definition 4.1.16 (Determinant). For an $n \times n$ matrix $A = [a_{ij}]$, the determinant of A is defined by $\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}$, where S_n is the symmetric group on $\{1, \dots, n\}$ and

$$\text{sgn}(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is even,} \\ -1, & \text{if } \sigma \text{ is odd.} \end{cases}$$

Note. If a permutation $\sigma \in S_n$ decomposes into d disjoint cycles, including fixed points as 1-cycles, then $\text{sgn}(\sigma) = (-1)^{n-d}$. This observation is particularly useful for adjacency matrices. Since the diagonal entries of the adjacency matrix of a simple graph are zero, any permutation having a fixed point contributes zero to the determinant expansion. Hence only fixed-point-free permutations can contribute nonzero terms.

Example 4.1.17. Evaluate

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

directly from the permutation definition of the determinant.

Solution. The group S_2 contains exactly two permutations: $S_2 = \{\text{id}, (12)\}$. The identity permutation is even, whereas the transposition (12) is odd. Therefore

$$\begin{aligned} \det(A) &= \text{sgn}(\text{id}) a_{11}a_{22} + \text{sgn}((12)) a_{12}a_{21} \\ &= a_{11}a_{22} - a_{12}a_{21}. \end{aligned}$$

■

Definition 4.1.18 (Elementary Spanning Subgraph). Let G be a graph. A spanning subgraph $H \subseteq G$ is called *elementary* if every connected component of H is either

1. a single edge K_2 , or
2. a cycle C_k with $k \geq 3$.

The importance of elementary spanning subgraphs comes from the permutation expansion of the determinant. A transposition in a permutation corresponds to an edge K_2 , while a permutation cycle of length at least 3 corresponds to a cycle in the graph. This leads to the following determinant formula.

Theorem 4.1.19 (Determinant Expansion). *Let A be the adjacency matrix of a simple graph G of order n . Then $\det(A) = \sum_H (-1)^{n-c_1(H)-c(H)} 2^{c(H)}$, where the sum is taken over all elementary spanning subgraphs H of G , $c_1(H)$ denotes the number of K_2 components of H , and $c(H)$ denotes the number of cycle components of length at least 3.*

Proof. By the permutation expansion of the determinant, $\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}$. We determine which permutations contribute nonzero terms.

A term $\prod_{i=1}^n a_{i,\sigma(i)}$ is nonzero only when $a_{i,\sigma(i)} = 1$ for every i . Since $a_{ii} = 0$, no fixed point of σ is allowed. Thus every cycle in the disjoint-cycle decomposition of σ has length at least 2.

A cycle of length 2, say (ij) , contributes $a_{ij}a_{ji}$, which is nonzero precisely when $v_i v_j$ is an edge of G . Thus a transposition corresponds to a K_2 component.

A cycle $(i_1 i_2 \cdots i_r)$, $r \geq 3$, contributes $a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_r i_1}$. This product is nonzero precisely when $v_{i_1} v_{i_2} \cdots v_{i_r} v_{i_1}$ is a cycle in G . Hence every nonzero determinant term corresponds to an elementary spanning subgraph H of G .

Suppose H has $c_1(H)$ components isomorphic to K_2 and $c(H)$ cycle components of length at least 3. Then the corresponding permutation contains $c_1(H) + c(H)$ disjoint cycles. Therefore its sign is $(-1)^{n-c_1(H)-c(H)}$.

It remains to count how many permutation terms correspond to the same elementary spanning subgraph. A K_2 component determines a unique transposition. On the other hand, every cycle component of length at least 3 may be traversed in either direction, giving two permutations. Hence an elementary spanning subgraph with $c(H)$ such cycle components gives $2^{c(H)}$ nonzero permutation terms, all having the same sign.

Summing the contributions of all elementary spanning subgraphs yields $\det(A) = \sum_H (-1)^{n-c_1(H)-c(H)} 2^{c(H)}$ ■

Example 4.1.20. For the graph shown in Figure 4.1, determine $\det(A)$ by listing all elementary spanning subgraphs.

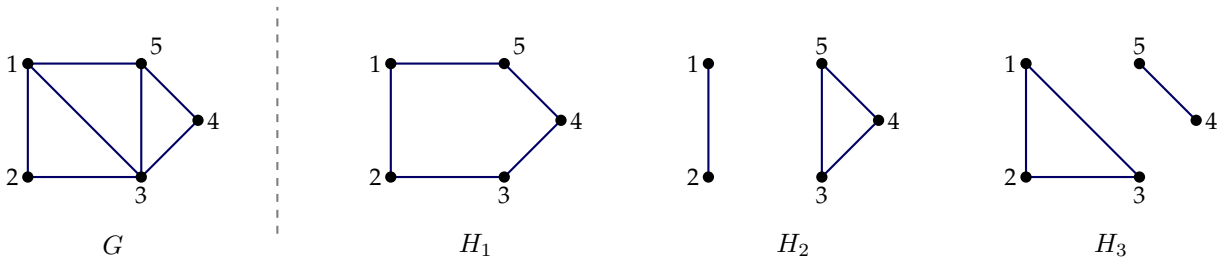


Figure 4.1: The elementary spanning subgraphs of G .

Solution. The graph has exactly three elementary spanning subgraphs, namely H_1 , H_2 , and H_3 shown in Figure 4.1.

For H_1 , there is one 5-cycle and no K_2 component. Hence $c_1(H_1) = 0$, $c(H_1) = 1$, so its contribution is $(-1)^{5-0-1} 2^1 = 2$.

For each of H_2 and H_3 , there is one K_2 component and one 3-cycle. Thus $c_1(H_i) = 1$, $c(H_i) = 1$ for $i = 2, 3$. Each therefore contributes $(-1)^{5-1-1} 2^1 = -2$.

Consequently, $\det(A) = 2 - 2 - 2 = -2$. Hence

$$\boxed{\det(A) = -2}.$$

The preceding determinant formula also gives a combinatorial interpretation of every coefficient of the characteristic polynomial.

Theorem 4.1.21 (Sachs' Theorem). Let $\phi(G, \lambda) = \det(\lambda I - A) = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_n$ be the characteristic polynomial of a simple graph G . For $1 \leq i \leq n$, $a_i = \sum_{H \in \mathcal{H}_i} (-1)^{p(H)} 2^{c(H)}$, where $\mathcal{H}_i$ is the set of elementary subgraphs of G having exactly i vertices, $p(H)$ is the number of connected components of H , and $c(H)$ is the number of cycle components of H .

Proof. A standard result from matrix theory states that the coefficient a_i of $\det(\lambda I - A)$ is $(-1)^i$ times the sum of all $i \times i$ principal minors of A .

Choose a set $S \subseteq V(G)$ with $|S| = i$. The corresponding principal submatrix of A is precisely the adjacency matrix of the induced subgraph $G[S]$. Therefore the corresponding principal minor is $\det A(G[S])$.

Applying Theorem 4.1.19 to the graph $G[S]$, which has order i , gives $\det A(G[S]) = \sum_H (-1)^{i-c_1(H)-c(H)} 2^{c(H)}$ where H ranges over the elementary spanning subgraphs of $G[S]$.

As S ranges over all i -element subsets of $V(G)$, these elementary spanning subgraphs are exactly the elementary subgraphs of G having i vertices. Hence $a_i = (-1)^i \sum_{H \in \mathcal{H}_i} (-1)^{i-c_1(H)-c(H)} 2^{c(H)}$. Combining the signs, $a_i = \sum_{H \in \mathcal{H}_i} (-1)^{2i-c_1(H)-c(H)} 2^{c(H)}$. Since $2i$ is even, $(-1)^{2i-c_1(H)-c(H)} = (-1)^{c_1(H)+c(H)}$. But $p(H) = c_1(H) + c(H)$, because every component of an elementary graph is either a K_2 or a cycle. Therefore $a_i = \sum_{H \in \mathcal{H}_i} (-1)^{p(H)} 2^{c(H)}$. This proves Sachs' theorem. ■

4.2 Isomorphism and Cospectral Graphs

Definition 4.2.1 (Cospectral Graphs). Two graphs G_1 and G_2 are **cospectral** if they have the same adjacency spectrum, including multiplicities. Equivalently, $\phi(G_1, \lambda) = \phi(G_2, \lambda)$.

If two graphs are isomorphic, then one adjacency matrix is obtained from the other by simultaneously permuting its rows and columns. Thus, if $G_1 \cong G_2$, there exists a permutation matrix P such that $A(G_2) = P^T A(G_1) P$. Hence the adjacency matrices are similar and have the same characteristic polynomial. Therefore isomorphic graphs are always cospectral. The converse, however, is false.

Example 4.2.2 (Non-isomorphic Cospectral Graphs). The graphs $K_{1,4}$ and $C_4 \cup K_1$ are non-isomorphic but cospectral. Indeed, $\text{Spec}(K_{1,4}) = \text{Spec}(C_4 \cup K_1) = \{2, -2, 0, 0, 0\}$. They are clearly non-isomorphic because $K_{1,4}$ is connected whereas $C_4 \cup K_1$ is disconnected.

The connected non-isomorphic cospectral pair shown in Figure 4.3 has the common characteristic polynomial $\phi(G, \lambda) = \lambda^6 - 7\lambda^4 - 4\lambda^3 + 7\lambda^2 + 4\lambda - 1$.



Figure 4.2: A pair of non-isomorphic cospectral graphs.



Figure 4.3: A smallest connected pair of non-isomorphic cospectral graphs.

4.3 Regular Graphs

Definition 4.3.1 (k -regular Graph). A graph G is **k -regular** if every vertex of G has degree k .

Theorem 4.3.2 (Spectral Properties of Regular Graphs). Let G be a k -regular graph of order n . Then:

1. k is an eigenvalue of $A(G)$ with eigenvector $\mathbf{1} = (1, \dots, 1)^T$;
2. every eigenvalue λ of $A(G)$ satisfies $|\lambda| \leq k$;
3. the multiplicity of the eigenvalue k equals the number of connected components of G .

In particular, k has multiplicity 1 if and only if G is connected.

Proof. Let $A = A(G)$. Since G is k -regular, every row of A contains exactly k entries equal to 1. Therefore every row sum is k , and hence $A\mathbf{1} = k\mathbf{1}$. Thus k is an eigenvalue of A .

Next, let $Ax = \lambda x$, where $x = (x_1, \dots, x_n)^T \neq 0$. Choose j such that $|x_j| = \max_{1 \leq i \leq n} |x_i|$. The j th coordinate of $Ax = \lambda x$ gives $\lambda x_j = \sum_{v_i \sim v_j} x_i$. Taking absolute values, $|\lambda| |x_j| \leq \sum_{v_i \sim v_j} |x_i| \leq k|x_j|$. Since $|x_j| > 0$, we obtain $|\lambda| \leq k$.

We now determine the multiplicity of the eigenvalue k . First suppose that G is connected and that $Ax = kx$. Choose j as above. Then $k|x_j| = \left| \sum_{v_i \sim v_j} x_i \right| \leq \sum_{v_i \sim v_j} |x_i| \leq k|x_j|$. Since the first and last quantities are equal, equality holds throughout. Hence every neighbor v_i of v_j satisfies $|x_i| = |x_j|$ and all such x_i have the same sign as x_j . Thus $x_i = x_j$ for every neighbor of v_j .

Applying the same argument successively along paths and using the connectedness of G , we obtain $x_1 = x_2 = \dots = x_n$. Therefore every eigenvector corresponding to k is a scalar multiple of $\mathbf{1}$, so the eigenspace of k is one-dimensional.

Finally, suppose G has r connected components $G_1, \dots, G_r$. Since G is k -regular, every component is also k -regular. For each component G_i , let u_i be its indicator vector. Then $Au_i = ku_i$. The vectors $u_1, \dots, u_r$ are linearly independent. By the connected case, the eigenspace corresponding to k on each component is one-dimensional, so these vectors span the entire eigenspace. Hence the multiplicity of k is exactly r . ■

Example 4.3.3. The cycle C_n is 2-regular. Therefore 2 is an eigenvalue of C_n . Since C_n is connected, the eigenvalue 2 has multiplicity 1. This agrees with $\lambda_j = 2 \cos\left(\frac{2\pi j}{n}\right)$, $j = 0, 1, \dots, n-1$, for which the value 2 occurs only when $j = 0$.

4.3.1 Spectrum of the Complement

Recall that the complement $\overline{G}$ of a graph G has the same vertex set as G , with two distinct vertices adjacent in $\overline{G}$ exactly when they are not adjacent in G .

Theorem 4.3.4 (Spectrum of the Complement). *Let G be a k -regular graph of order n . Choose an orthonormal eigenbasis $x_1, x_2, \dots, x_n$ of $A(G)$ such that $x_1 = \frac{1}{\sqrt{n}}\mathbf{1}$, with corresponding eigenvalues $k, \lambda_2, \dots, \lambda_n$. Then $\text{Spec}(\overline{G}) = \{n - k - 1, -\lambda_2 - 1, \dots, -\lambda_n - 1\}$.*

Proof. Let A and $\overline{A}$ denote the adjacency matrices of G and $\overline{G}$, respectively. Since every pair of distinct vertices is adjacent in exactly one of these two graphs, $A + \overline{A} = J - I$, or equivalently, $\overline{A} = J - I - A$.

For $x_1 = \frac{1}{\sqrt{n}}\mathbf{1}$, we have $Jx_1 = nx_1$ and $Ax_1 = kx_1$. Therefore $\overline{A}x_1 = (J - I - A)x_1 = (n - 1 - k)x_1$. Thus $n - k - 1$ is an eigenvalue of $\overline{G}$.

Now let $i \geq 2$. Since the eigenbasis is orthonormal, $x_i \perp x_1$, and hence $x_i \perp \mathbf{1}$. Therefore the sum of the coordinates of x_i is zero, which implies $Jx_i = 0$. Since $Ax_i = \lambda_i x_i$, $\overline{A}x_i = (J - I - A)x_i = -x_i - \lambda_i x_i = (-1 - \lambda_i)x_i$. Hence the remaining eigenvalues of $\overline{G}$ are $-\lambda_2 - 1, \dots, -\lambda_n - 1$. ■

Example 4.3.5. The cycle C_4 is 2-regular and $\text{Spec}(C_4) = \{2, 0, 0, -2\}$. Therefore $\text{Spec}(\overline{C_4}) = \{4 - 2 - 1, -0 - 1, -0 - 1, -(-2) - 1\} = \{1, 1, -1, -1\}$. Since $\overline{C_4} \cong 2K_2$, this agrees with the fact that each copy of K_2 has eigenvalues 1 and -1 .

4.3.2 Spectrum of Line Graphs

Let $V(G) = \{v_1, \dots, v_n\}$, $E(G) = \{e_1, \dots, e_m\}$. The **line graph** G_ℓ has one vertex for each edge of G . Two vertices of G_ℓ are adjacent exactly when the corresponding edges of G have a common endpoint.

If G is k -regular, then every edge of G is incident with two vertices, and at each endpoint there are $k - 1$ other incident edges. Hence every vertex of G_ℓ has degree $(k - 1) + (k - 1) = 2k - 2$. Thus G_ℓ is $(2k - 2)$ -regular.

Lemma 4.3.6. *Let G be a simple graph, let A and B be the adjacency matrices of G and G_ℓ , respectively, and let M be the unoriented vertex-edge incidence matrix of G . Then $M^T M = B + 2I$. Moreover, $MM^T = D + A$, where D is the degree matrix of G . In particular, if G is k -regular, then $MM^T = A + kI$.*

Proof. The matrix M has one row for each vertex and one column for each edge. Its (v, e) -entry is 1 if the vertex v is incident with the edge e , and 0 otherwise.

Consider $M^T M$. Each column of M contains exactly two 1's because every edge has two endpoints. Hence every diagonal entry of $M^T M$ is 2.

For $i \neq j$, the (i, j) -entry of $M^T M$ is the number of vertices incident with both e_i and e_j . Since G is simple, this number is 1 precisely when e_i and e_j share an endpoint, and is 0 otherwise. But this is exactly the adjacency relation in the line graph. Therefore $M^T M = B + 2I$.

Now consider MM^T . Its diagonal entry corresponding to v_i counts the edges incident with v_i , so it equals $d(v_i)$. For $i \neq j$, the (i, j) -entry is 1 precisely when v_i and v_j are adjacent. Hence $MM^T = D + A$. If G is k -regular, then $D = kI$, giving $MM^T = A + kI$. ■

Corollary 4.3.7. *Every adjacency eigenvalue μ of a line graph satisfies $\mu \geq -2$.*

Proof. Let B be the adjacency matrix of G_ℓ . By Lemma 4.3.6, $B + 2I = M^T M$. For every vector x , $x^T M^T M x = (Mx)^T (Mx) = \|Mx\|^2 \geq 0$. Thus $M^T M$ is positive semidefinite, so all its eigenvalues are nonnegative.

If μ is an eigenvalue of B , then $\mu + 2$ is an eigenvalue of $B + 2I = M^T M$. Hence $\mu + 2 \geq 0$, and therefore $\mu \geq -2$. ■

Theorem 4.3.8 (Characteristic Polynomial of a Line Graph). *Let G be a k -regular graph of order n , where $k \geq 2$. Then $\phi(G_\ell, \lambda) = (\lambda + 2)^{\frac{n}{2}(k-2)} \phi(G, \lambda + 2 - k)$.*

Proof. Let $m = |E(G)|$, and let A and B denote the adjacency matrices of G and G_ℓ , respectively. Let M be the $n \times m$ unoriented incidence matrix of G .

Write the adjacency eigenvalues of G as $k = \lambda_1, \lambda_2, \dots, \lambda_n$. By Lemma 4.3.6, $MM^T = A + kI$. Adding kI shifts each eigenvalue of A by k . Hence the eigenvalues of MM^T are $2k, \lambda_2 + k, \dots, \lambda_n + k$.

The matrices MM^T and $M^T M$ have the same nonzero eigenvalues with the same multiplicities. Since G is k -regular, $2m = nk$. Because $k \geq 2$, we have $m \geq n$. Thus, when the n eigenvalues of MM^T are viewed as part of the spectrum of the $m \times m$ matrix $M^T M$, there are an additional $m - n$ zero eigenvalues.

Since $M^T M = B + 2I$, subtracting $2I$ shows that the eigenvalues of B are $2k - 2, \lambda_2 + k - 2, \dots, \lambda_n + k - 2$, together with -2 repeated $m - n$ additional times.

Therefore

$$\begin{aligned} \phi(G_\ell, \lambda) &= (\lambda + 2)^{m-n} (\lambda - (2k - 2)) \prod_{i=2}^n (\lambda - (\lambda_i + k - 2)) \\ &= (\lambda + 2)^{m-n} \phi(G, \lambda + 2 - k). \end{aligned}$$

Finally, $m - n = \frac{nk}{2} - n = \frac{n}{2}(k - 2)$, and hence $\phi(G_\ell, \lambda) = (\lambda + 2)^{\frac{n}{2}(k-2)} \phi(G, \lambda + 2 - k)$. ■

Example 4.3.9 (Spectrum of the Line Graph of a Cycle). For $n \geq 3$, the cycle C_n is 2-regular, and its line graph is again a cycle: $(C_n)_\ell \cong C_n$. The preceding theorem gives

$$\phi((C_n)_\ell, \lambda) = (\lambda + 2)^{\frac{n}{2}(2-2)} \phi(C_n, \lambda + 2 - 2) = \phi(C_n, \lambda),$$

which agrees with $(C_n)_\ell \cong C_n$.

4.4 The Laplacian Spectrum of a Graph

Let $G = (V, E)$ be a simple undirected graph with $V(G) = \{v_1, \dots, v_n\}$ and m edges. Recall that $A(G)$ denotes the adjacency matrix of G .

Definition 4.4.1 (Degree Matrix). The **degree matrix** of G is the diagonal matrix $D(G) = \text{diag}(d(v_1), d(v_2), \dots)$.

Definition 4.4.2 (Laplacian Matrix). The **Laplacian matrix** of G is $L(G) = D(G) - A(G)$. Thus

$$L_{ij} = \begin{cases} d(v_i), & i = j, \\ -1, & i \neq j \text{ and } v_i \sim v_j, \\ 0, & \text{otherwise.} \end{cases}$$

Since both $D(G)$ and $A(G)$ are real symmetric matrices, $L(G)$ is also real and symmetric.

Definition 4.4.3 (Laplacian Spectrum). The **Laplacian spectrum** of G , denoted by $\text{Spec}_L(G)$, is the multiset of eigenvalues of $L(G)$. Since $L(G)$ is real and symmetric, all its eigenvalues are real. We order them as $\mu_1 \geq \mu_2 \geq \dots \geq \mu_n$.

The following quadratic-form identity is fundamental in the study of the Laplacian.

Theorem 4.4.4 (Positive Semidefinite Property). *For every $x = (x_1, \dots, x_n)^T \in \mathbb{R}^n$, $x^T L(G)x = \sum_{v_i v_j \in E(G)} (x_i - x_j)^2$. Consequently, $L(G)$ is positive semidefinite and every Laplacian eigenvalue is nonnegative.*

Proof. Using $L = D - A$, we obtain $x^T Lx = x^T Dx - x^T Ax$. Now $x^T Dx = \sum_{i=1}^n d(v_i)x_i^2$. Each edge $v_i v_j$ contributes x_i^2 once through the degree of v_i and x_j^2 once through the degree of v_j . Hence $x^T Dx = \sum_{v_i v_j \in E(G)} (x_i^2 + x_j^2)$. Since A is symmetric, each edge contributes twice to $x^T Ax$, so $x^T Ax = 2 \sum_{v_i v_j \in E(G)} x_i x_j$. Therefore

$$\begin{aligned} x^T Lx &= \sum_{v_i v_j \in E(G)} (x_i^2 + x_j^2 - 2x_i x_j) \\ &= \sum_{v_i v_j \in E(G)} (x_i - x_j)^2. \end{aligned}$$

Every term in the last sum is nonnegative. Thus $x^T Lx \geq 0$ for every $x \in \mathbb{R}^n$, so $L(G)$ is positive semidefinite. Hence all eigenvalues of $L(G)$ are nonnegative. ■

Remark 4.4.5. The identity $x^T Lx = \sum_{v_i v_j \in E(G)} (x_i - x_j)^2$ shows that the Laplacian measures how much the values assigned to adjacent vertices differ. The quadratic form vanishes exactly when x is constant along every edge.

Theorem 4.4.6 (Smallest Laplacian Eigenvalue). *The smallest Laplacian eigenvalue of every graph is 0.*

Proof. For every vertex v_i , the sum of the entries in the i th row of $L(G)$ is $d(v_i) - d(v_i) = 0$. Therefore $L(G)\mathbf{1} = 0$, where $\mathbf{1} = (1, \dots, 1)^T$. Thus 0 is a Laplacian eigenvalue. Since all Laplacian eigenvalues are nonnegative, 0 must be the smallest one. ■

Theorem 4.4.7 (Multiplicity of the Zero Eigenvalue). *The multiplicity of 0 as a Laplacian eigenvalue equals the number of connected components of G . In particular,*

$$G \text{ is connected} \iff 0 \text{ is a simple Laplacian eigenvalue.}$$

Proof. Suppose first that $Lx = 0$. Then $x^T Lx = 0$. By the quadratic-form identity, $0 = \sum_{uv \in E(G)} (x_u - x_v)^2$. Every term in this sum is nonnegative, so each term must be zero. Hence $x_u = x_v$ for every edge uv .

If two vertices lie in the same connected component, they are joined by a path. Repeatedly applying the equality along that path shows that x has the same value at every vertex of the component. Thus every vector in $\ker L$ is constant on each connected component.

Conversely, suppose x is constant on every connected component. If v_i lies in a component on which the constant value is c , then all neighbors of v_i also have value c . Hence $(Lx)_i = d(v_i)c - \sum_{v_j \sim v_i} c = d(v_i)c - d(v_i)c = 0$. Therefore $Lx = 0$.

If G has r connected components, the characteristic vectors of these components form a basis for $\ker L$. Hence $\dim(\ker L) = r$. Since L is symmetric, the dimension of the nullspace is the multiplicity of the eigenvalue 0. Therefore the multiplicity of 0 is exactly the number of connected components. ■

Theorem 4.4.8 (Sum of Laplacian Eigenvalues). *If G has m edges, then $\sum_{i=1}^n \mu_i = 2m$.*

Proof. The sum of the eigenvalues of a matrix equals its trace. Therefore $\sum_{i=1}^n \mu_i = \text{tr}(L) = \sum_{i=1}^n d(v_i)$. By the Handshaking Lemma, $\sum_{i=1}^n d(v_i) = 2m$. Hence $\sum_{i=1}^n \mu_i = 2m$. ■

Remark 4.4.9. The average of the Laplacian eigenvalues is therefore $\frac{1}{n} \sum_{i=1}^n \mu_i = \frac{2m}{n}$, which is exactly the average degree of G .

Example 4.4.10 (Laplacian Spectrum of P_3). For the path $P_3 = v_1 v_2 v_3$, the degree matrix and adjacency matrix are

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Hence

$$L = D - A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Its characteristic polynomial is $\det(\lambda I - L) = \lambda(\lambda - 1)(\lambda - 3)$. Therefore $\text{Spec}_L(P_3) = \{3, 1, 0\}$.

This agrees with the general results above: P_3 is connected, so 0 occurs once, and $3 + 1 + 0 = 4 = 2|E(P_3)|$.

4.5 Laplacian Energy

Before defining Laplacian energy, recall the usual notion of graph energy.

Definition 4.5.1 (Graph Energy). If $\lambda_1, \dots, \lambda_n$ are the adjacency eigenvalues of G , then the **energy** of G is $E(G) = \sum_{i=1}^n |\lambda_i|$.

Graph energy was introduced by Gutman in 1978. Laplacian energy was later introduced by Gutman and Zhou as an analogous quantity based on the Laplacian eigenvalues.

Since $\sum_{i=1}^n \mu_i = 2m$, the average Laplacian eigenvalue is $\frac{2m}{n}$, which is also the average degree of the graph.

Definition 4.5.2 (Laplacian Energy). Let $\mu_1, \dots, \mu_n$ be the Laplacian eigenvalues of a graph G with n vertices and m edges. The **Laplacian energy** of G is $LE(G) = \sum_{i=1}^n \left| \mu_i - \frac{2m}{n} \right|$.

Thus Laplacian energy measures the total deviation of the Laplacian eigenvalues from their mean value $2m/n$.

Theorem 4.5.3 (Laplacian Energy of Regular Graphs). *If G is k -regular, then $LE(G) = E(G)$.*

Proof. For a k -regular graph, $D = kI$, and therefore $L = kI - A$. If $Ax = \lambda_i x$, then $Lx = (kI - A)x = (k - \lambda_i)x$. Hence the Laplacian eigenvalues are $\mu_i = k - \lambda_i$.

Since G is k -regular, $2m = nk$, so $\frac{2m}{n} = k$. Therefore

$$\begin{aligned} LE(G) &= \sum_{i=1}^n |\mu_i - k| \\ &= \sum_{i=1}^n |k - \lambda_i - k| \\ &= \sum_{i=1}^n |\lambda_i| \\ &= E(G). \end{aligned}$$

■

Example 4.5.4 (Energy of K_3). The adjacency spectrum of K_3 is $\text{Spec}(K_3) = \{2, -1, -1\}$. Hence $E(K_3) = |2| + |-1| + |-1| = 4$.

Since K_3 is 2-regular, its Laplacian eigenvalues are obtained from $\mu_i = 2 - \lambda_i$, giving $\text{Spec}_L(K_3) = \{3, 3, 0\}$. Also, $\frac{2m}{n} = \frac{2 \cdot 3}{3} = 2$. Therefore $LE(K_3) = |3 - 2| + |3 - 2| + |0 - 2| = 4$. Thus $LE(K_3) = E(K_3)$, as predicted by the preceding theorem.

Example 4.5.5 (Laplacian Energy of P_3). The path P_3 is not regular, so its Laplacian energy need not equal its ordinary graph energy.

From the preceding example, $\text{Spec}_L(P_3) = \{3, 1, 0\}$. Since P_3 has $n = 3$ vertices and $m = 2$ edges, $\frac{2m}{n} = \frac{4}{3}$. Hence

$$\begin{aligned} LE(P_3) &= \left| 3 - \frac{4}{3} \right| + \left| 1 - \frac{4}{3} \right| + \left| 0 - \frac{4}{3} \right| \\ &= \frac{5}{3} + \frac{1}{3} + \frac{4}{3} \\ &= \frac{10}{3}. \end{aligned}$$

This example illustrates that the equality $LE(G) = E(G)$ is a special property of regular graphs.

Theorem 4.5.6 (Basic Properties of Laplacian Energy). *For every graph G :*

- (i) $LE(G) \geq 0$;
- (ii) $LE(G) = 0$ if and only if G has no edges.

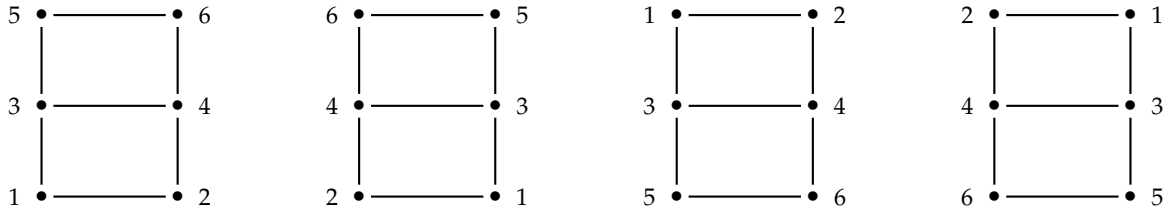
Proof. Since Laplacian energy is a sum of absolute values, $LE(G) = \sum_{i=1}^n |\mu_i - \frac{2m}{n}| \geq 0$. This proves (i).

Suppose first that G has no edges. Then $m = 0$ and $L = 0$. Thus every Laplacian eigenvalue is 0, and consequently $LE(G) = 0$.

Conversely, suppose $LE(G) = 0$. A sum of nonnegative terms can equal zero only when every term is zero. Hence $\mu_i = \frac{2m}{n}$ for every i . But 0 is always a Laplacian eigenvalue. Therefore $\frac{2m}{n} = 0$, which implies $m = 0$. Thus G has no edges. ■

4.5.1 Laplacian Spectra of Some Standard Graphs

Theorem 4.5.7 (Laplacian Spectrum of K_n). *For $n \geq 2$, $\text{Spec}_L(K_n) = \{0^{(1)}, n^{(n-1)}\}$.*

Figure 4.4: Automorphism group of $G_{2,3}$.

Proof. For the complete graph K_n , every vertex has degree $n-1$. Hence $D = (n-1)I$. Also, $A = J - I$, so $L = D - A = (n-1)I - (J - I) = nI - J$.

The all-ones vector $\mathbf{1}$ satisfies $J\mathbf{1} = n\mathbf{1}$. Therefore $L\mathbf{1} = (nI - J)\mathbf{1} = 0$, so 0 is an eigenvalue.

If $x \perp \mathbf{1}$, then the sum of the coordinates of x is zero, so $Jx = 0$. Hence $Lx = (nI - J)x = nx$. The subspace orthogonal to $\mathbf{1}$ has dimension $n-1$. Therefore n is an eigenvalue with multiplicity $n-1$, and $\text{Spec}_L(K_n) = \{0^{(1)}, n^{(n-1)}\}$. ■

Theorem 4.5.8 (Laplacian Spectrum of $K_{r,s}$). For $r, s \geq 1$, $\text{Spec}_L(K_{r,s}) = \{0^{(1)}, (r+s)^{(1)}, r^{(s-1)}, s^{(r-1)}\}$.

Proof. Let the two partite sets be $V_1 = \{u_1, \dots, u_r\}$, $V_2 = \{v_1, \dots, v_s\}$. Every vertex in V_1 has degree s , while every vertex in V_2 has degree r .

Since $K_{r,s}$ is connected, 0 is a simple Laplacian eigenvalue, with eigenvector $\mathbf{1}$.

Consider a vector x supported on V_1 whose coordinates on V_1 sum to zero. Thus $x = (x_1, \dots, x_r, 0, \dots, 0)^T$. For a vertex $u_i \in V_1$, all neighbors lie in V_2 , where the coordinates of x are zero. Hence $(Lx)_{u_i} = sx_i$. For a vertex $v_j \in V_2$, $(Lx)_{v_j} = -\sum_{i=1}^r x_i = 0$. Thus $Lx = sx$. The space of vectors on V_1 whose coordinates sum to zero has dimension $r-1$. Therefore s is an eigenvalue with multiplicity $r-1$.

Similarly, vectors supported on V_2 whose coordinates sum to zero form an $(s-1)$ -dimensional eigenspace corresponding to the eigenvalue r . Hence r has multiplicity $s-1$.

It remains to find one eigenvalue. Consider the vector

$$z = (\underbrace{s, \dots, s}_{r \text{ entries}}, \underbrace{-r, \dots, -r}_{s \text{ entries}})^T.$$

For $u_i \in V_1$, $(Lz)_{u_i} = s(s) - \sum_{v_j \in V_2} (-r) = s^2 + rs = (r+s)s$. For $v_j \in V_2$, $(Lz)_{v_j} = r(-r) - \sum_{u_i \in V_1} s = -r^2 - rs = (r+s)(-r)$. Therefore $Lz = (r+s)z$. Thus $r+s$ is an eigenvalue.

The multiplicities now sum to $1 + (r-1) + (s-1) + 1 = r+s$, which is the order of $K_{r,s}$. Hence all Laplacian eigenvalues have been accounted for: $\text{Spec}_L(K_{r,s}) = \{0^{(1)}, (r+s)^{(1)}, r^{(s-1)}, s^{(r-1)}\}$. ■

4.6 Automorphism Groups of Graphs

Definition 4.6.1. A graph automorphism of G is a permutation φ on the set of vertices $V(G)$ that satisfies the property that $\{u_i, u_j\} \in E(G)$ if and only if $\{\varphi(u_i), \varphi(u_j)\} \in E(G)$.

Example 4.6.2. For example, the grid graph $G_{2,3}$ has four automorphisms: $(1, 2, 3, 4, 5, 6)$, $(2, 1, 4, 3, 6, 5)$, $(5, 6, 3, 4, 1, 2)$, $(6, 5, 4, 3, 2, 1)$. These correspond to the graph itself, the graph flipped from left to right, the graph flipped up down, and the graph flipped left to right and up down, respectively.

The star graph S_4 has six automorphisms: $(1, 2, 3, 4)$, $(1, 3, 2, 4)$, $(2, 1, 3, 4)$, $(2, 3, 1, 4)$, $(3, 1, 2, 4)$, $(3, 2, 1, 4)$.

Theorem 4.6.3. The set $\text{Aut}(G)$ of all graph automorphisms of a graph G forms a group under function composition.

Definition 4.6.4 (Identity Graph). A graph G whose automorphism group is $\text{Aut}(G) \cong S_1$ is called the identity graph.

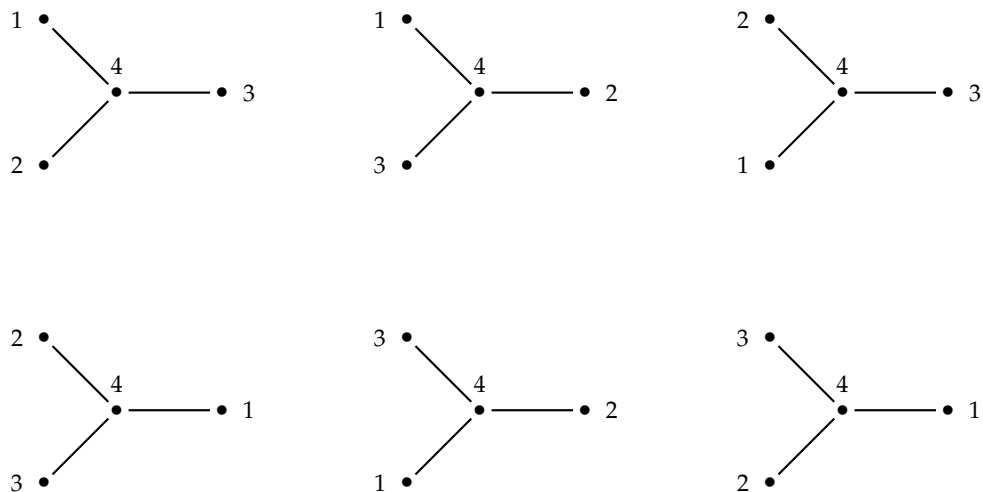
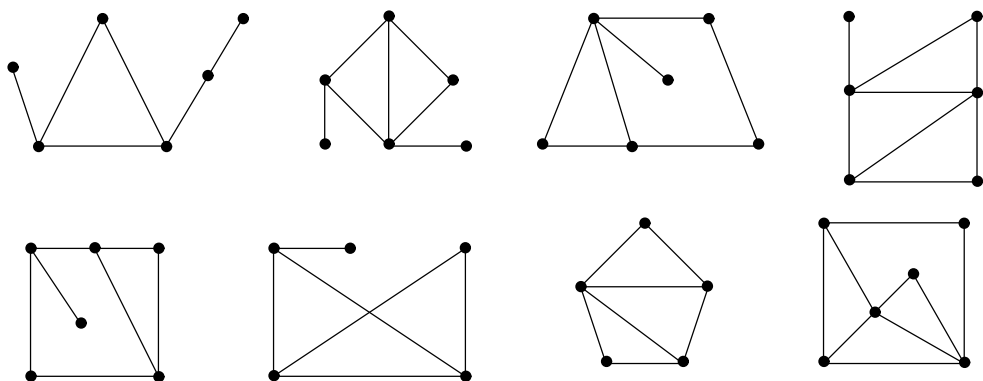
Figure 4.5: Automorphisms of star graph S_4 .

Figure 4.6: The Eight Identity Graphs of order Six.

The elements of $Aut(G)$ induce a group of permutations on the edge set E of G in the obvious way:

If $e = uv \in E$ and $\alpha \in Aut(G)$, then $\alpha_1(e) = \alpha_0(u)\alpha_0(v)$ where α_0 and α_1 denote the action of α on V and E respectively. This permutation group on E with degree $m = |E|$ and order $|Aut(G)|$ is called the induced edge group of G and is denoted by $Aut_I(G)$.

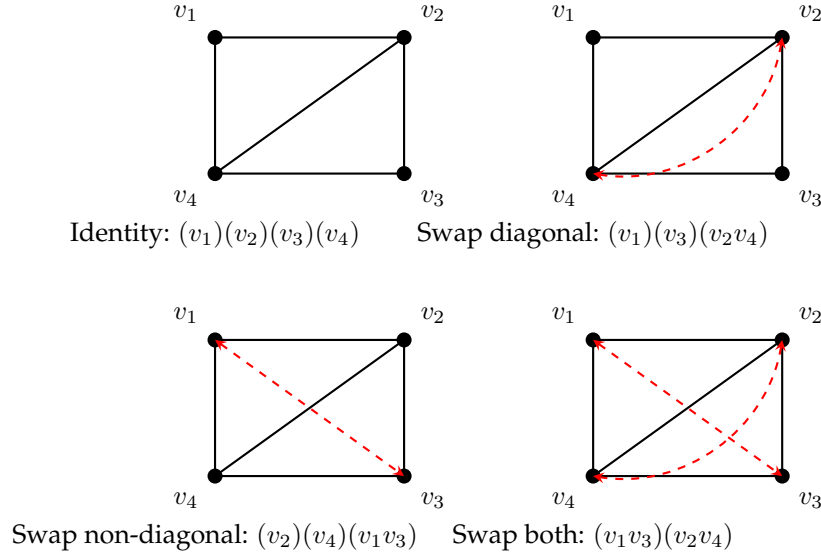


Figure 4.7: Automorphisms of $K_4 - e$

Example 4.6.5. Consider the graph $K_4 - e$ with vertices, Figure 4.7, labeled v_1, v_2, v_3, v_4 and edges e_1, e_2, e_3, e_4, e_5 . The vertex group $Aut(K_4 - e)$ consists of the four permutations:

$$(v_1)(v_2)(v_3)(v_4), \quad (v_1)(v_3)(v_2v_4), \quad (v_2)(v_4)(v_1v_3), \quad (v_1v_3)(v_2v_4)$$

or

$$\begin{pmatrix} v_1 & v_2 & v_3 & v_4 \\ v_1 & v_2 & v_3 & v_4 \end{pmatrix}, \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_1 & v_3 & v_4 & v_2 \end{pmatrix}, \begin{pmatrix} v_2 & v_4 & v_1 & v_3 \\ v_2 & v_4 & v_3 & v_1 \end{pmatrix}, \begin{pmatrix} v_1 & v_3 & v_2 & v_4 \\ v_3 & v_1 & v_4 & v_2 \end{pmatrix}.$$

The identity permutation of the vertex-group induces the identity permutation on the edges, while $(v_1)(v_3)(v_2v_4)$ induces a permutation on the edges which fixes e_5 , interchanges e_1 with e_4 , and e_2 with e_3 . In this way, the edge-group consists of the following permutations, induced respectively by the above members of the vertex-group.

$$(e_1)(e_2)(e_3)(e_4)(e_5), \quad (e_1e_4)(e_2e_3)(e_5), \quad (e_1e_2)(e_3e_4)(e_5), \quad (e_1e_3)(e_2e_4)(e_5)$$

or

$$\begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_1 & e_2 & e_3 & e_4 & e_5 \end{pmatrix}, \begin{pmatrix} e_1 & e_4 & e_2 & e_3 & e_5 \\ e_4 & e_1 & e_3 & e_2 & e_5 \end{pmatrix}, \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_2 & e_1 & e_4 & e_3 & e_5 \end{pmatrix}, \begin{pmatrix} e_1 & e_2 & e_3 & e_4 & e_5 \\ e_3 & e_1 & e_4 & e_2 & e_5 \end{pmatrix}.$$

Of course, the edge-group and the vertex-group of $K_4 - e$ are isomorphic. But they are not identical permutation groups since $Aut_I(K_4 - e)$ has degree 5 and $Aut(K_4 - e)$ has degree 4. Note that edge e_5 is fixed by every member of the edge-group. Even the permutation group obtained from $Aut_I(K_4 - e)$ by restricting its object set to e_1, e_2, e_3, e_4 is not identical with $Aut(K_4 - e)$, since these two isomorphic permutation groups of the same degree have different cycle structure.

Theorem 4.6.6. The vertex automorphism group $Aut(G)$ and the induced edge automorphism group $Aut_I(G)$ of a graph G are isomorphic if and only if G contains at most one isolated vertex and no connected component isomorphic to K_2 .

Proof. Let $\phi : \text{Aut}(G) \rightarrow \text{Aut}_I(G)$ be the natural homomorphism defined by $\phi(\alpha) = \alpha_I$, where α_I is the edge automorphism induced by α , that is,

$$\alpha_I(\{u, v\}) = \{\alpha(u), \alpha(v)\}.$$

By the definition of $\text{Aut}_I(G)$, the map ϕ is surjective. Hence $\text{Aut}(G) \cong \text{Aut}_I(G)$ if and only if ϕ is injective, or equivalently, if and only if $\ker \phi = \{\text{id}\}$.

Suppose first that G contains two distinct isolated vertices u and v . Let α be the permutation of $V(G)$ that interchanges u and v and fixes every other vertex. Since u and v are isolated, α is a nonidentity automorphism of G and fixes every edge of G . Thus $\alpha \in \ker \phi$, so ϕ is not injective.

Similarly, suppose that G has a connected component isomorphic to K_2 , with vertices u and v and edge $\{u, v\}$. The automorphism that interchanges u and v and fixes every other vertex fixes the edge $\{u, v\}$ as an unordered pair and fixes all other edges. Hence it is a nonidentity element of $\ker \phi$. Therefore, if ϕ is injective, then G has at most one isolated vertex and no connected component isomorphic to K_2 .

Conversely, suppose that G has at most one isolated vertex and no connected component isomorphic to K_2 . Let $\alpha \in \ker \phi$. Then every edge is fixed as an unordered pair, so for each $\{x, y\} \in E(G)$,

$$\{\alpha(x), \alpha(y)\} = \{x, y\}.$$

We show that $\alpha = \text{id}$. Suppose, to the contrary, that $\alpha(u) = v \neq u$ for some vertex u . Since automorphisms preserve degree, $\deg(u) = \deg(v)$. If $\deg(u) = 0$, then u and v are two distinct isolated vertices, contrary to the hypothesis. Hence $\deg(u) \geq 1$.

Choose a neighbor w of u . Since the edge $\{u, w\}$ is fixed,

$$\{\alpha(u), \alpha(w)\} = \{u, w\}.$$

Using $\alpha(u) = v$, we obtain

$$\{v, \alpha(w)\} = \{u, w\}.$$

Therefore $v \in \{u, w\}$. Since $v \neq u$, it follows that $v = w$. Hence $\{u, v\} \in E(G)$. Substituting $w = v$ in the preceding equality gives

$$\{v, \alpha(v)\} = \{u, v\},$$

and therefore $\alpha(v) = u$.

We now show that u has no neighbor other than v . Suppose that $z \neq v$ is another neighbor of u . Since $\{u, z\}$ is fixed,

$$\{\alpha(u), \alpha(z)\} = \{u, z\}.$$

Thus

$$\{v, \alpha(z)\} = \{u, z\}.$$

The equality of these two sets implies $v \in \{u, z\}$, which is impossible because $v \neq u$ and $v \neq z$. Hence v is the only neighbor of u , so $\deg(u) = 1$. Since $\deg(v) = \deg(u)$, we also have $\deg(v) = 1$. Thus $\{u, v\}$ is a connected component isomorphic to K_2 , contrary to the hypothesis.

Therefore no vertex is moved by α , so $\alpha = \text{id}$. Hence $\ker \phi = \{\text{id}\}$, and ϕ is injective. Since ϕ is also surjective, it is an isomorphism. Therefore,

$$\text{Aut}(G) \cong \text{Aut}_I(G).$$

■

Corollary 4.6.7. *If G is a connected graph with $n \geq 3$, then $\text{Aut}(G) \cong \text{Aut}_I(G)$.*

Theorem 4.6.8. *Given any graph G , $\text{Aut}(G) = \text{Aut}(\overline{G})$.*

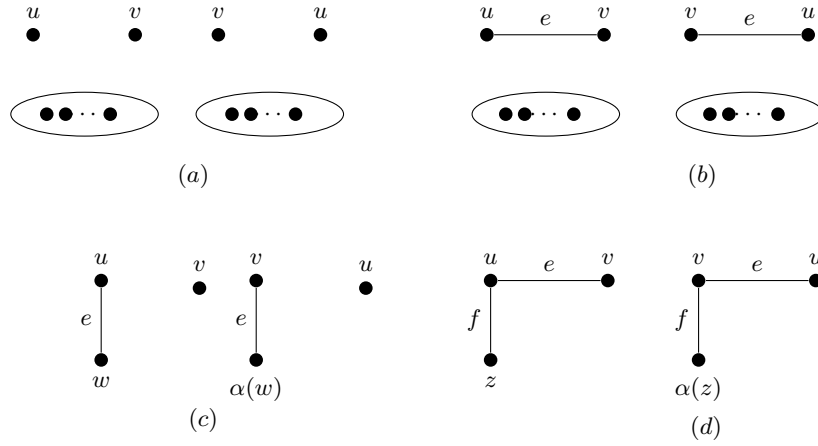


Figure 4.8: Illustration for Theorem 4.6.6. In (a) two isolated vertices may be interchanged; in (b) the two vertices of a K_2 -component may be interchanged; in (c) the fixed edge $\{u, w\}$ together with $\alpha(u) = v$ forces $w = v$; and in (d) the existence of another neighbour $z \neq v$ leads to a contradiction.

Proof. Let V be the vertex set of G . Recall that for the complement graph $\overline{G}$, an edge $\{u, v\}$ exists in $E(\overline{G})$ if and only if $\{u, v\} \notin E(G)$.

Let σ be a permutation of V . We show that $\sigma \in \text{Aut}(G)$ if and only if $\sigma \in \text{Aut}(\overline{G})$. By the definition of an automorphism, $\sigma \in \text{Aut}(G)$ if and only if for all distinct vertices $u, v \in V$:

$$\{u, v\} \in E(G) \iff \{\sigma(u), \sigma(v)\} \in E(G).$$

Taking the logical negation of both sides of the equivalence, we obtain:

$$\{u, v\} \notin E(G) \iff \{\sigma(u), \sigma(v)\} \notin E(G).$$

By the definition of the complement graph, $\{x, y\} \notin E(G)$ is equivalent to $\{x, y\} \in E(\overline{G})$. Substituting this into the expression above yields:

$$\{u, v\} \in E(\overline{G}) \iff \{\sigma(u), \sigma(v)\} \in E(\overline{G}).$$

This is precisely the condition for σ to be an automorphism of $\overline{G}$. Thus, $\sigma \in \text{Aut}(G) \iff \sigma \in \text{Aut}(\overline{G})$, proving that $\text{Aut}(G) = \text{Aut}(\overline{G})$. ■

Definition 4.6.9. Let S_n be the symmetric group of degree n , and let $\mathbb{Z}_2 = \{0, 1\}$ be the cyclic group of order 2 (under addition modulo 2).

The **semidirect product** $S_n \rtimes_{\varphi} \mathbb{Z}_2$ is the set of ordered pairs $S_n \times \mathbb{Z}_2$ equipped with a group operation defined by a homomorphism:

$$\varphi : \mathbb{Z}_2 \rightarrow \text{Aut}(S_n)$$

Let $\varphi_0 = \text{id}_{S_n}$ and $\varphi_1 = \theta$, where θ is an automorphism of S_n such that $\theta^2 = \text{id}$ (an involution).

The group operation for two elements (σ, a) and (τ, b) in $S_n \rtimes_{\varphi} \mathbb{Z}_2$ is defined as:

$$(\sigma, a) \cdot (\tau, b) = (\sigma \circ \varphi_a(\tau), a + b)$$

where $\circ$ denotes permutation multiplication and $+$ denotes addition modulo 2.

Note:

- If φ is the trivial homomorphism (maps everything to the identity automorphism), the group becomes the **Direct Product** $S_n \times \mathbb{Z}_2$.
- To obtain a non-trivial semidirect product, θ is typically chosen to be an inner automorphism (conjugation by an element of order 2) or, for $n = 6$, an outer automorphism.

Theorem 4.6.10. *The following hold.*

1. For $n \geq 1$, $\text{Aut}(K_n) \cong S_n$.
2. For $n \geq 3$, $\text{Aut}(K_n - e) \cong S_{n-2} \times S_2$.
3. For $n \geq 4$, the automorphism group of K_n with two adjacent edges removed is isomorphic to $S_{n-3} \times S_2$.
4. For $n \geq 4$, the automorphism group of K_n with two non-adjacent edges removed is isomorphic to $S_{n-4} \times D_4$, where S_0 is understood to be the trivial group.
5. For $n \geq 3$, $\text{Aut}(C_n) \cong D_n$.
6. For $m, n \geq 1$,

$$\text{Aut}(K_{m,n}) \cong \begin{cases} S_m \times S_n, & m \neq n, \\ (S_n \times S_n) \rtimes \mathbb{Z}_2, & m = n. \end{cases}$$

7. For $n \geq 2$, $\text{Aut}(P_n) \cong S_2$.

Proof. We prove the assertions separately.

1. Since every pair of distinct vertices of K_n is adjacent, every permutation of its vertex set preserves adjacency. Conversely, every automorphism of K_n is a permutation of its n vertices. Hence

$$\text{Aut}(K_n) \cong S_n.$$

2. Let $G = K_n - e$, where $e = \{u, v\}$. In G , the vertices u and v have degree $n - 2$, while each of the remaining $n - 2$ vertices has degree $n - 1$. Since automorphisms preserve degrees, every automorphism of G preserves the set $\{u, v\}$ and its complement $W = V(G) \setminus \{u, v\}$. The vertices of W may be permuted arbitrarily, giving a subgroup isomorphic to S_{n-2} , while u and v may either be fixed or interchanged, giving a subgroup isomorphic to S_2 . These two actions are independent. Therefore

$$\text{Aut}(K_n - e) \cong S_{n-2} \times S_2.$$

3. Let G be obtained from K_n by deleting the adjacent edges $\{u, v\}$ and $\{v, w\}$. Then

$$\deg(v) = n - 3, \quad \deg(u) = \deg(w) = n - 2,$$

while every other vertex has degree $n - 1$. Hence every automorphism fixes v , preserves the set $\{u, w\}$, and preserves the remaining $n - 3$ vertices as a set. The vertices u and w may be interchanged, giving S_2 , while the remaining $n - 3$ vertices may be permuted arbitrarily, giving S_{n-3} . These actions are independent. Thus

$$\text{Aut}(G) \cong S_{n-3} \times S_2.$$

4. Let G be obtained from K_n by deleting two non-adjacent edges $\{u, v\}$ and $\{x, y\}$. The four vertices u, v, x, y have degree $n - 2$, while every other vertex has degree $n - 1$. Therefore every automorphism preserves the sets

$$X = \{u, v, x, y\}, \quad Z = V(G) \setminus X.$$

The vertices of Z may be permuted arbitrarily, giving S_{n-4} . The subgraph induced by X is K_4 with two independent edges removed, and hence is a cycle C_4 . Its automorphism group is D_4 , the dihedral group of order 8. Since the actions on X and Z are independent,

$$\text{Aut}(G) \cong S_{n-4} \times D_4.$$

5. Let the vertices of C_n be $v_0, v_1, \dots, v_{n-1}$ in cyclic order. An automorphism is completely determined by the image of v_0 and by the choice of which neighbor of its image is the image of v_1 . There are n choices for the image of v_0 and, for each such choice, two choices for the image of v_1 . Thus $\text{Aut}(C_n)$ has $2n$ elements, namely the n rotations and n reflections of the cycle. Therefore

$$\text{Aut}(C_n) \cong D_n,$$

where D_n denotes the dihedral group of order $2n$.

6. Let $K_{m,n}$ have partite sets V_1 and V_2 , with $|V_1| = m$ and $|V_2| = n$.

Suppose first that $m \neq n$. Every vertex of V_1 has degree n , whereas every vertex of V_2 has degree m . Since these degrees are different, every automorphism preserves V_1 and V_2 setwise. The vertices within each part may be permuted arbitrarily, and any such pair of permutations preserves adjacency. Hence

$$\text{Aut}(K_{m,n}) \cong S_m \times S_n.$$

Now suppose that $m = n$. Let $G = K_{n,n}$ with partite sets V_1 and V_2 . Since a permutation of the vertex set is an automorphism of a graph if and only if it is an automorphism of its complement,

$$\text{Aut}(G) = \text{Aut}(\overline{G}).$$

The complement $\overline{G}$ is the disjoint union of two copies of K_n , one induced by V_1 and the other by V_2 . We may independently permute the vertices within these two components, giving the subgroup $S_n \times S_n$. In addition, the two isomorphic components may be interchanged. This interchange has order 2 and swaps the two factors of $S_n \times S_n$. Therefore

$$\text{Aut}(K_{n,n}) \cong (S_n \times S_n) \rtimes \mathbb{Z}_2,$$

which is also the wreath product $S_n \wr S_2$.

7. Let the vertices of P_n be $v_1, v_2, \dots, v_n$ in their natural order. The vertices v_1 and v_n are the only vertices of degree 1, so every automorphism preserves the set $\{v_1, v_n\}$.

If an automorphism fixes v_1 , then it must fix its unique neighbor v_2 . Continuing along the path, it follows successively that every vertex is fixed. Hence the automorphism is the identity.

If an automorphism sends v_1 to v_n , then adjacency forces

$$v_2 \mapsto v_{n-1}, \quad v_3 \mapsto v_{n-2}, \quad \dots, \quad v_i \mapsto v_{n-i+1}.$$

Thus the only nonidentity automorphism is the reflection σ defined by

$$\sigma(v_i) = v_{n-i+1}.$$

Therefore

$$\text{Aut}(P_n) = \{\text{id}, \sigma\}, \quad \sigma^2 = \text{id},$$

and hence

$$\text{Aut}(P_n) \cong S_2.$$

■

4.7 Edge Isomorphisms and Automorphisms

Definition. Two graphs $G(V, E)$ and $H(U, F)$ are said to be **edge-isomorphic** if there exists a bijection $\eta : E \rightarrow F$ that preserves edge-adjacency and edge-non-adjacency. Specifically, two edges $e, f \in E$ share a common vertex in G if and only if their images $\eta(e), \eta(f) \in F$ share a common vertex in H . Such a map η is called an *edge-isomorphism*.

An edge-isomorphism from a graph to itself is called an **edge-automorphism**. The set of all edge-automorphisms of a graph G forms a group under composition, known as the *edge-automorphism group* of G , denoted by $\text{Aut}_E(G)$.

It is important to observe that for a given graph G , the vertex automorphism group $\text{Aut}(G)$ and the edge-automorphism group $\text{Aut}_E(G)$ need not be isomorphic. This discrepancy arises because there may exist edge-automorphisms that are not induced by any vertex automorphism (i.e., they are not in the induced edge group $\text{Aut}_I(G)$).

Figure 1 illustrates three specific cases— $K_{1,3} + e$, $K_4 - e$, and K_4 —where each graph admits an edge-automorphism η that cannot be induced by any permutation of the vertices.

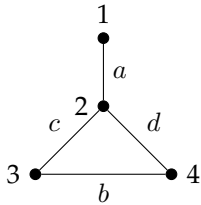


Figure 4.9: *
 $K_{1,3} + e$
 $\eta = (a\ b)(c\ d)$

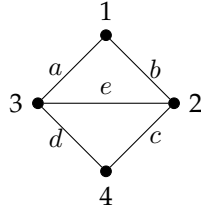


Figure 4.10: *
 $K_4 - e$
 $\eta = (a\ b\ c\ d)(e)$

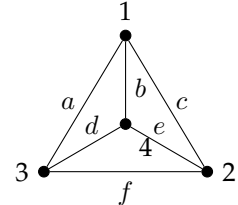


Figure 4.11: *
 K_4
 $\eta = (b\ f)(d)(e)(a)(c)$

Figure 4.12: Edge-automorphisms not induced by vertex automorphisms.

Theorem 4.7.1. Let $\eta : E(G) \rightarrow E(H)$ be an edge-isomorphism from a connected graph G onto a connected graph H . If G is not isomorphic to any of the graphs

$$K_3, K_{1,3}, K_{1,3} + e, K_4 - e, K_4,$$

then η is induced by a vertex isomorphism from G to H .

Proof. For $v \in V(G)$, let

$$E_G(v) = \{e \in E(G) : e \text{ is incident with } v\}.$$

Since η is an edge-isomorphism, two edges of G are adjacent if and only if their images under η are adjacent in H .

We shall use the following simple fact: if at least four edges of a simple graph are pairwise adjacent, then they have a common end-vertex. For three pairwise adjacent edges there are only two possibilities: either they have a common end-vertex or they are the three edges of a triangle.

Let $v \in V(G)$ with $\deg(v) \geq 3$. The edges in $E_G(v)$ are pairwise adjacent, so the edges in $\eta(E_G(v))$ are also pairwise adjacent. If $\deg(v) \geq 4$, they must therefore have a common end-vertex in H .

Now suppose that $\deg(v) = 3$. Again the three edges in $\eta(E_G(v))$ are pairwise adjacent. If they do not have a common end-vertex, they must form a triangle in H . We show that this can occur only when G is one of the excluded graphs.

Write $N_G(v) = \{v_1, v_2, v_3\}$. Suppose that some v_i is incident with an edge $f \neq vv_i$ whose other end is not one of v_1, v_2, v_3 . Then f is adjacent to exactly one of the three edges vv_1, vv_2, vv_3 . Hence $\eta(f)$ would have to be adjacent to exactly one edge of the triangle $\eta(E_G(v))$. This is impossible, because an edge outside a triangle is adjacent to either none or exactly two of its edges. Therefore every edge incident with one of v_1, v_2, v_3 , apart from vv_i , must join two of the vertices v_1, v_2, v_3 .

Since G is connected, no other vertices can occur. Thus G consists of the vertex v , its three neighbors v_1, v_2, v_3 , and some of the possible edges among v_1, v_2, v_3 . According as there are 0, 1, 2, or 3 such edges, G is isomorphic to $K_{1,3}$, $K_{1,3} + e$, $K_4 - e$, or K_4 , respectively. All of these are excluded. Hence, for every vertex v with $\deg(v) \geq 3$, the edges in $\eta(E_G(v))$ have a common end-vertex.

If $\deg(v) = 2$, the two edges in $E_G(v)$ are adjacent, and therefore their images have a unique common end-vertex. Thus, for every vertex v with $\deg(v) \geq 2$, define $\varphi(v)$ to be the common end-vertex of the edges in $\eta(E_G(v))$.

We next show that if $uv \in E(G)$ and $\deg(u), \deg(v) \geq 2$, then $\varphi(u) \neq \varphi(v)$. Suppose, to the contrary, that $\varphi(u) = \varphi(v)$. Choose an edge $e \neq uv$ incident with u and an edge $f \neq uv$ incident with v . Since $\eta(e)$ and $\eta(f)$ are both incident with $\varphi(u)$, they are adjacent in H . Hence e and f are adjacent in G . It follows that $e = uw$ and $f = vw$ for some vertex w , so u, v, w form a triangle.

If either u or v had another incident edge, the same argument would force that edge to be adjacent to both uw and vw , which is impossible in a simple graph. Hence $\deg(u) = \deg(v) = 2$. If w had an edge other than wu and wv , then the images of all edges incident with w would have to meet at the same vertex $\varphi(u)$, making the image of that additional edge adjacent to $\eta(uv)$. Its preimage would then have to be adjacent to uv , which is impossible. Thus $\deg(w) = 2$. Since G is connected, $G \cong K_3$, contrary to the hypothesis. Therefore $\varphi(u) \neq \varphi(v)$.

Now let $uv \in E(G)$ with $\deg(u), \deg(v) \geq 2$. The edge $\eta(uv)$ belongs to both $\eta(E_G(u))$ and $\eta(E_G(v))$. Hence it is incident with both $\varphi(u)$ and $\varphi(v)$. Since these vertices are distinct,

$$\eta(uv) = \{\varphi(u), \varphi(v)\}.$$

It remains to define φ on vertices of degree 1. Let u be such a vertex and let v be its unique neighbor. If $G \cong K_2$, then $H \cong K_2$ and the result is immediate. Otherwise $\deg(v) \geq 2$. Since $\eta(uv)$ is incident with $\varphi(v)$, define $\varphi(u)$ to be the other end-vertex of $\eta(uv)$. Then again

$$\eta(uv) = \{\varphi(u), \varphi(v)\}.$$

Thus, for every edge $uv \in E(G)$,

$$\eta(uv) = \{\varphi(u), \varphi(v)\}.$$

Hence φ preserves adjacency and induces η .

Finally, φ is bijective. Indeed, if two distinct vertices had the same image, two distinct incident edges in H would have adjacent preimages in a way incompatible with the construction above. Also, every vertex of H is incident with some edge because H is connected and nontrivial, and every edge of H is the image under η of an edge of G . Hence every vertex of H lies in the image of φ . Therefore φ is a vertex isomorphism from G onto H , and η is induced by φ . ■

Theorem 4.7.2. *For a non-empty graph G , $\Gamma_I(G) \cong \Gamma_E(G)$ if and only if the following conditions hold:*

- (a) K_3 and $K_{1,3}$ do not both occur as components of G ;
- (b) none of the graphs $K_{1,3} + e$, $K_4 - e$, and K_4 is a component of G .

Proof. (See Figure 4.13.) We first prove necessity.

Suppose that both K_3 and $K_{1,3}$ occur as components of G . In both graphs every pair of edges is adjacent, so their edge sets are edge-isomorphic. Hence an edge-automorphism of G can interchange the edge sets of these two components while fixing all other edges. Such an edge-automorphism cannot be induced by a vertex automorphism, because a vertex automorphism maps a connected component only onto an isomorphic connected component, whereas $K_3 \not\cong K_{1,3}$. Thus condition (a) is necessary.

Now suppose that $K_{1,3} + e$ is a component of G . Let v be its vertex of degree 3, let u_1, u_2, u_3 be its neighbors, and suppose that u_1u_2 is the additional edge. The edge permutation which interchanges vu_3 and u_1u_2 and fixes vu_1 and vu_2 preserves edge adjacency, but it cannot be induced by a vertex automorphism. Hence $K_{1,3} + e$ cannot be a component.

If $K_4 - e$ is a component, let x, y be its vertices of degree 3 and let u, v be its vertices of degree 2. The four edges xu, yu, yv, xv form a 4-cycle in the edge-adjacency graph, while the edge xy is adjacent to all four. A rotation of this 4-cycle through one step, fixing xy , gives an edge-automorphism which is not induced by any vertex automorphism. Hence $K_4 - e$ cannot be a component.

If K_4 is a component, label its vertices 1, 2, 3, 4. The edge permutation

$$12 \leftrightarrow 34, 13 \leftrightarrow 24, 14 \leftrightarrow 23$$

preserves edge adjacency but is not induced by any vertex automorphism of K_4 . Therefore K_4 cannot be a component. This proves necessity of condition (b).

Conversely, assume that conditions (a) and (b) hold, and let $\alpha \in \Gamma_E(G)$ be an arbitrary edge-automorphism. We show that α is induced by a vertex automorphism of G .

Let C be a connected component of G containing at least one edge. Since α preserves adjacency between edges, the set $\alpha(E(C))$ is the edge set of some connected component C' of G . Thus

$$\alpha|_{E(C)} : E(C) \rightarrow E(C')$$

is an edge-isomorphism.

If $C \cong K_3$ or $C \cong K_{1,3}$, then condition (a) ensures that C' is of the same type as C . Indeed, K_3 and $K_{1,3}$ are the only two nonisomorphic connected graphs among the present exceptions whose edge-adjacency graphs are both K_3 . Every permutation of the three edges of K_3 is induced by a vertex automorphism of K_3 , and every permutation of the three edges of $K_{1,3}$ is induced by a permutation of its three leaves. Hence $\alpha|_{E(C)}$ is induced by a vertex isomorphism from C to C' .

If C is not isomorphic to K_3 or $K_{1,3}$, then condition (b) ensures that it is also not isomorphic to $K_{1,3} + e$, $K_4 - e$, or K_4 . Therefore the preceding theorem applies, and $\alpha|_{E(C)}$ is induced by a vertex isomorphism

$$\varphi_C : C \rightarrow C'.$$

Thus, on every component of G containing an edge, the restriction of α is induced by a vertex isomorphism. Since distinct components are disjoint, these componentwise isomorphisms combine to give a vertex automorphism on the union of all nontrivial components of G . The isolated vertices of G are not incident with any edge, so they may be fixed, or permuted arbitrarily among themselves. Extending the componentwise maps in this way gives a vertex automorphism φ of G whose induced edge-automorphism is exactly α .

Hence every edge-automorphism of G is induced by a vertex automorphism. Therefore

$$\Gamma_I(G) \cong \Gamma_E(G).$$

■

Corollary 4.7.3. *If G is a connected graph of order $n \geq 3$, then $\text{Aut}(G)$, $\text{Aut}_I(G)$ and $\text{Aut}_E(G)$ are all isomorphic if and only if G is none of the graphs $K_{1,3} + e$, $K_4 - e$, K_4 .*

Proof. Since G is connected and has order $n \geq 3$, it has no isolated vertices and no component isomorphic to K_2 . Hence, by the preceding theorem,

$$\text{Aut}(G) \cong \text{Aut}_I(G).$$

Also, since G is connected, K_3 and $K_{1,3}$ cannot both occur as components of G . Therefore, by the preceding theorem,

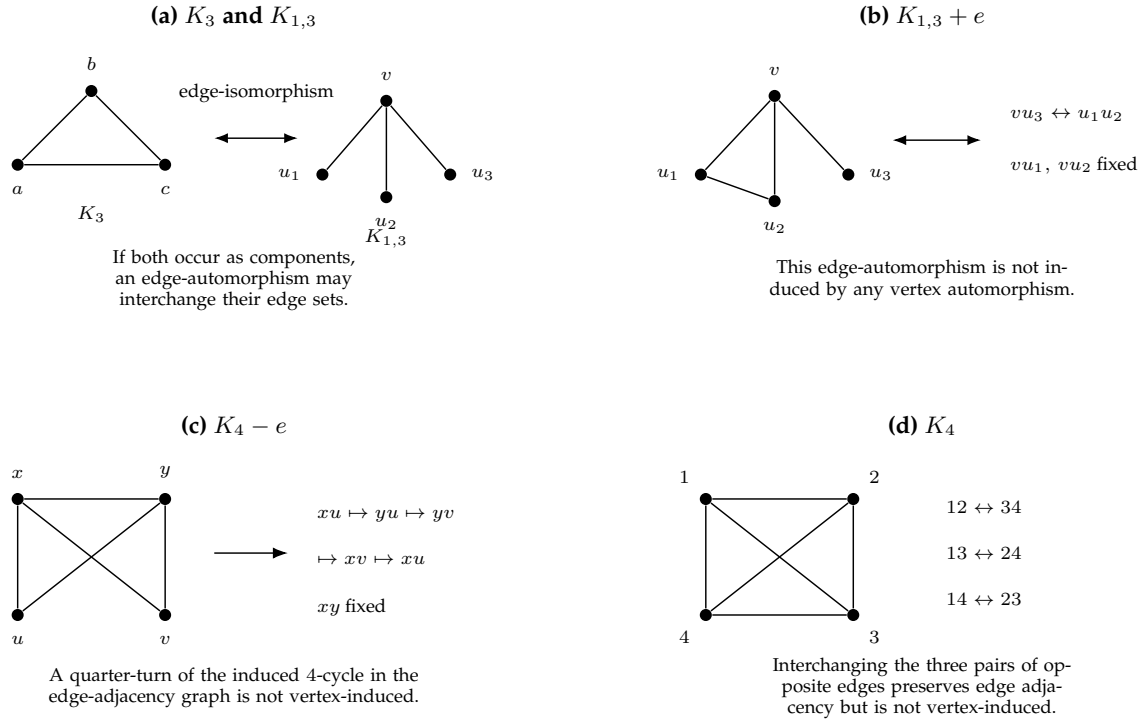
$$\text{Aut}_I(G) \cong \text{Aut}_E(G)$$

if and only if G is none of the graphs $K_{1,3} + e$, $K_4 - e$, and K_4 . Consequently,

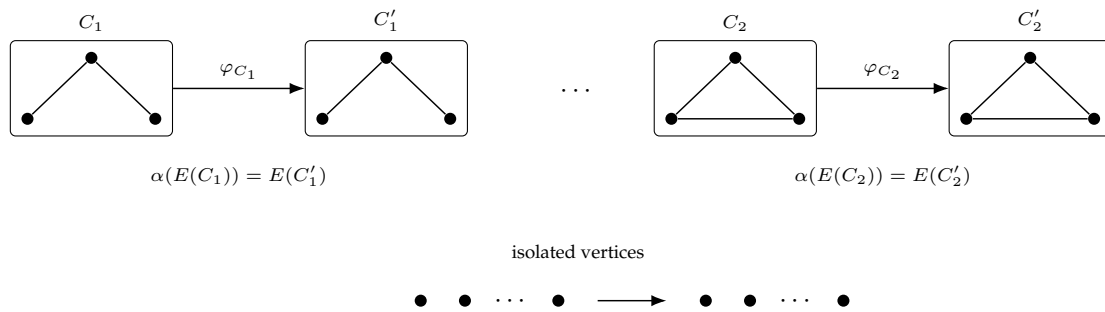
$$\text{Aut}(G) \cong \text{Aut}_I(G) \cong \text{Aut}_E(G)$$

if and only if G is none of the graphs $K_{1,3} + e$, $K_4 - e$, and K_4 .

■



(e) Sufficiency: componentwise construction



Each restriction $\alpha|_{E(C)}$ is induced by a vertex isomorphism $\varphi_C : C \rightarrow C'$. Since the components are disjoint, the maps φ_C combine; the isolated vertices may be permuted arbitrarily.

Figure 4.13: The proof of the criterion for $\Gamma_I(G) \cong \Gamma_E(G)$. Panels (a)–(d) illustrate the exceptional edge-automorphisms appearing in the necessity part. Panel (e) shows the componentwise construction used in the sufficiency part.

4.8 Cayley Graphs of Groups

Let G be a group and let $S \subseteq G$ be a generating set of G . The **Cayley graph** $\Gamma = \text{Cay}(G, S)$ is a graph constructed as follows:

1. The set of vertices $V(\Gamma)$ corresponds to the elements of the group G . That is, $V(\Gamma) = G$.
2. The set of edges $E(\Gamma)$ is defined such that there is a directed edge from g to h if and only if $h = gs$ for some $s \in S$.

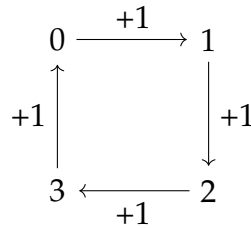
Notation: An edge is typically denoted as (g, gs) and acts as a directed edge colored or labeled by the generator s .

- Usually, the generating set S is chosen such that the identity element $1 \notin S$.
- If S is symmetric (i.e., $s \in S \implies s^{-1} \in S$), the graph is viewed as undirected, because an edge (g, gs) implies the existence of an edge $(gs, gss^{-1}) = (gs, g)$.

Example 4.8.1 (The Cyclic Group $\mathbb{Z}_4$). Let $G = \mathbb{Z}_4 = \{0, 1, 2, 3\}$ under addition modulo 4. Let the generating set be $S = \{1\}$.

- **Vertices:** 0, 1, 2, 3.
- **Edges:** Since the operation is $+$, edges are $(g, g + 1)$.

$$0 \rightarrow 1, \quad 1 \rightarrow 2, \quad 2 \rightarrow 3, \quad 3 \rightarrow 0$$

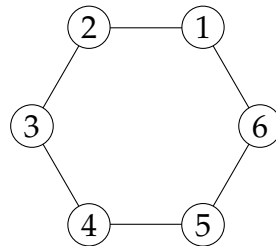


This forms a directed cycle C_4 .

Example 4.8.2 (Cyclic Group $\mathbb{Z}_6$). Let $G = \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ (under addition modulo 6). Let the generating set be $S = \{1\}$.

- **Vertices:** $\{0, 1, 2, 3, 4, 5\}$
- **Edges:** $(i, i + 1 \pmod{6})$.

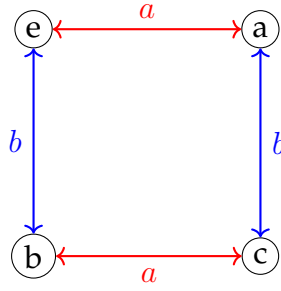
This forms a simple directed cycle C_6 .



Example 4.8.3 (The Klein 4-Group (V_4)). Let $G = V_4 = \{e, a, b, c\}$ where $a^2 = b^2 = c^2 = e$ and $ab = c$. Let the generating set be $S = \{a, b\}$.

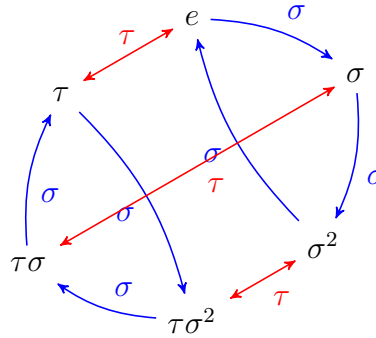
- Edges generated by a (Red): $(e, a), (a, e), (b, c), (c, b)$.
- Edges generated by b (Blue): $(e, b), (b, e), (a, c), (c, a)$.

Since $a = a^{-1}$ and $b = b^{-1}$, for every arrow $x \rightarrow y$, there is an arrow $y \rightarrow x$. We often draw these as undirected edges.



Example 4.8.4 (Symmetric Group S_3). Let $G = S_3$, the permutations of $\{1, 2, 3\}$. Let $S = \{\tau, \sigma\}$ where $\tau = (12)$ and $\sigma = (123)$.

- σ has order 3 (forms hexagons).
- τ has order 2 (forms bi-directional swaps).



Note: The red edges represent τ . Since $\tau^2 = e$, these are technically bidirectional arrows $x \leftrightarrow y$.

4.9 Undirected Cayley Graphs

While the definition implies a digraph (directed graph), we often treat them as undirected graphs under specific conditions.

Definition 4.9.1. $\text{Cay}(G, S)$ is considered an undirected graph if S is inverse-closed. That is, for every $s \in S$, we have $s^{-1} \in S$.

In this case, if there is an edge $g \rightarrow gs$, there is also an edge $gs \rightarrow (gs)s^{-1} = g$. These pairs of directed edges are treated as a single undirected edge.

Theorem 4.9.2. If S generates the group G , then the Cayley graph $\text{Cay}(G, S)$ is connected.

Proof. Let $u, v \in G$ be any two vertices in the graph. We wish to show there is a path from u to v .

Consider the element $g = u^{-1}v$. Since S generates G , the element g can be written as a product of generators and their inverses (if we consider the undirected case or if S is closed under inverses).

$$g = s_1 s_2 \dots s_k \quad \text{where } s_i \in S \cup S^{-1}$$

Therefore, $v = u(s_1 s_2 \dots s_k)$. We can construct a walk starting at u :

$$\begin{aligned} w_0 &= u \\ w_1 &= u s_1 \\ w_2 &= u s_1 s_2 \\ &\vdots \\ w_k &= u s_1 s_2 \dots s_k = v \end{aligned}$$

Since there is a sequence of edges connecting u to v , the graph is connected. ■

Theorem 4.9.3. *The Cayley graph $\Gamma = \text{Cay}(G, S)$ is $|S|$ -regular (in terms of out-degree).*

Proof. Let $g \in G$ be any vertex. The edges originating from g are determined by the definition:

$$E_g = \{(g, gs) \mid s \in S\}$$

Since group cancellation holds ($gs_1 = gs_2 \implies s_1 = s_2$), the number of distinct edges starting at g is exactly equal to the number of distinct elements in S . Thus, the out-degree of every vertex is $|S|$. ■

This is the most fundamental property of Cayley graphs. It implies that the graph "looks the same" from every vertex.

Definition 4.9.4. A graph Γ is said to be *vertex-transitive* if, for every pair of vertices $u, v \in V(\Gamma)$, there exists an automorphism $\phi \in \text{Aut}(\Gamma)$ such that $\phi(u) = v$.

Theorem 4.9.5. *Every Cayley graph $\text{Cay}(G, S)$ is vertex-transitive.*

Proof. Let $\Gamma = \text{Cay}(G, S)$. Thus the vertices of Γ are the elements of the group G , and two vertices $x, y \in G$ are adjacent whenever $y = xs$ for some $s \in S$.

Take arbitrary vertices $u, v \in G$. We shall construct an automorphism of Γ which maps u to v . Put $h = vu^{-1}$ and define $\lambda_h : G \rightarrow G$ by $\lambda_h(x) = hx$. Thus λ_h is simply left multiplication by the group element h .

The map λ_h is a bijection, since its inverse is $\lambda_{h^{-1}}$. We now show that it preserves adjacency. Suppose that x and y are adjacent in Γ . By the definition of a Cayley graph, $y = xs$ for some $s \in S$. Therefore $\lambda_h(y) = hy = h(xs) = (hx)s = \lambda_h(x)s$. Hence $\lambda_h(x)$ and $\lambda_h(y)$ are also adjacent. Thus λ_h is an automorphism of Γ .

Finally, $\lambda_h(u) = hu = (vu^{-1})u = v$. Thus, for every pair of vertices $u, v \in G$, there exists an automorphism of Γ mapping u to v . Hence $\text{Cay}(G, S)$ is vertex-transitive. ■

Definition 4.9.6. Let G be a group and let $\Lambda \subseteq G$ be a generating set. The *Cayley digraph* $\Gamma = D(G, \Lambda)$ has vertex set $V(\Gamma) = G$, and there is an arc from g to k if and only if $k = gh$ for some $h \in \Lambda$. Equivalently, (g, k) is an arc of Γ if and only if $g^{-1}k \in \Lambda$.

If (g, k) is an arc of Γ , then the element $g^{-1}k \in \Lambda$ is called the *color* of the arc (g, k) . Thus, the arc from g to gh , where $h \in \Lambda$, has color h .

An automorphism $\phi \in \text{Aut}(\Gamma)$ is said to be *color-preserving* if it preserves the color of every arc. More precisely, whenever (g, k) is an arc of color h , the image arc $(\phi(g), \phi(k))$ also has color h . Thus, $\phi(k) = \phi(g)h$ then $h = \phi^{-1}(g)\phi(k)$, that is,

$$g^{-1}k = \phi(g)^{-1}\phi(k)$$

for every arc (g, k) of Γ .

Theorem 4.9.7 (Automorphism Theorem for Colored Cayley Digraphs). *Let $D(\Gamma, \Lambda)$ be a Cayley digraph, where Λ generates Γ . Then the group of all color-preserving automorphisms of $D(\Gamma, \Lambda)$ is isomorphic to Γ .*

Proof. Let $C(D)$ denote the group of all color-preserving automorphisms of $D(\Gamma, \Lambda)$. We shall show that the elements of $C(D)$ are precisely the left multiplications by elements of Γ .

For each $g \in \Gamma$, define $\alpha_g : \Gamma \rightarrow \Gamma$ by $\alpha_g(x) = gx$. Since left multiplication by g is a bijection, α_g is a permutation of the vertex set of D .

We first show that α_g preserves the color of every arc. Suppose that (u, v) is an arc of color $h \in \Lambda$. Then $u^{-1}v = h$, or equivalently, $v = uh$. Under α_g , the arc (u, v) is mapped to (gu, gv) . The color of the image arc is

$$(gu)^{-1}(gv) = u^{-1}g^{-1}gv = u^{-1}v = h.$$

Thus the color of every arc is preserved. Hence $\alpha_g \in C(D)$ for every $g \in \Gamma$.

If

$$L(\Gamma) = \{\alpha_g : g \in \Gamma\},$$

then we have $L(\Gamma) \subseteq C(D)$.

We now prove the reverse inclusion. Let $\phi \in C(D)$ be arbitrary, let e denote the identity element of Γ , and suppose that $\phi(e) = g$. We claim that $\phi(x) = gx$ for every $x \in \Gamma$.

Since Λ generates Γ , every vertex can be reached from e by following arcs whose colors are elements of Λ . We show that once the image of one vertex is known, the image of the next vertex along such an arc is completely determined.

Suppose that $\phi(u) = gu$ and that (u, v) is an arc of color $h \in \Lambda$. Since $v = uh$ and ϕ preserves colors, the image arc $(\phi(u), \phi(v))$ also has color h . Therefore $\phi(u)^{-1}\phi(v) = h$. Substituting $\phi(u) = gu$, we obtain $(gu)^{-1}\phi(v) = h$, and hence $\phi(v) = guh = gv$.

Starting with $\phi(e) = g = ge$ and applying this argument successively along a path from e to an arbitrary vertex x , we obtain $\phi(x) = gx$. Therefore $\phi = \alpha_g$, and hence $C(D) \subseteq L(\Gamma)$.

Consequently,

$$C(D) = L(\Gamma).$$

It remains to identify this group with Γ . Define $\Psi : \Gamma \rightarrow C(D)$ by $\Psi(g) = \alpha_g$. The map Ψ is bijective. Moreover, for any $g, k, x \in \Gamma$,

$$(\alpha_g \circ \alpha_k)(x) = \alpha_g(kx) = gkx = \alpha_{gk}(x).$$

Thus $\alpha_g \circ \alpha_k = \alpha_{gk}$, and therefore $\Psi(gk) = \Psi(g)\Psi(k)$. Hence Ψ is a group isomorphism. Consequently,

$$C(D) \cong \Gamma.$$

■

Remark 4.9.8. The essential point of the preceding theorem is that a color-preserving automorphism is completely determined by the image of the identity element. Indeed, if $\phi(e) = g$, then necessarily $\phi(x) = gx$ for every $x \in \Gamma$. Thus the color-preserving automorphisms are exactly the left translations of the group.

Theorem 4.9.9 (Frucht, 1939). *Let Γ be a finite group. Then there exists a finite undirected graph G whose automorphism group is isomorphic to Γ ; that is,*

$$\text{Aut}(G) \cong \Gamma.$$

Proof. Let $\Lambda = \{h_1, \dots, h_n\}$ be a generating set for the finite group Γ , and let $D = D(\Gamma, \Lambda)$ be the corresponding Cayley digraph. Its colour-preserving automorphism group $C(D)$ is isomorphic to Γ . Thus it is enough to construct a finite undirected graph G such that $\text{Aut}(G) \cong C(D)$.

For each directed arc $g_i \rightarrow g_j$ of colour h_k , replace the arc by a path

$$g_i - u - v - g_j.$$

To record the direction and the colour of the arc, attach to u a pendant path of length $2k$ and to v a pendant path of length $2k + 1$. Carrying out this replacement for every arc of D produces a finite undirected graph G .

Every colour-preserving automorphism of D extends naturally to an automorphism of G , since arcs of the same colour are replaced by identical subgraphs. Hence

$$C(D) \leq \text{Aut}(G).$$

Conversely, every automorphism of G preserves the set of original vertices of D , since these vertices are structurally distinguishable from the newly introduced vertices. Moreover, in each replacement subgraph the two pendant paths have different lengths, namely $2k$ and $2k + 1$. Therefore

an automorphism cannot interchange the two ends of the replaced arc, so the direction is preserved. The pair of lengths also determines the index k , and hence the colour h_k . Thus every automorphism of G restricts to a colour-preserving automorphism of D .

It follows that

$$\text{Aut}(G) \cong C(D).$$

Since $C(D) \cong \Gamma$, we conclude that

$$\text{Aut}(G) \cong \Gamma.$$

Hence every finite group is isomorphic to the automorphism group of a finite undirected graph. ■

Detailed Explanation of the Above Proof. Let $\Lambda = \{h_1, h_2, \dots, h_n\}$ be a generating set for the group Γ , and let $D = D(\Gamma, \Lambda)$ be the corresponding Cayley digraph. We have already shown that the group $C(D)$ of colour-preserving automorphisms of D is isomorphic to Γ . Thus it is enough to construct a finite undirected graph G whose full automorphism group is precisely $C(D)$.

The idea is to replace every directed, coloured arc of the Cayley digraph by an undirected subgraph whose shape records both the direction and the colour of that arc.

Consider an arc (g_i, g_j) of D having colour h_k . Thus $g_i^{-1}g_j = h_k$. Replace this arc by a path $g_i - u - v - g_j$ of length three. To distinguish the two internal vertices and to record the colour h_k , attach to u a pendant path P_{2k} of length $2k$, and attach to v a pendant path P_{2k+1} of length $2k + 1$. Performing this replacement for every arc of D produces a finite undirected graph G . The replacement corresponding to an arc $g_i \xrightarrow{h_k} g_j$ is shown in Figure 4.9.

We first show that every colour-preserving automorphism of D induces an automorphism of G . Let $\alpha \in C(D)$. If α maps the arc (g_i, g_j) of colour h_k to the arc (g_x, g_y) , then the latter arc also has colour h_k , since α preserves colours. Consequently, the subgraph replacing (g_i, g_j) and the subgraph replacing (g_x, g_y) have exactly the same structure: in both cases the first internal vertex has an attached path of length $2k$ and the second has an attached path of length $2k + 1$.

Hence α extends naturally to the new vertices of G by mapping the internal vertices and the corresponding pendant paths of one replacement subgraph onto those of the other. This extension preserves all adjacencies and is therefore an automorphism of G . Thus $C(D)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Conversely, let $\beta \in \text{Aut}(G)$. We show that its action on the original vertices induces a colour-preserving automorphism of D .

The original vertices of the Cayley digraph can be distinguished from the newly introduced vertices by their structural position in G . The internal vertices u and v of each replacement subgraph have degree three, while the vertices lying on the attached pendant paths have degree one or two. The original group vertices, on the other hand, are precisely the vertices at which the replacement subgraphs corresponding to the incident arcs meet. Hence every automorphism of G must preserve the set of original vertices.

Now consider the subgraph replacing an arc $g_i \xrightarrow{h_k} g_j$. The vertex adjacent to g_i has an attached path of length $2k$, whereas the vertex adjacent to g_j has an attached path of length $2k + 1$. Since these two path lengths are different, an automorphism cannot interchange the two ends of the replacement subgraph. Thus the asymmetry of the gadget preserves the direction of the original arc.

The same construction also preserves its colour. Indeed, the lengths $2k$ and $2k + 1$ uniquely determine the index k . Therefore a replacement subgraph corresponding to the colour h_k can be mapped only to another replacement subgraph corresponding to the same colour h_k .

It follows that the restriction of β to the original vertices maps arcs of D to arcs of the same direction and the same colour. Hence this restriction is a colour-preserving automorphism of D , and so it belongs to $C(D)$.

Therefore every automorphism of G arises uniquely from a colour-preserving automorphism of D . Consequently,

$$\text{Aut}(G) \cong C(D).$$

Since $C(D) \cong \Gamma$, we conclude that

$$\text{Aut}(G) \cong \Gamma.$$

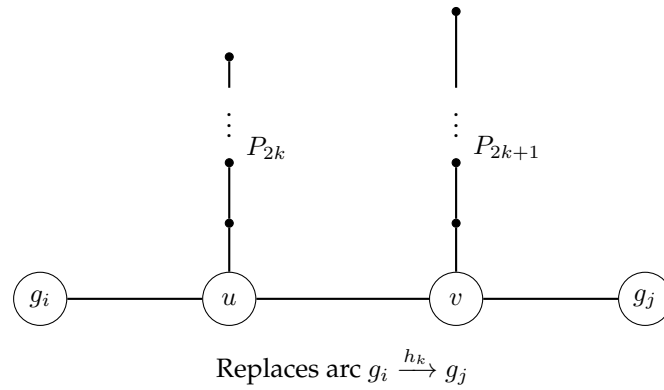


Figure 4.14: Frucht Graph: The subgraph replacing the directed edge from g_i to g_j with color h_k . The asymmetry of the paths P_{2k} and P_{2k+1} enforces the direction and color preservation.

Corollary 4.9.10. Let Γ be a finite group of order n , and let Λ be a generating set of size $s = |\Lambda|$. The Frucht graph $G(\Gamma)$ (Fig 4.13) has exactly $n(s+1)(2s+1)$ vertices.

Proof. To determine the order of the Frucht graph $G(\Gamma)$, we consider the vertices contributed by the original group elements and those introduced by the topological construction. The underlying Cayley digraph $D(\Gamma, \Lambda)$ contains n vertices, and each vertex possesses exactly s outgoing arcs corresponding to the k generators indexed $k = 1$ to s . In the transition to $G(\Gamma)$, each arc associated with the k -th generator is replaced by a subgraph that introduces $4k+1$ new internal vertices. Consequently, for every element in the group, the set of outgoing arcs contributes a total of $\sum_{k=1}^s (4k+1)$ additional vertices. Evaluating this sum yields $4\frac{s(s+1)}{2} + s = 2s^2 + 3s$ new vertices per group element. Across the entire group of n elements, the total number of added vertices is $n(2s^2 + 3s)$. Adding the n original vertices to this count, we find that the total order of the graph is $n(2s^2 + 3s + 1)$. Factoring the quadratic expression, we conclude that $|V(G(\Gamma))| = n(s+1)(2s+1)$. ■

Example 4.9.11. Frucht Graph Order for $n = 3$, and $s = 2$.

Solution. First, calculating directly using the formula:

$$|V(G)| = 3(2+1)(2(2)+1) = 3(3)(5) = 45.$$

We now construct the count step-by-step based on the topological definition of the graph.

Every vertex in Γ has $s = 2$ outgoing arcs, one for each generator. The construction replaces these arcs with "gadgets" of different sizes depending on the index k . The number of internal vertices added for the k -th generator is $4k+1$.

- For generator h_1 ($k = 1$): The gadget adds $4(1) + 1 = 5$ vertices.
- For generator h_2 ($k = 2$): The gadget adds $4(2) + 1 = 9$ vertices.

Since every group element (e, x, y) has exactly one outgoing arc for h_1 and one outgoing arc for h_2 , the total number of vertices added **per group element** is:

$$\text{Added per vertex} = 5 + 9 = 14.$$

Therefore,

$$\begin{aligned}\text{Total Vertices} &= n + n \sum_{k=1}^2 (4k + 1) \\ &= 3 + 3[(4(1) + 1) + (4(2) + 1)] \\ &= 3 + 3[5 + 9] \\ &= 3 + 42 \\ &= 45.\end{aligned}$$

■